

## **UNIT-1**

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**Lathe:** Classification, Principle of operation, Parts of a centre Lathe, lathe specification, Lathe operations: Turning, facing, knurling, thread cutting, Taper Turning by swivelling compound rest.

**Drilling machine:** Principle and Classification of drilling machines, Bench drilling machine, Radial drilling machine, Operations on drilling machine-Drilling, Boring, Reaming, Tapping, Counter Sinking, Counter boring and Spot facing.

**Computer Numerical Control (CNC):** Introduction, Components of CNC, open loop and closed loop systems, advantages and disadvantages of CNC.

**Robotics:** Introduction, Classification based on robot configuration: Polar, Cylindrical, Cartesian and jointed arm configuration. Application, Advantages and disadvantages of robots

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## 1.1 Lathe

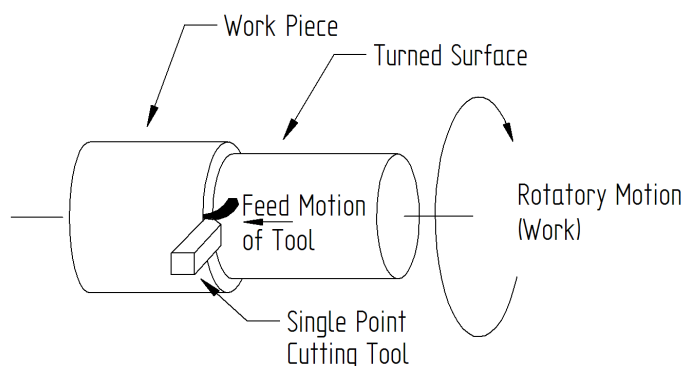
### 1.1.1 Machining: Introduction

Machining is a manufacturing process that involves removing materials using cutting tools to get rid of unwanted materials from some workpiece and converting it into the shape you desire. A large piece of stock is used for cutting the workpiece. The large stock might be in any shape such as solid bar, flat sheet, beam or even hollow tubes. The process can also be performed on some existing part like forging or casting. Machining is necessary where tight tolerances on dimensions and finishes are required. Machining processes include Turning (using Lathe), Drilling, Milling, Grinding etc.

### 1.1.2 Lathe

A lathe is defined as a machine tool primarily used to produce circular objects and is used to remove excess material by forcing a cutting tool against a rotating workpiece. Lathes are also called turning machines, since the workpiece is turned or rotated between two centres. Since it is so versatile, that almost all the machining operations which are performed on other machine tools like, drilling, grinding, shaping, milling, etc., can be performed on it. It is said to be the *mother* of all the machine tools. The lathe is the oldest of all machine tools and the most basic tool used in industries.

### 1.1.3 Working principle of lathe



In a lathe, the workpiece is held in chuck and rotates about its axis by means of power. A single point cutting tool is mounted in tool post. The tool moves parallel to the

axis of rotation of work piece to produce a cylindrical surface, whereas the tool moves perpendicular to the work piece to produce a flat surface. The tool moves at an angle to the axis of work piece to produce a turn surface. The material is removed in the form of chip from the work piece by giving proper feed and depth of cut. So, the required size and shape of the work is obtained.

#### 1.1.4 Definition of Lathe

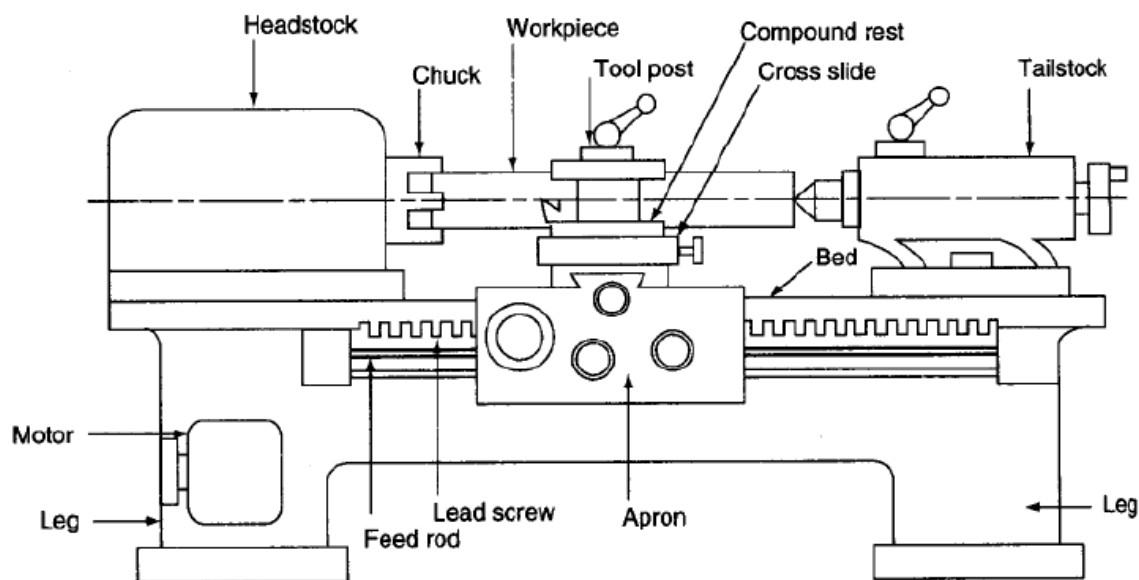
Lathe is a machine tool used to remove metal from the work piece, to a required shape and size.

#### 1.1.5 Lathe Classification

1. **Bench lathe:** This is a very small machine mostly used by wrist watch manufacturers. It is mount on a cabinet or bench, used for small and precision work that requires high accuracy. It contains all parts the larger lathe carries and will perform almost the same operation with the larger ones.
2. **Speed lathes:** As the name implies, speed lathes are named because of the very high speed at which the spindle rotates. These types of lathe machine are used in wood turning, polishing, centering and metal spinning. They're either available with supporting legs, fitted to the bed and the one mounted on bench.
3. **Engine lathe/ Center lathes:** Engine lathe is one of the most used lathes. The name is quite confusing right, after all lathes are available with a motor drive. Well, the engine lathe carries a great historical significance because in the very early days it is driven by steam engine. Because of this, it is popularly known engine lathe today.  
The construction of engine lathe is bigger in size and more robust, and a large mechanism for providing multiple speed to the lathe spindle.
4. **Tool room lathe:** These types of lathe machine are the same with the engine lathe but has some extra features. One of this feature is to make it suitable for a relatively more accurate angle of speeds and feeds. However, the ideal features of the tool room lathe include, taper turning, follower rest, collets, chucks, etc.
5. **Capstan and turret lathe:** These types of lathes are widely used in mass production and it's very important due to its features. They are actually semi-automatic type and various operations can be performed on them.

6. **Automatic lathe:** Automatic lathe is known for their enhancement in quality and the quantity of production. This machine is so perfect that the complete manufacturing process is automatically achieved. This means, no participation of the operator is required during the operation.

### 1.1.6 Parts of an Engine Lathe (Centre Lathe)



#### Bed

The bed is the foundation part of a lathe. It is usually made from **Gray cast iron**. All the major components are mounted on the lathe bed, like tail stock, headstock, carriage, etc. The top of the bed is formed by machine guideways. **Tailstock and headstock** are mounted on the *inner guide ways*. The *outer guideways guide* the longitudinal movement of the **carriage assembly**.

#### Headstock

Headstock is mounted on the left-hand side of the lathe bed. The head stock is hollow accommodates gear box, which helps to vary the spindle speed. The gear box also transmits the power to other parts like feed rod and lead screw. The chuck or face plate is attached to the spindle which provides mechanical means clutching and rotating the work piece. The headstock contains speed and feed changing levers. Head stock is also known as *live center*

#### Tailstock

The tailstock is mounted on the right hand side of the lathe bed. The function of the tailstock is to support the work piece, and to accommodate different tools like drill, reaming, boring and tapping, etc. The tailstock moves on the guide ways over the bed, to accommodate for different length of work piece. Tailstock is known as dead center.

## **Carriage**

The carriage is mounted on the lathe bed, which slides on the outer guide ways of the bed. The carriage has various other parts like, saddle, cross slide, compound rest, tool post and apron.

### **i) Saddle**

The saddle is an H shaped casting that slides over the outer set of guide ways and serves as the base for cross slide. The movement of the saddle is parallel along the axis of the lathe, which is known as feed.

### **ii) Cross slide**

The cross slide is mounted on the top of the saddle. This moves the tool at perpendicular to the work piece or machine axis. The cross slide can be moved either by rotating the cross-slide hand wheel or engaged with the apron mechanism (Automatic movement). The perpendicular distance moved by the cross slide is proportional to the amount of metal removed and it is known as depth of cut.

### **iii) Compound slide**

The compound slide (compound rest) is mounted on the top of the cross slide. The rest part of the compound slide has graduations in degree. Compound slide is used to obtain taper on the work piece, even this helps to fix the tool right angle to machine axis.

### **iv) Tool post**

The tool post is mounted on top of the compound slide. The tool post holds the tool rigidly.

### **v) Apron**

The apron is fastened to the saddle and hangs over the front of the bed. It is fitted with mechanism for both manual and powered movement of the

saddle and the carriage. Split nut engages the Apron with lead screw, which is used for thread cutting operations.

**vi) Feed rod**

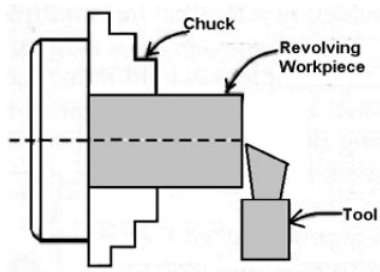
Feed rod is a long shaft extending from the feed box. The power is transmitted from a set of gears from headstock. The feed rod is used to move the carriage or cross slide for turning, boring and facing operations.

**vii) Lead screw**

The lead screw is a long-threaded shaft connected to the headstock. The lead screw is used only when thread cutting operation is to be carried out on the work piece. For normal turning operations the lead screw is disengaged.

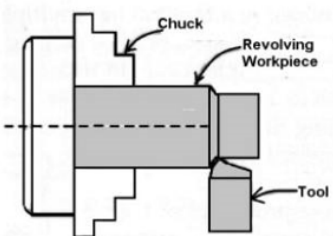
### 1.1.7 Lathe Operations

#### Facing



Facing is the process of removing metal from the end of a work piece to produce a **flat surface**. Facing is also performed to **reduce the length** of the workpiece. The work piece rotates about its axis and the facing tool is fed perpendicular to the axis of lathe.

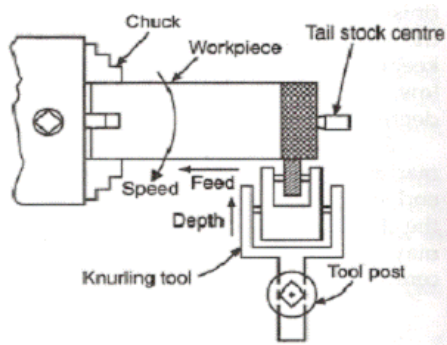
#### Turning/Straight Turning /Cylindrical Turning/Plain turning



Turning is the operation of removing the excess material from the workpiece **to produce a cylindrical surface** to the desired length. The job held between the centre or a chuck and rotating at a required speed. The tool moves in a longitudinal direction to give the feed towards the headstock with proper depth of cut.

Turning is used to **reduce the diameter of the work piece**, usually to a specified dimension, and to produce a smooth finish on the metal. Often the work piece will be turned so that adjacent sections have different diameters.

### Knurling



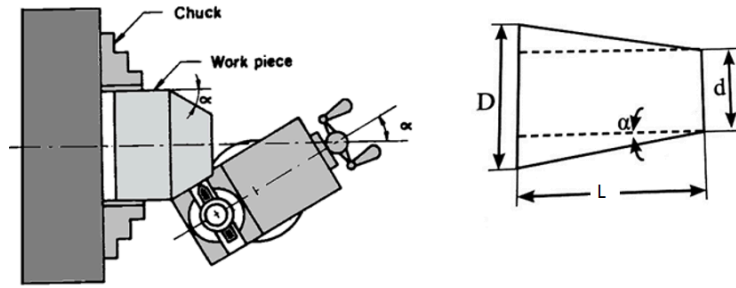
Knurling is the process of **embossing a required shaped pattern** (Straight, Diamond shaped etc.) on the surface of the work piece. Knurling provides an **effective gripping** surface on a job to prevent it from slipping when operated by hand. The lathe is set so that the chuck revolves at a low speed. The knurling tool is then pressed against the rotating work piece and pressure is slowly increased until the tool produces a pattern on the work piece.

### Taper turning

A taper may be defined as a uniform increase or decrease in diameter of work piece measured along its length.

#### Taper Turning by Swivelling Compound Rest

In this method of taper the half taper angle is calculated. The compound rest has rotating base graduated in degrees, which can be rotated to any angle (according to the taper angle). In this method the tool is advanced by rotating the compound rest and hand wheel so that the tool moves according to set taper angle.



The angle at which the compound rest to be swiveled is calculated using the equation given by  $\alpha = \left( \frac{D-d}{2L} \right)$

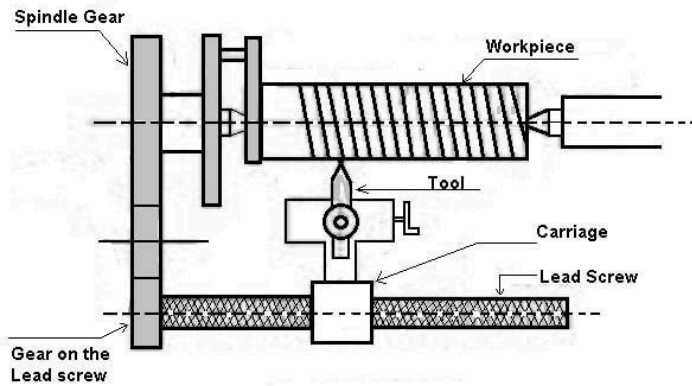
where  $\alpha$  = Half taper angle, D= Large diameter, d= Small diameter, L= Length of taper

### Thread Cutting

Thread cutting is the operation of producing helical groove on a cylindrical surface. Threads may be square or v threads can be cut on a cylindrical work piece. The threads of any pitch, shape and size can be cut on a lathe. A single point cutting tool (V-tool or square tool) is used to cut threads on the work piece.

For thread cutting operation, the tool is moved automatically in longitudinal direction. The longitudinal feed should be equal to the pitch of the thread to be cut per revolution of the work. The lead screw has a fixed pitch. So, ratio between the rotation of work piece and the longitudinal feed is found out. The work piece and lead screw are connected by a set of gears. The lead screw is engaged to the carriage by closing the half nut lever.

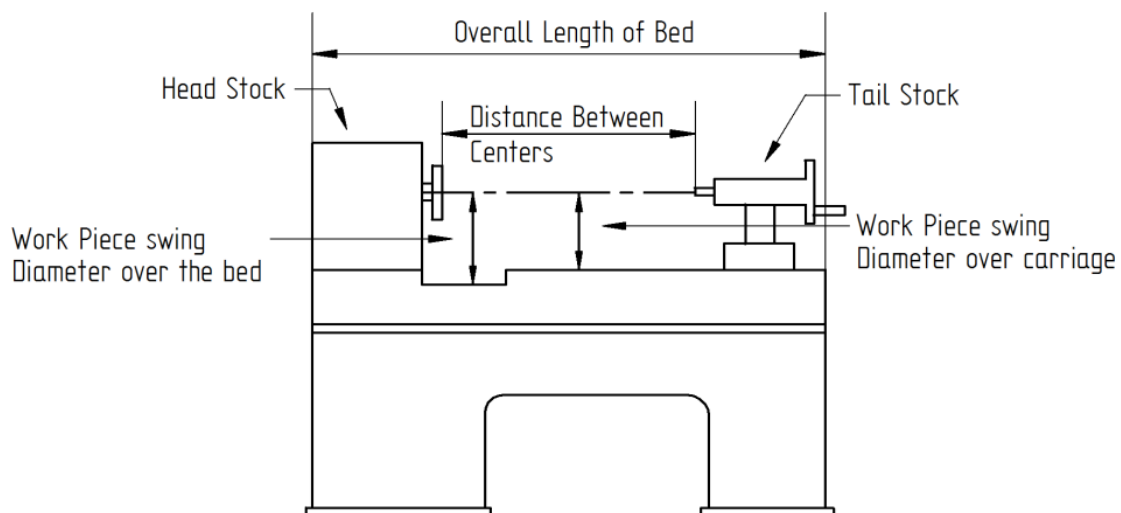




## Thread Cutting

During thread cutting, both work piece and lead screw rotate at the same speed. The pitch of the lead screw is equal to pitch of work piece. To cut the threads, the tool is brought in contact with work piece. The carriage is brought in contact with lead screw by operating half nut lever. The tool is moved along the axis, generates the threads on the work piece. This process is repeated several times till the required depth, pitch and finish is obtained.

### 1.1.8 Specification of a Lathe



A Lathe is generally specified by:

1. **Swing diameter-** It is the maximum diameter of the workpiece that can revolve without touching the guideways. Some manufacturers specify Height of centers instead of Swing diameter.
2. **The distance between the headstock and tailstock center:** It is the maximum length of the job that can be held between the centers.
3. **Length of the bed in a meter:** It indicates the approximate floor space occupied by the lathe.
4. The pitch of the lead screw.
5. Horsepower of the machine.
6. Speed range and the number of speeds of HS spindle.
7. The weight of the machine in a tonne

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## 1.2 Drilling

### 1.2.1 Introduction

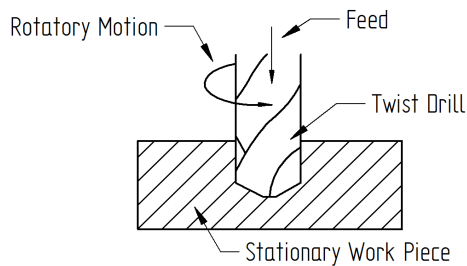
Drilling is a metal cutting process carried out by a rotating cutting tool to make circular holes in solid materials. The tool which makes the hole is called a drill bit. It is generally called as twist drill, since it has a sharp twisted edge formed around a cylindrical tool provided with a helical groove along its length to allow the cut material to escape through the sharp edges of the conical surfaces ground at the lower end of the rotating twist drill cuts the material by peeling it circularly layer by layer when forced against a work piece. The removed material chips get curled and escapes through the helical groove provided in the drill. A liquid coolant is generally used while drilling to remove the heat of friction and obtain a better finish for the hole.

**Note:** Drill bit/twist drill is a multipoint cutting tool

**Working principle:** The rotating edge of the drill bit exerts a large force on the work-piece and the hole is generated. Here the work-piece is rigidly clamped and the drill bit is forced through the work-piece to cut required hole. The material removed in form of chips gets curled and escapes through the helical grooves on the drill bit. The removal of metal in a drilling operation is by shearing and extrusion



**Definition:** Drilling is the operation of making holes in a work piece using a drill bit or twist drill. The hole generated by the rotating drill, which exerts large force on the work piece clamped rigidly on the machine table. The figure below illustrates a cross section of a hole being cut by a common twist drill.



### 1.2.2 Types of Drilling Machines

Drilling machines are manufactured in different types and sizes according to the type of operation, amount of feed, depth of cut, spindle speeds, method of spindle movement and the required accuracy. The different types of drilling machines are:

1. Portable drilling machine (or) Hand drilling machine
2. Sensitive drilling machine (or) Bench drilling machine
3. Upright drilling machine
4. Radial drilling machine
5. Gang drilling machine
6. Multiple spindle drilling machine
7. Deep hole drilling machine

**Portable drilling machine:** Portable drilling machine can be carried and used anywhere in the workshop. It is used for drilling holes on workpieces in any position, which is not possible in a standard drilling machine. The entire drilling mechanism is compact and small in size and so can be carried anywhere. This type of machine is widely adapted for automobile built-up work. The motor is generally universal type.

These machines can accommodate drills from 12mm to 18 mm diameter. Portable drilling machines are operated at higher speeds.

**Sensitive drilling machine:** It is designed for drilling small holes at high speeds in light jobs. High speed and hand feed are necessary for drilling small holes. The base of the machine is mounted either on a bench or on the floor by means of bolts and nuts. It can handle drills up to 15.5mm of diameter

**Upright drilling machine:** The upright drilling machine is designed for handling medium sized workpieces. Though it looks like a sensitive drilling machine, it is larger and heavier than a sensitive drilling machine. Holes of diameter upto 50mm can be made with this type of machine.

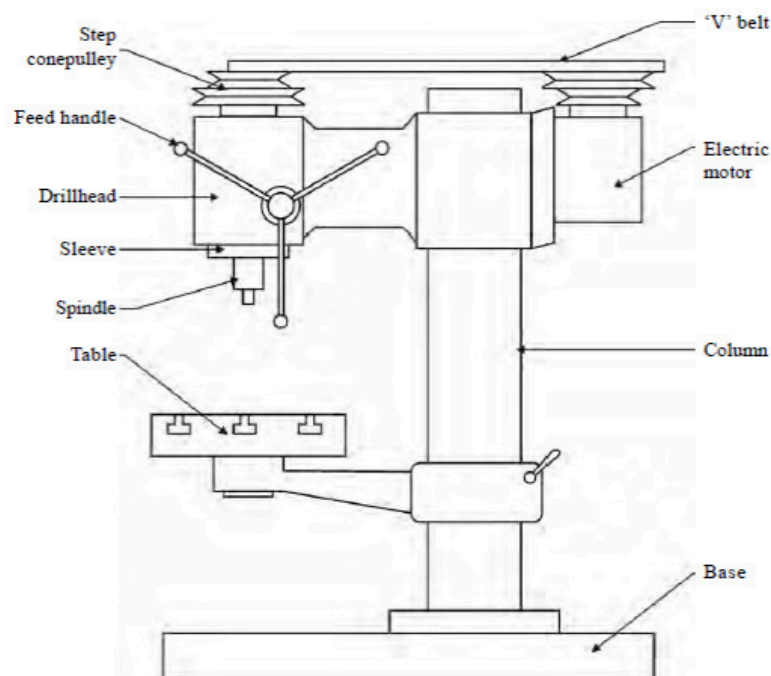
**Radial drilling machine:** The radial drilling machine is intended for drilling on medium to large and heavy workpieces. It has a heavy round column mounted on a large base. The column supports a radial arm, which can be raised or lowered to enable the table to accommodate workpieces of different heights. The arm, which has the drill head on it, can be swung around to any position

**Gang drilling machine:** Gang drilling machine has a long common table and a base. Four to six drill heads are placed side by side. The drill heads have separate driving motors. This machine is used for production work. A series of operations like drilling, reaming, counter boring and tapping may be performed on the work by simply shifting the work from one position to the other on the work table. Each spindle is set with different tools for different operations.

**Multiple spindle drilling machine:** This machine is used for drilling a number of holes in a workpiece simultaneously and for reproducing the same pattern of holes in a number of identical pieces. A multiple spindle drilling machine also has several spindles. A single motor using a set of gears drives all the spindles. All the spindles holding the drills are fed into the work at the same time. The distances between the spindles can be altered according to the locations where holes are to be drilled. Drill jigs are used to guide the drills.

**Deep hole drilling machine:** A special machine and drills are required to drill deeper holes in barrels of gun, spindles and connecting rods. The machine designed for this purpose is known as deep hole drilling machine. High cutting speeds and less feed are necessary to drill deep holes. A non-rotating drill is fed slowly into the rotating work at high speeds. Coolant should be used while drilling in this machine. There are two different types of deep hole drilling machines

### 1.2.3 Bench Drilling Machine (Sensitive Drilling Machine)



It is a lightweight-drilling machine mounted on a bench used to carry out light drilling operations. These machines are also known as sensitive drilling machine. Bench drilling machines can drill holes in the range of 0.35 mm to 15 mm in diameter. The drill bit is fed into the work piece by hand using feed handle. The operator gets to know or he can feel the travel of the drill into the work piece. Hence, the machine is known as a sensitive drilling machine.

#### Construction

The main parts of the machine are;

**Base:** The base is made of cast iron and so can withstand vibrations. It may be mounted on a bench or on the floor. It supports all the other parts of the machine on it.

**Column:** The column stands vertically on the base at one end. It supports the work table and the drill head. The drill head has drill spindle and the driving motor on either side of the column.

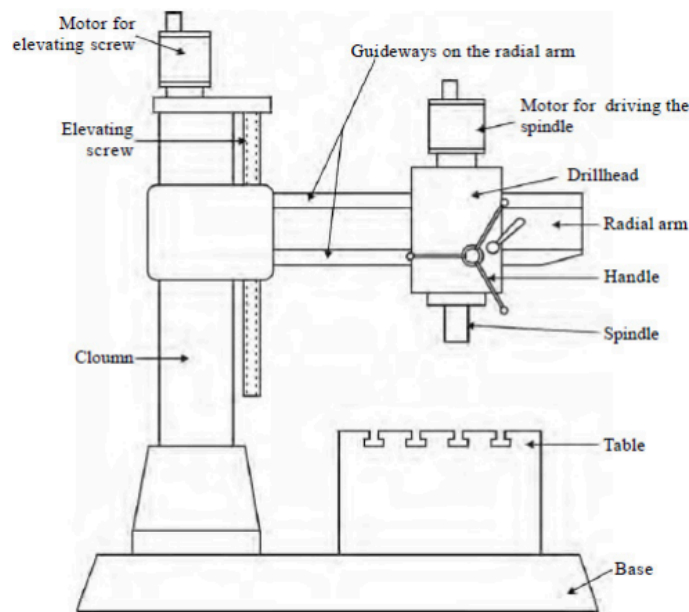
**Worktable:** The worktable is mounted on the vertical column and can be adjusted up and down on it. The table has "T"-slots on it for holding the workpieces or to hold any other work holding device. The table can be adjusted vertically to accommodate workpieces of different heights and can be clamped at the required position.

**Drill head:** Drill head is mounted on the top side of the column. The drill spindle and the driving motor are connected by means of a V-belt and cone pulleys. The motion is transmitted to the spindle from the motor by the belt. The pinion attached to the handle meshes with the rack on the sleeve of the spindle for providing the drill the required down feed. There is no power feed arrangement in this machine. The spindle rotates at a speed ranging from 50 to 2000 rpm.

### **Operation/Working**

In operation, the workpiece is clamped rigidly on the worktable. With the help of center punch, an indentation mark is made on the workpiece at the location where the hole is to be drilled. The drill bit is made to rotate at a suitable speed by adjusting the v-belt on the stepped cone pulley. With the help of hand feed lever, the drill bit is moved downwards till the drill bit touches the indentation mark. Gradual feed to the drill bit is given by operating feed lever till the desired depth of the hole is achieved. The drill bit is withdrawn slowly from the workpiece by operating the hand feed lever in the reverse direction.

### 1.2.4 Radial Drilling Machine



The radial drilling machine is intended for drilling on medium to large diameter holes of up to 50mm in heavy workpieces. It has a heavy round column mounted on a large base. The column supports a radial arm, which can be raised or lowered to enable the table to accommodate workpieces of different heights. The arm, which has the drill head on it, can be swung around to any position. The drill head can be made to slide on the radial arm. The machine is named so because of this reason.

#### Construction

The main parts of the machine are;

**Base:** Base is the main foundation; a heavy circular column is mounted on the base. The table is mounted on the top face of the base. The base should be strong enough to withstand the cutting forces, which are transmitted from the table to the base during operation.

**Column:** The column supports for the radial arm. The cylindrical portion of the column should have very good surface finish. On the cylindrical portion the column the radial arm can be lowered or raised, and even the radial arm can be swing to any desire angle on the column. On the top of the column a motor is mounted which helps to raise or to lower the radial arm on the vertical column.

**Radial arm:** The radial arm should be strong enough to carry drill head, motor and other accessories mounted on the radial arm. The drill head can be moved to and fro on the face (guide ways) of the radial arm. The drill head contains a spindle which can be raised or lowered with respect to the work piece.

**Table:** The table is mounted on the base of the machine. The work piece may be mounted on the table or when the work is very large it may be placed on the floor.

**Drill head:** Drill head is mounted on radial arm and carries a driving motor and a mechanism for revolving and feeding the drill bit into the workpiece.

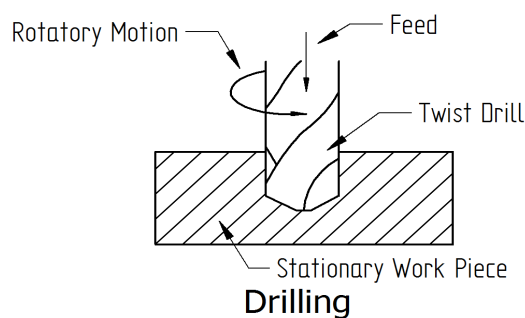
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With the combination of the movements of radial arm and the drill head, it is possible to move the drill bit and hence generate a hole at any desired position without moving the workpiece.

## 1.2.5 Drilling Operations

### Drilling



Drilling is a cutting process that uses a drill bit to produce a hole of circular cross section in solid materials. The drill bit is a rotary cutting tool, often multipoint. Before drilling a hole, the center point of the hole has to be marked on the work piece. The center point of the hole is marked by just drawing two cross lines or by using instruments. The mark is indented using a center punch. The hole to be drilled may be a





to increase the size of a hole at one end only through a small hole. The counter boring forms a larger sized recess or a shoulder to the hole. The counter boring tool will have a small cylindrical projection known as pilot or pilot drill. The diameter of the pilot will always be equal to the diameter of the pre-drilled hole. Interchangeable pilots of different diameters are available for counter boring holes of different diameters. The speeds for counter boring are the same as of the drilling speed the corresponding size of the drilled hole. Counter boring is done on the holes to accommodate the socket head or round head bolts.



g Tool

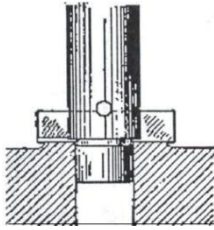
ted Hole

rk Piece

the operation of making a conical shaped at the top of the hole  
indrical hole. Countersinking is done to fit in a screw or a  
of the hole is conical in shape compared to square shape in  
hole is drilled in the work piece using a drill bit. Then the  
d to make cone shaped hole at the top.

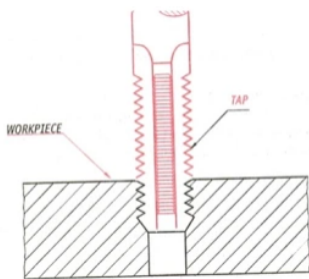
1. **Introduction**  
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 6. **Conclusion**  
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Spot facing is the operation of smoothing and squaring the surface around a hole for the seat for bolt head or nut. This operation is done with a special tool called facing tool.

## Tapping



The tapping is the process of cutting internal threads with a thread cutting tool called tap. A tap is a fluted threaded tool used for cutting internal threads. Before tapping, a hole which is slightly smaller than the size of the tap is drilled. For cutting the threads, the tap is fitted in the tapping attachment which in turn is mounted in the drilling machine spindle, and the threads are cut in the same way as drilling. While tapping in a drilling machine the spindle has to rotate at very slow speeds.

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## 1.3 Computer Numerical Control (CNC)

**1.3.1 Introduction** Modern precision manufacturing demands extreme dimensional accuracy and surface finish. Such performance is very difficult to achieve manually, if not impossible, even with expert operators. In cases where it is possible, it takes much higher time due to the need for frequent dimensional measurement to prevent overcutting. It is thus obvious that automated motion control would replace manual “hand wheel” control in modern manufacturing. Development of computer numerically controlled (CNC) machines has also made possible the automation of the

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In the 1940s when the U.S. Air Force perceived the need to manufacture complex parts for highspeed aircraft. This led to the development of computer-based automatic machine tool controls also known as the Numerical Control (NC) systems. Commercial production of NC machine tools started around the fifties and sixties around the world. Note that at this time the microprocessor has not yet been invented.

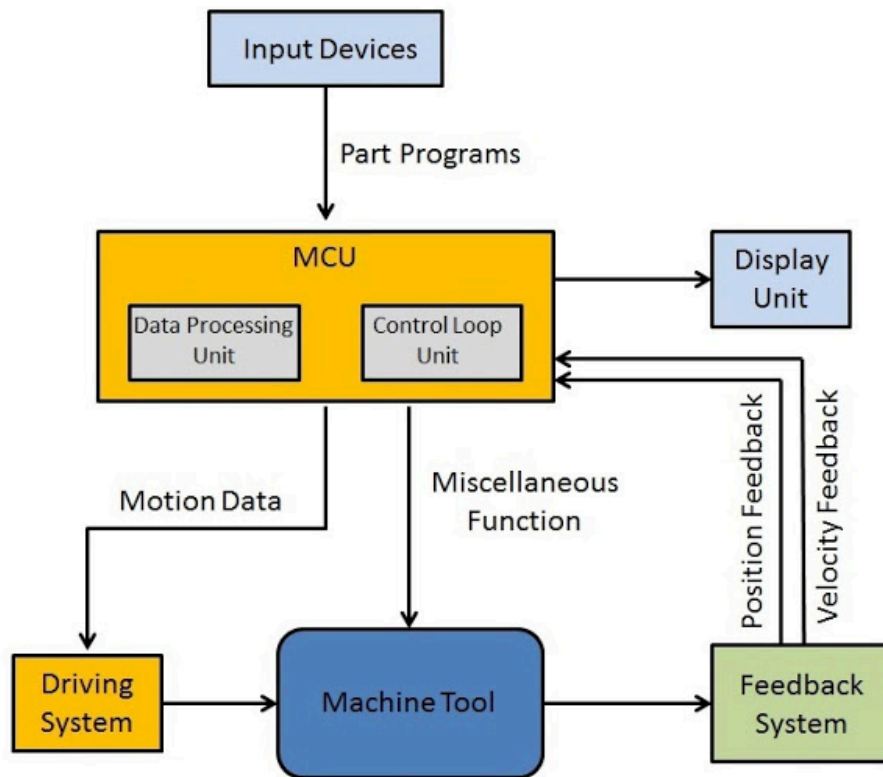
Initially, the CNC technology was applied on lathes, milling machines, etc. which could perform a single type of metal cutting operation. Later, attempt was made to handle a variety of workpieces that may require several different types machining operations and to finish them in a single set-up. Thus, CNC machining Centres capable of performing multiple operations were developed. To start with, CNC machining centres were developed for machining prismatic components combining operations like milling, drilling, boring and tapping. Gradually machines for manufacturing cylindrical components, called turning centers were developed.

### 1.3.2 What is CNC?

CNC means **Computer Numerical Control**. This means a computer converts the design produced by Computer Aided Design software (CAD), into numbers. The numbers can be considered to be the coordinates of a graph and they control the movement of the cutter. In this way the computer controls the cutting and shaping of the material.



### 1.3.3 Components of CNC



Any CNC machine tool essentially consists of the following parts:

#### 1. Part program:

A part program is a series of coded instructions required to produce a part. It controls the movement of the machine tool and on/off control of auxiliary functions such as spindle rotation and coolant. The coded instructions are composed of letters, numbers and symbols.

#### 2. Program input device:

The program input device is the means for part program to be entered into the CNC control. Three commonly used program input devices are punch tape reader, magnetic tape reader, and computer.

#### 3. Machine Control Unit:

The machine control unit (MCU) is the heart of a CNC system. It is used to perform the following functions:

- To read the coded instructions.
- To decode the coded instructions.

- c. To implement interpolations (linear, circular, and helical) to generate axis motion commands.
- d. To feed the axis motion commands to the amplifier circuits for driving the axis mechanisms.
- e. To receive the feedback signals of position and speed for each drive axis.
- f. To implement auxiliary control functions such as coolant or spindle on/off and tool change.

#### 4. **Drive System:**

A drive system consists of amplifier circuits, drive motors, and ball lead-screws. The MCU feeds the control signals (position and speed) of each axis to the amplifier circuits. The control signals are augmented to actuate drive motors which in turn rotate the ball lead-screws to position the machine table.

#### 5. **Machine Tool:**

CNC controls are used to control various types of machine tools. Regardless of which type of machine tool is controlled, it always has a slide table and a spindle to control of position and speed. The machine table is controlled in the X and Y axes, while the spindle runs along the Z axis.

#### 6. **Feed Back System:**

The feedback system is also referred to as the measuring system. It uses position and speed transducers to continuously monitor the position at which the cutting tool is located at any particular instant. The MCU uses the difference between reference signals and feedback signals to generate the control signals for correcting position and speed errors.

### 1.3.4 Open Loop Systems

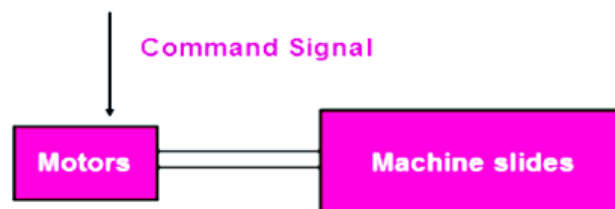


Figure: Open Loop System

The term open-loop means that there is no feedback, and in open loop systems the motion controller produces outputs depending only on its set points, without feedback information about the effect that the output produces on the motion axes. We have already seen that the effects of controller outputs on the plant may not be the same always, since it depends on factors such as loads, parameter variations in the plant etc. In open loop systems, the set points are computed from the instructions in the Part program and fed to the controller, which may reside in a different microprocessor, through an interface. These motion commands may be in the form of electrical pulses (typical for step motor drives) or analog or digital signals, and converted to speed or current set points by the controller. These setpoints, in turn, are sent to the power electronic drive system that applies the necessary voltage/current to the motors.

The primary drawback of open-loop system is that there is no feedback system to check whether the commanded position and velocity has been achieved. If the system performance is affected by load, temperature or friction then the actual output could deviate from the desired output.

For these reasons, the open-loop system is generally used in point-to-point systems where the accuracy requirements are not critical. Contouring systems do not use open-loop control.

### 1.3.5 Closed loop systems:

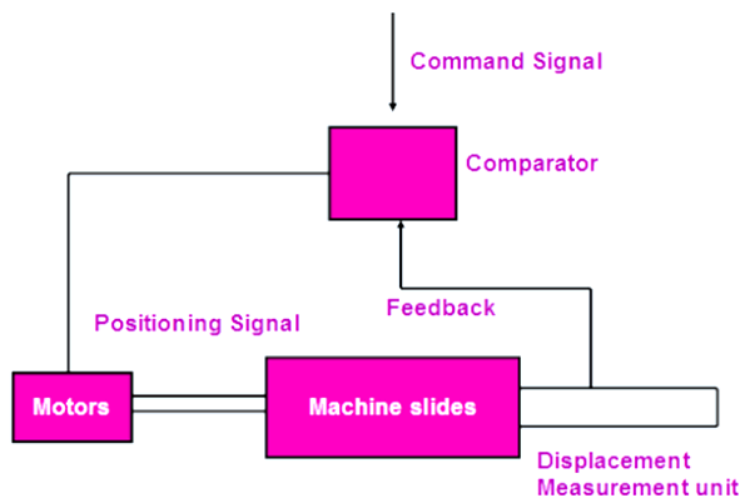


Figure: Closed Loop System

The closed-loop system has a feedback subsystem to monitor the actual output and correct any discrepancy from the programmed input. These systems use position and velocity feedback. The feedback system could be either analog or digital. The analog systems measure the variation of physical variables such as position and velocity in terms of voltage levels. Digital systems monitor output variations by means of electrical pulses. To control the dynamic behaviour and the final position of the machine slides, a variety of position transducers are employed. Majority of CNC systems operate on servo mechanism, a closed loop principle. If a discrepancy is revealed between where the machine element should be and where it actually is, the sensing device signals the driving unit to make an adjustment, bringing the movable component to the required location.

Closed-loop systems are very powerful and accurate because they are capable of monitoring operating conditions through feedback subsystems and automatically compensating for any variations in real-time.

### 1.3.6 Advantages of a CNC Machine

CNC machines offer the following advantages in manufacturing.

1. **Higher flexibility:** This is essentially because of programmability, programmed control and facilities for multiple operations in one machining center
2. **Increased productivity:** Due to low cycle time achieved through higher material removal rates and low set up times achieved by faster tool positioning, changing, automated material handling etc.
3. **Improved quality:** Due to accurate part dimensions and excellent surface finish that can be achieved due to precision motion control and improved thermal control by automatic control of coolant flow.
4. **Reduced scrap rate:** Use of Part programs that are developed using optimization procedures





5. **Reliable and Safe operation:** Advanced engineering practices for design and manufacturing, automated monitoring, improved maintenance and low human interaction
6. **Smaller footprint:** Due to the fact that several machines are fused into one.

### 1.3.7 Disadvantages of a CNC Machine

1. Relatively higher cost compared to manual versions
2. More complicated maintenance due to the complex nature of the technologies
3. Need for skilled part programmers.

The above disadvantages indicate that CNC machines can be gainfully deployed only when the required product quality and average volume of production demand it.

\*\*\*\*\*

## 1.4 Robotics

**1.4.1 Introduction** Robot is defined as "a reprogrammable, multifunctional manipulator designed to move material, parts, tools, or specialized devices through various programmed motions for the performance of a variety of tasks"

Robots designed for industrial purposes are called industrial robots

An industrial robot is a manipulator designed to move materials, parts and tools, and perform a variety of programmed tasks in manufacturing and production settings.

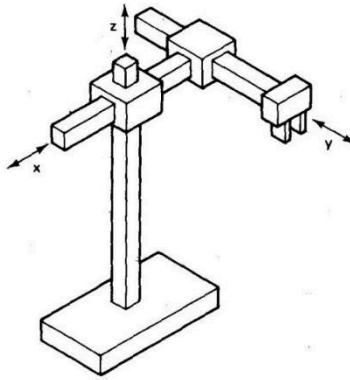
### 1.4.2 Classification of Robots based on their configuration

Robots may be classified based on their physical configuration as follows

1. Cartesian or rectangular co-ordinate configuration robots
2. Polar or Spherical configuration robots
3. Cylindrical configuration robots
4. Articulating or jointed-arm configuration robots

#### 1.4.2.1 Cartesian or rectangular co-ordinate configuration robots





Cartesian robot is formed by **3 prismatic joints**, whose axes are coincident with the X, Y and Z planes. Cartesian robots are also called as **Rectilinear robots or XYZ robots**. It consists of 3 sliding joints along the X, Y and Z directions in three-dimensional space. There are **two orthogonal joints**.

In the Cartesian coordinate configuration shown in figure, the three orthogonal directions are X,Y and Z. X-coordinate axis may represent left and right motion; Y-coordinate axis may describe forward and backward motion; Z-coordinate axis may be used to represent up and down motions. The manipulator can reach any point in a cubic volume of space. It allows three DOFs (x, y, z) in translation only.

#### **Advantages:**

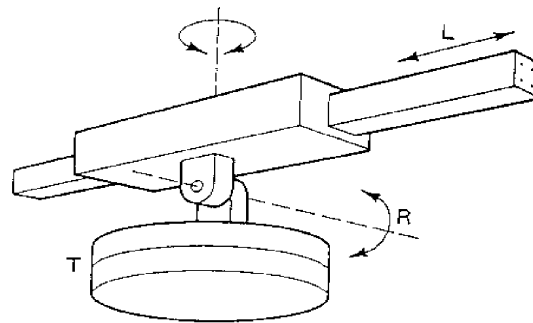
- 1) 3 linear axes.
- 2) Easy to visualize, ability to do straight line insertions into furnaces.
- 3) Most rigid structure for given length.
- 4) Easy computation and programming.

#### **Disadvantages:**

- 1) Can only reach front of it.
- 2) Requires large floor space.
- 3) Axes hard to seal.

**Applications:** Pick and Place operations, Assembly and Sub-Assembly (Mostly Straight), automated loading CNC Lathe and Milling operations, Nuclear Material handling, Welding etc.

#### 1.4.2.2 Polar or Spherical configuration robots



Polar robots are robot configurations with a combined **linear joint and two rotary joints**, with an arm connected to a robotic base and a twisting joint. Also known as spherical robots, the axes create a spherical work envelope and a polar coordinate system. The Polar configuration robots consists of a sliding arm (L-joint) that is actuated relative to the body and a rotational base along with a pivot, which can rotate about a horizontal axis (R-joint) and the vertical axis (T-joint)

##### **Advantages:**

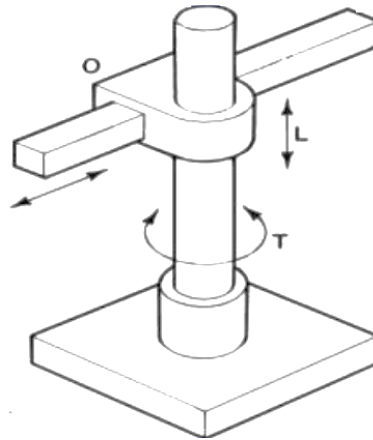
- 1) **1 linear + 2 rotating axes.**
- 2) Large working envelopes.

##### **Disadvantages:**

- 1) Can't reach around obstacles.
- 2) Low accuracy.
- 3) Complex coordinates more difficult to visualize, control, and program.

Applications: Die Casting, Dip Coating, Forging, Glass Handling, Heat Treating, Injection Moulding, Machine Tool Handling, Material Transfer, Parts cleaning, Press Loading etc.

#### 1.4.2.3 Cylindrical configuration robots



Cylindrical robot is able to rotate along his main axes forming a cylindrical shape. In the cylindrical coordinate configuration shown in figure, consists of a vertical column, relative to which an arm assembly is moved up or down. The arm can be moved in or out relative to the column.

#### **Advantages:**

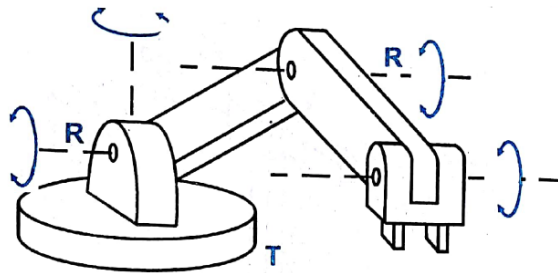
- 1) **2 linear axes +1 rotating.**
- 2) Can reach all around itself.
- 3) Reach and height axes rigid.
- 4) Rotational axis easy to seal.
- 5) Relatively easy programming.

#### **Disadvantages:**

- 1) Can't reach above itself.
- 2) Base rotation axis as less rigid.
- 3) Linear axes are hard to seal.
- 4) Won't reach around obstacles.

**Applications:** Assembly, Coating Applications, Conveyor Pallet Transfer, Die Casting, Forging Applications, Inspection Moulding, Investment Casting, Machine Loading and Unloading etc.

#### 1.4.2.4 Articulating or jointed-arm configuration robots



This type of configuration resembles the human arm where the column swivels about a base (the column and the base forms a T-joint), the column top connects to the shoulder through a shoulder joint (which is the R-joint) and the shoulder connecting to the elbow through an elbow joint (which is also an R-joint). Thus, this configuration has the capability to be controlled at any adjustments in the workspace.

##### **Advantages:**

- 1) All rotary joints allows for maximum capacity.
- 2) Any point in total volume can be reached.
- 3) All joints can be sealed from the environment.

##### **Disadvantages:**

- 1) Extremely difficult to visualize, control, and program.
- 2) Low accuracy.

**Applications:** Assembly operations, Welding, Spray painting, Weld sealing etc.

#### 1.4.3 Applications of Robots

**1. Industrial robots** – These robots bring into play in an industrialized manufacturing atmosphere. Typically, these are articulated arms particularly created for applications like- material handling, painting, welding and others. If we evaluate merely by application then this sort of robots can also consist of some automatically guided automobiles and other robots.

**2. Domestic or household robots** – Robots which are used at home. This sort of robots consists of numerous different gears for example- robotic pool cleaners, robotic sweepers, robotic vacuum cleaners, robotic sewer cleaners and other robots that can perform different household tasks. Also, a number of scrutiny and tele-presence robots can also be considered as domestic robots if brought into play in that sort of environment.

**3. Medical robots** – Robots employed in medicine and medicinal institutes. First & foremost surgical treatment robots. Also, a number of robotic directed automobiles and perhaps lifting supporters.

**4. Service robots** – Robots that cannot be classed into any other types by practice. These could be various data collecting robots, robots prepared to exhibit technologies, robots employed for research, etc.

**5. Military robots** – Robots brought into play in military & armed forces. This sort of robots consists of bomb discarding robots, various shipping robots, exploration drones. Often robots at the start produced for military and armed forces purposes can be employed in law enforcement, exploration and salvage and other associated fields.

**6. Entertainment robots** – These types of robots are employed for entertainment. This is an extremely wide-ranging category. It begins with model robots such as robosapien or the running photo frames and concludes with real heavy weights like articulated robot arms employed as movement simulators.

**7. Space robots** – This type of robots would consist of the robots employed on Canadarm that was brought into play in space Shuttles, the International Space Station, together with Mars explorers and other robots employed in space exploration & other activities.

**8. Hobby and competition robots** – Robots that is created by students. Sumo-bots, Line followers, robots prepared merely for learning, fun and robots prepared for contests.

**9. Repetitive operations:** such as placing a hot block on table of press by pick up and correct placing as the robot can repeat these functions which are very simple and easy, where as human effort is subjected to fatigue, monotonous.

**10. Hazardous environment:** Movements of unsafe, unhealthy or uncomfortable actions are potential use of a robot. For example, loading fuel rods into nuclear reactor core or loading and unloading material into and out of a hot furnace or spray painting or spot welding.

**11. Handling hot or heavy jobs:** such as transferring, positioning heavy jobs on the machines, hot forging or furnace loading

#### 1.4.4 Advantages of robots

Robots can carry out simple and repetitive work for human beings.

1. Robots do not take rest and so can work continuously for a longer period. The productivity is raised.
2. They can be used in assembly work with high precision and density. The quality of products is enhanced.
3. Robots can replace workers without laboring cost, and robots can achieve stable production rate as robots will not be late to work, resign, take day off or strike.
4. The program can be modified easily so robots can increase the flexibility in manufacturing.
5. They can work in unfavorable circumstances, e.g. removing the high-temperature artifact from a casting machine, welding task, spraying task, transportation of chemicals, etc., such work will affect the health of the workers.
6. The manufacturing speed can be increased and the cost be reduced when robots are used.
7. Using robots may minimize the wasting of materials.
8. Using robots can improve the quality of a product by its ability to assemble precise components.

#### 1.4.5 Disadvantages of robots

1. Very high initial cost
2. Needs skilled programmers
3. Leads to loss due to operational and technical problems
4. Robots are suitable only for large mass production industries
5. High maintenance cost

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Department of Mechanical Engineering, DSCE





## Unit 2

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### Joining Process

**Soldering:** Principle of soldering, Surface preparation, Methods of soldering, Applications

**Brazing:** Principle of brazing, methods of brazing, Applications

**Welding:** Definition, Classification, Applications of welding, Flux and its functions, Description of arc welding, Electrodes used in arc welding, Description of oxyacetylene welding, Types of flames produced in gas welding, Comparison between welding, soldering and Brazing, Welding Defects.

### Additive Manufacturing:

Basic principle, need and advantages of additive manufacturing, Procedure for product development in additive manufacturing, Difference between Additive and Subtractive Manufacturing, Classification of additive manufacturing process, Materials and softwares used, Applications and Limitations, Principle and Applications of 3D Printing

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## MODULE 02

# 2.1 JOINING PROCESS

### 2.1.1 Introduction:

Joining process is where two or more pieces of parts are joined together to produce a single product of required shape and size. The parts required for joining are produced by any other manufacturing technique. There are different methods used to join the parts. The joining process can be classified as

- a) Permanent joining process
- b) Semi – permanent or temporary joining process

Permanent joining process is done by fusing the metal together. In this process the metal is heated to its melting state and then it is fused to become one. Some of the examples for permanent joining process are welding soldering and brazing

Temporary joining process is where the metal is not heated. The joining process is carried out at room temperature. Temporary joining process can be done using nuts, bolts, screws and adhesives.

#### 2.1.1.1 Soldering

It is defined as “a joining process wherein coalescence is produced by heating to a suitable temperature and by using a filler metal having a melting point not exceeding 427°C and below the solidification temperature of the base metals”. The filler metal fills in the gap of the joint by capillary action. Soldering uses fusible alloys to join metals known as solder. Ordinary gas flames or electric soldering iron is used to supply the heat to melt the solder. Fluxes are used with solder in soldering process.

Fluxes are defined as any solid, liquid or gaseous material when heated accelerates the wetting of metal with the solder. Due to wetting molten solder flows into the joint and fills the space between the two pieces to be soldered. At elevated temperature flux is highly reducing in nature preventing the formation of metal oxides. Fluxes that are generally used in soldering are Rosin, Zinc Chloride and Aluminum Chloride.



**Hard solders** are called spelter and hard soldering process is called silver solder brazing. The hard solder has lead and silver as its constituents. The melting point of the hard solder is in the range of 350 and above. This process gives greater strength and will stand more heat than soft solder.

### 2.1.1.2 Soldering Methods

### 1. Hand Soldering or Soldering iron method

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- Diagram illustrating the soldering process. A soldering iron is used to heat a copper pad on a circuit board. Solder is applied to the heated pad, and an LED component is inserted into the pad.

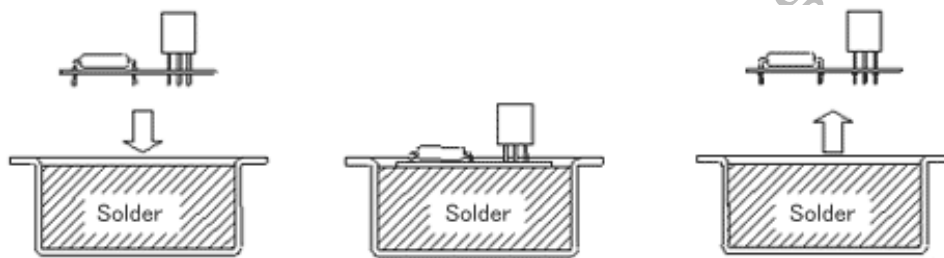
A detailed illustration showing a hand holding a thin metal rod, which is being heated by a blue flame from a torch. The torch is connected to a gas supply line. The background is dark, emphasizing the bright light of the flame.

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- The torch mixes a fuel gas like acetylene with oxygen in suitable proportions, and burn the mixture at its tip.
- **Flame generated** at the torch tip is directed at the workpieces **with a flux** applied on their surfaces.
- When the workpieces are heated to a suitable temperature, the solder is fed into the joint region to melt and flow into the gap between the two workpieces
- The solder cools down and solidifies forming a joint.

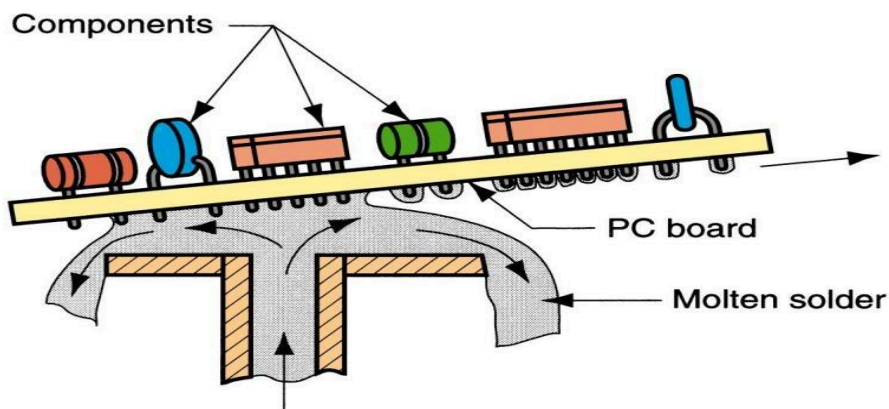
### 3. Dip Soldering:

- Bath of molten solder is kept at the desired temperature
- Usually this process is adopted to solder an entire unit comprising a large number of joints which can be soldered in one operation.



### 4. Wave Soldering

- Work pieces are moved touching a wave of molten solder which is produced by pumping it through a narrow slot as shown in Fig.



## 5. Reflow Soldering

- This process is also widely used in electronics to assemble surface mount components to printed circuit boards.
- In the process, a **solder paste consisting of solder powders in a flux binder** is **applied to spots** on the board where electrical contacts are to be made between surface mount components and the copper circuit.
- The components are then placed on the paste spots, and the **board is heated to melt the solder**, forming mechanical and electrical bonds between the component leads and the copper on the circuit board.

### 2.1.1.3 Applications, Advantages, Disadvantages of soldering

#### Applications

- Printed Circuit Boards (PCB's)
- Making connections between copper pipes in plumbing system
- Drain water gutters
- Radiator brass tubes for motor car
- Joints in sheet metal objects such as food cans, metal containers

#### Advantages

1. The process is done at low temperatures hence, no metallurgical damage to the base metal.
2. The soldering joint can be dismantled by simple heating of the solder.
3. It is cost effective

#### Disadvantage

1. The strength of the joint is not good compared to welding.
2. Flux material has to be cleaned after soldering, as most of the fluxes are corrosive in nature.

### 2.1.2 Brazing

Brazing is the method of joining two similar or dissimilar metals using a special fusible alloy. Joints formed by brazing are stronger than that of soldering. During the brazing, the base metal of the two pieces to be joined is not melted. The filler metal must have ability to wet the surfaces of the base metal to which it is applied. Some diffusion or alloying of the filler metal with base metal takes place even though the base metal does not reach its melting temp. The materials used in brazing are copper base and silver base alloy.

The fluxes used in brazing are borax, boric acid, borates, chlorides and fluorides. Some of the filler metal alloys used for brazing are Aluminum – Silicon, magnesium, copper and copper zinc etc.

#### 2.1.2.1 Brazing procedure

- ❖ Before brazing, the surfaces of the parts are cleaned to remove oxides and grease.
- ❖ After cleaning, a flux is applied at the place of the joint. Commonly, borax and boric acid are used as fluxes.
- ❖ After the flux is applied, the joint and the filler material are heated by oxy-acetylene welding torch to the temperature above melting point of the filler material. The molten filler material flows by capillary action, into the joint space and after cooling, produces a strong joint

#### Advantages of Brazing

1. It is easy to learn.
2. It is possible to join virtually any dissimilar metals.
3. The bond line is very neat aesthetically.
4. Joint strength is strong enough for most non-heavy-duty type of application.

#### Disadvantages of Brazing

1. Brazed joints can be damaged under high temp.
2. Brazed joint requires a high degree of cleanliness.
3. The joint color is often different from that of the base metal.

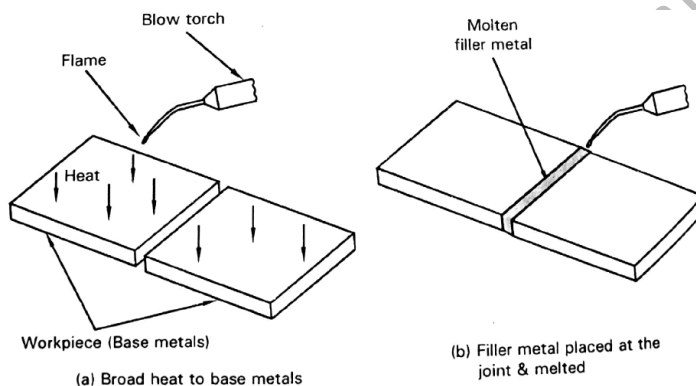
## Applications of Brazing

- Parts of bicycle such as frame and rims
- Pipe joints subjected to vibrations
- Exhaust pipes in motor engines
- Band saws
- Tipped tools
- Nipples and unions to M.S. and copper tubing

### 2.1.2.2 Brazing Methods

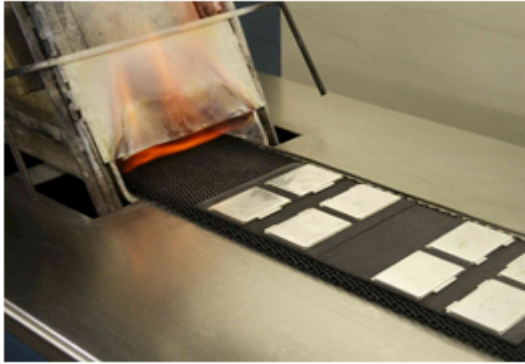
#### 1. Torch brazing

Torch brazing is the most commonly used brazing process wherein the two metals are joined by the heat obtained from a gas flame and by using a non-ferrous filler metal having a melting temperature of above 450°C, but below the melting temperature of the base metal.



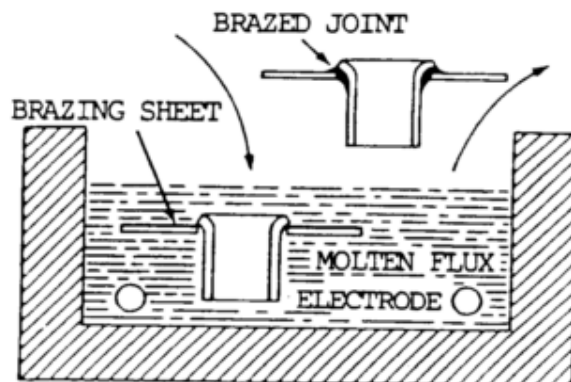
#### 2. Furnace brazing

- Bonding is produced by using the furnace heat and a non-ferrous filler metal having a melting temperature of above 450°C, but below the melting.
- Pre-fluxed or pre-cleaned parts with filler metal pre-placed at the joints are heated in furnaces.



### 3. Dip brazing

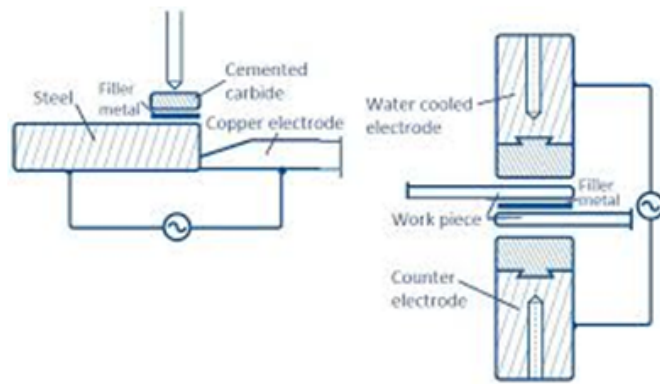
- The molten bath serves as both heating medium and fluxing agent.
- Dip brazing is a process in which two pieces of metal are joined together with a filler metal when heated to the melting point of the filler metal in the molten chemical salt bath.



### 4. Resistance brazing

- Resistance brazing is a variation of the resistance welding process. ... The goal is to heat the braze material until it melts and bonds to both mating substrates. Then the current is turned off and the braze material solidifies. The result is a brazed joint bonding the parts together
- The heat required for brazing is produced due to the electrical resistance offered by the joint members to the flow of current through them.





### 2.1.3 Welding Process

At one time, the simple definition of welding was "joining metals through heating them to a molten state and fusing them together." As technical progress in welding processes has advanced, the definition has had to change.

Welding is defined as "a localized coalescence of metals, wherein coalescence is obtained by heating to suitable temperature, with or without the application of pressure and with or without the use of filler material. This filler material has the melting point same as the base material." It is also known as a *metallurgical joining process* of two metal pieces, to produce a single piece of product. .

#### 2.1.3.1 Welding Terminologies

Base Metal – Work pieces that are needed to be joined

Weld Bead - A weld bead is a weld deposit produced by a single pass of the welding processes. It is also known as bead

Puddle – The molten base metal at the joint during the welding operation

Weld pass – Movement of weld torch from one end of the joint to the other end.

Tack Weld – it is the temporary joint done before welding to keep the work pieces to be welded in place during welding. The joints are done at the ends of the work piece.

### 2.1.3.2 Classification of Welding Process

The welding process can be classified based on the source of energy to heat the metal and the state of metal at the joint.

1. Pressure Welding
2. Fusion Welding

#### 2.1.3.2.1 Pressure Welding

The surfaces of the joint to be welded is heated to a plastic state and forced together with external pressure to finish the joint.

Pressure welding depends on the application of pressures and temperatures, resulting in a plastic state with local deformation of the pieces to be joined in the weld area so that a bond between both pieces is made.

ISO – 857 defines Pressure welding process as “Welding in which sufficient outer force is applied to cause more or less plastic deformation of both the facing surfaces, generally without the addition of filler metal. Usually, but not necessarily, the facing surfaces are heated in order to permit or to facilitate bonding”

Some of the common welding processes that can be grouped under this category are

1. Resistance welding
2. Friction Welding
3. High Frequency welding
4. Ultrasonic Welding
5. Explosion Welding
6. Magnetic Pulse welding
7. Cold pressure welding
8. Diffusion Welding etc.,

#### 2.1.3.2.2 Fusion Welding

The piece to be joined is heated to molten state and allowed to solidify to form joint with or without the addition of filler material. The process is carried out with out the application of pressure.



Some of the fusion welding processes are

- ## Arc Welding

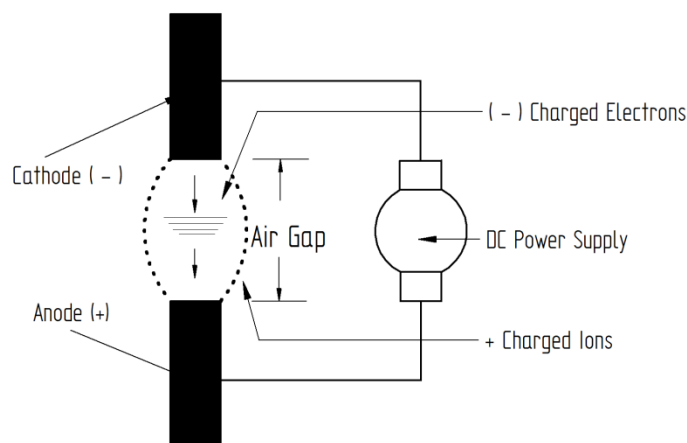
A cross-sectional diagram of the SMAW process. A welding torch is shown at an angle, with a flux-coated electrode (labeled 'Flux Coating' and 'Core') entering the workpiece. A 'Gaseous Shield' of sparks surrounds the electrode tip. 'Metal Transfer' is indicated by small droplets moving from the electrode to the 'Molten Metal Pool' at the bottom. The pool is covered by 'Slag Covering'. The base material is labeled 'Base Metal', and the solidified part is 'Weld Metal'.

Welding arc can be defined as a “Sustained electrical discharge through an ionized gas”.

## Ion exchange theory method

In arc welding the electrons are emitted from the cathode and ions from anode. The electrons at cathode get accelerated and gain energy due to passage of high current and low voltage. As these electrons enter the arc column, they lose their energy by colliding with gas molecules in the air gap between the cathode (electrode) and the anode (work piece). Due to the collision, electrons give out their kinetic energy and break the gas atoms into electrons and positive ions. The electrons and ions move towards cathode and anode respectively, concentrate at anode and cathode regions, get condensed and absorbed. This produces good amount of heat energy which is employed for joining various metals and alloys by fusion.

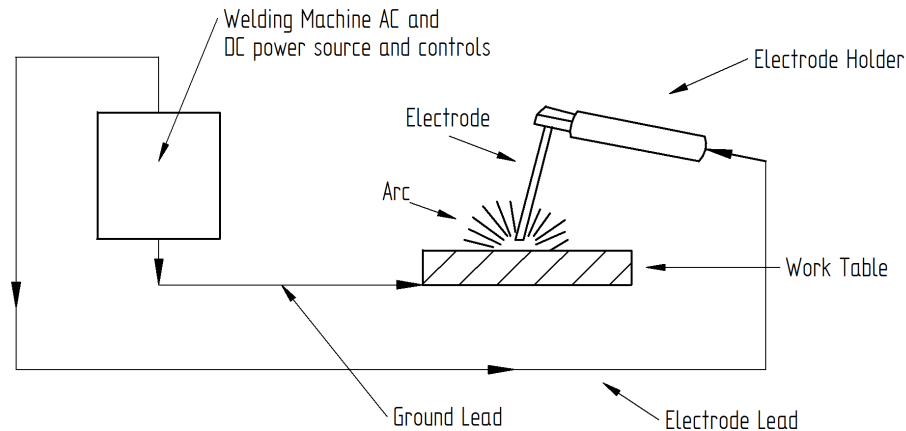
In **electron theory**, when an electric arc is struck between the anode and the cathode, flow of electrons takes place from cathode to anode. The electrons mass is very less and they attain high velocities. High velocity electrons strike the anode giving out their kinetic energy into heat energy. The positive charge ions move from the anode to cathode encircling the stream of electrons at the center. The maximum heat energy released from an electric arc column is at the anode when DC arc welding is done.



*Figure Electron theory of arc column*

### Arc Welding Equipment

For Arc welding DC and AC welding supply, electrode holder, and welding cables are used



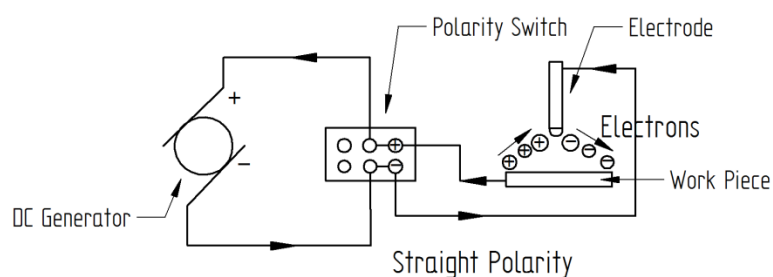
## DC Power Supply

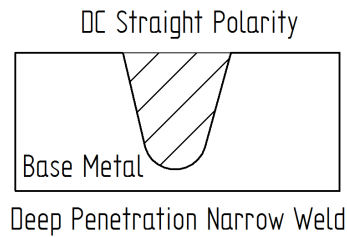
DC arc welding equipment has AC transformer and a rectifier or a DC generator. In DC supply the open circuit voltage ranges from 30 to 75 volts and 150 to 600 Amps current. There are 6 standardized rating for general purpose. The machines with 150 and 200 amps with a voltage of 30 volts are used for general purpose job shop work. The machines rated with 200, 300 and 400 amps with 40 volts are used for general welding purpose. Machines rated with 600 amps and 40 volts are generally used in specialized arc welding.

## DC Welding

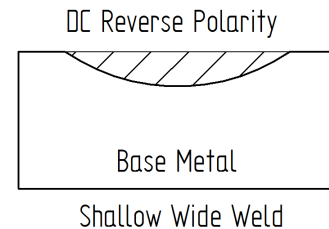
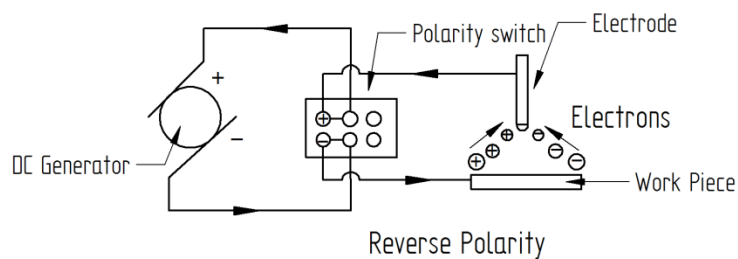
In DC arc welding, the welding current circuit may be hooked up as either straight polarity (DCSP) or reverse polarity (DCRP).

In DCSP (Direct Current Straight Polarity) the machine connections are electrode negative and work piece is positive. The electrons flow from electrode to work piece. Electrons hitting the plate are at high velocity; they exert a considerable heating effect on the plate. In DCSP the weld would be narrow and deep in nature when compared to DCRP.





In DCRP (Direct Current Reverse Polarity) the machine connections are positive and work piece negative. The electrons move from the work piece to the electrode. The electrodes get extra heat; this melts the electrode leading to faster consumption. Hence, bigger diameter electrodes are used when DCRP welding is done. DCRP gives a wide and relatively shallow weld to that of DCSP.

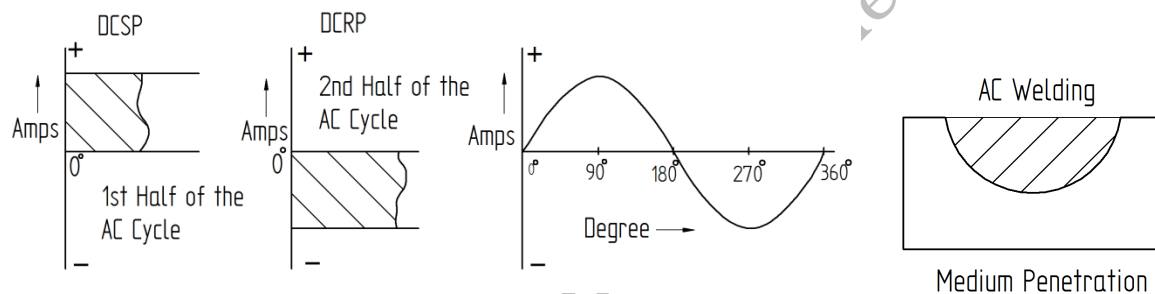


## AC Power Supply

The most commonly used power source for welding is AC supply. The equipment used is a Transformer, which changes high voltage, low amperage power to low voltage, high amperage welding power. High current supply is used in the circuit. The transformer used can be connected to single, two and three phase connection. When connected to single phase the current in the circuit is less hence used for welding thinner sections of work piece with smaller diameter electrode. Two phase and three phase supply, transformers can operate at high current hence used for welding thicker sections of the work piece. The transformers are equipped with arc stabilizing capacitors. They are either air-cooled or oil cooled depending upon their ampere rating. Generally current is in the range of 150 – 400 Amps with open circuit voltage varying from 30 to 100 volts.

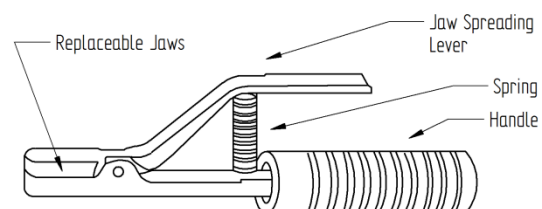
## AC welding

AC welding is theoretically a combination of DCSP and DCRP welding. The current rises from zero to maximum (DCSP), return to zero, increase in opposite direction to maximum and return to zero (DCRP). By this it completes one cycle which is of  $360^\circ$ . The maximum value that it reaches is  $90^\circ$  on one side and  $270^\circ$  on the opposite side. With the alternate flow of current, the arc is extinguished during each half – cycle as the current reduces to zero and re-ignites as the voltage increases in the circuit. By this the polarity of the electrode and the base metal change in every cycle. The arc energy liberated is equal on work surface and the electrode. The penetration of the weld in AC welding is approximately midway between the two DC types.



## Electrode Holder

It is a device use to hold the electrode and conduct electricity from the power cables to the electrode. The holder has jaws in which the electrode can be fixed. The jaws may not be insulated. Electrode holders are available from a working range of 100 to 600 amps. The holder should have a heat and electricity resistant handle for working. The holder should be light, sturdy and easy to fix electrode.



## **Welding Cables**

Cables conduct current from the power source through the electrode holder, the arc, and the work piece and back to the welding power source. Cables are of copper or aluminum. The cables are well insulated with rubber and woven, fabric reinforced layer. The cable that connects the electrode holder to the power source is known as the electrode lead and the cable that connects the work piece to the power source is known as ground lead.

## **Welding Electrodes**

It is a piece of rod or wire, which is made of metal or alloy, with or without flux coated. The electrode one end is connected to the electrode holder and at other end arc is set up. Electrodes come in different diameter and length. The diameter and length depends upon the amount of weld to be deposited and the type of joint to be welded. Generally thin and large diameter electrodes are of short in lengths and medium size electrodes have bigger lengths. The types of electrodes that are generally used for welding are

**Consumable Electrode** – it melts and supplies filler metal to the weld

**Non – Consumable Electrode** - it does not melt but separate filler metal is to be added to the weld.

## **Consumable Electrode**

It is made of low melting point metal or alloy. When arc is struck electrode starts to melt and transfer to the work piece in the form of droplets. These droplets fill the space between the two metals to be joined. The thermal efficiency of consumable electrode system is about 85 % when compared to non – consumable electrode of 55 %. Consumable electrodes can be further classified into bare electrodes and flux covered electrodes.

**Bare electrodes** are those, which consist of metal or alloy wire without any flux coating.



**Flux covered electrodes** are those which have flux covering the metal or alloy. The flux acts as a shield to protect the arc and also to protect the weld pool. Flux when it gets melted during arc, forms an inert gas shield around the arc and the chemical present reacts with the molten metal to form slag. .

### **Non – Consumable Electrodes**

These electrodes are made up of high melting point metals like carbon, pure tungsten etc. These electrodes do not melt during welding process. They generate arc, which melts the work piece. Generally these metals do not undergo vaporization and oxidation during welding. In this kind of welding filler material addition depends up on the thickness of the plate to be welded.

### **Gas Welding**

Gas welding is one of the oldest methods of fusion welding process and, for many years, was the most widely used method of metal melting. The equipment is relatively simple and cheap. Heat is generated by the combustion of combustible gas with oxygen. Required gas ratio is mixed in the hand held torch. Combustion takes place at the nozzle or the outlet of the torch. This process of welding is generally known as Oxygen – Fuel Gas welding (OFG). Some of the commercial gas that is used for welding is acetylene, hydrogen, propane, butane and commercial LPG.

### **Oxy – Acetylene Gas welding**

This is a common gas welding process. Acetylene is the fuel gas used. Acetylene produces high heat content in the range of 3200° C than other fuel gases. Acetylene gas has more available carbon (92.3 %) and hydrogen (7.7 %) by weight. The heat is released when the carbon breaks away from hydrogen to combine with O<sub>2</sub> and burn.



Depending up on the gas pressure required for welding or cutting oxy acetylene welding is classified into two

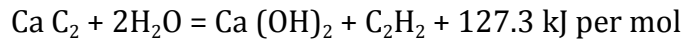
Low pressure System and

High-pressure system

### **Low-pressure system**



Acetylene required for welding is manufactured at the place of welding using an acetylene generator. Acetylene is produced by reacting calcium carbide with water.



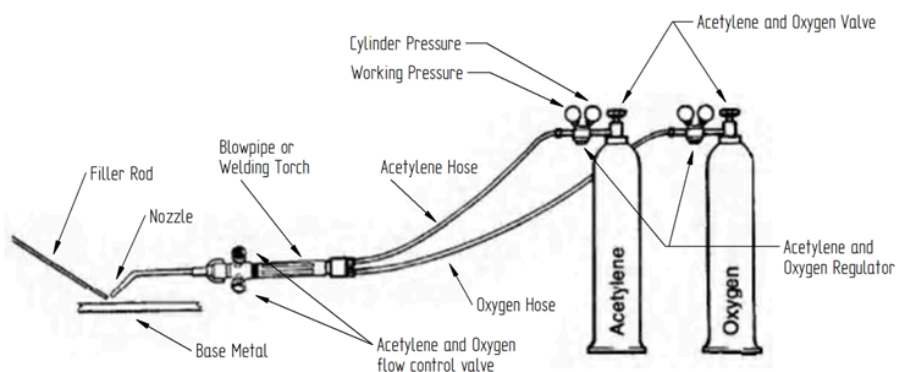
Acetylene produced is stored in the generator; it is then passed through a purifier and sent to welding torch at a low pressure. The pressure range at the nozzle is in the range of 0.5 to 3.5 bars

### High Pressure System

$\text{O}_2$  and  $\text{C}_2\text{H}_2$  are supplied in high-pressure cylinders. Acetylene is chemically unstable under pressure, even without the presence of air and, under certain conditions; it can explosively decompose to its constituents (carbon and hydrogen). Acetylene ignites very easily, and produces an explosive mixture in air over a wide range of concentrations. Pure acetylene cannot be compressed to a pressure more than 0.1 Pa above atmosphere. To enable the gas to be stored, the cylinders are filled with a porous mass, saturated with acetone, which absorbs the gas when it is filled. The pressure in the bottles is 2 MPa. However, explosive decomposition can occur in the pipes from the bottle if the pressure exceeds 1.5 MPa.

### Gas welding equipment

The equipments for a gas welding are gas cylinders, pressure regulating valves, gas hoses and fittings, flash back arrestors and blowpipes or welding torch,



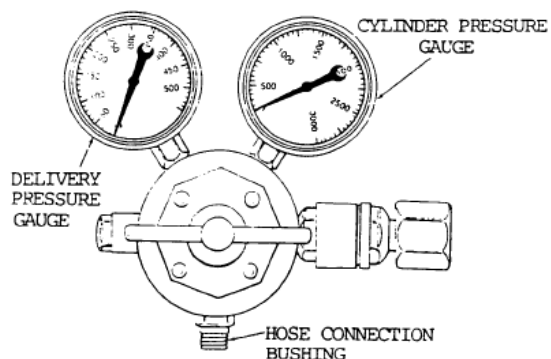
### Cylinders

**Acetylene cylinders** are filled with porous materials such as balsa wood, charcoal, corn pith, or Portland cement. **Acetone**, a colorless, flammable liquid, is added to the cylinder until about 40 percent of the porous material is saturated. The porous material acts as a large sponge which absorbs the acetone, which then absorbs the acetylene. In this process, the volume of acetone increases as it absorbs the acetylene, while acetylene, being a gas, decreases in volume. Acetylene cylinder is painted in Maroon and it is maintained at 2MPa pressure. The outlet pressure should be below 1.5MPa

Oxygen cylinder is made of steel and it is stored at a pressure of 20MPa. The cylinder consists of an outlet valve, a removable metal cap for the protection of the valve, and a low melting point safety fuse plug and disk. Generally oxygen cylinders are painted in black.

### Pressure Regulating Valve

The purpose of the pressure regulator is to reduce the high and variable pressure in the cylinder to a suitable working pressure. The pressure in an oxygen cylinder can be as high as 2200 psi (15,169 kPa), which must be reduced to a working pressure of 1 to 25 psi (6.90 to 172.38 kPa). The pressure of acetylene in an acetylene cylinder can be as high as 250 psi (1724 kPa) and must be reduced to a working pressure of from 1 to 12 psi (6.90 to 82.74 kPa). Regulators keep the gas flow rate constant throughout the life of the cylinder charge, despite any variations in back pressure caused by the heating of the welding torch.



## Hose and Hose fittings

The cylinder regulators and torch are usually connected together by double line rubber hoses. Double line hose is known as type VD. The rubber hose is used for flexibility. The hose has to be of highest quality. The Oxygen line is green in color and the Acetylene line is red in color.



The hose fittings are provided at the end of the hose for necessary attachment of the hose to the blowpipe and to the pressure regulating valve. The nuts provided in both the hose are different to each other. The oxygen hose connection nut is a right hand threaded and where as acetylene hose connection nut is left handed.

## Flashback Arrestor:

A flashback means that the flame burns backwards into the torch with a popping sound. It occurs if the combustion speed of the flame exceeds the speed at which the gas is being supplied, so that the flame front moves backwards. A flashback arrester fitted at the regulator prevents a flashback from going any further back. The reason for a flashback occurring is that a mixture of oxygen and acetylene has occurred in the hoses, e.g. by oxygen having entered the acetylene hose and formed an explosive mixture. The flashback arrester prevents the flame from reaching the acetylene bottle and triggering an explosive decomposition.

## Blowtorch or Welding Torch

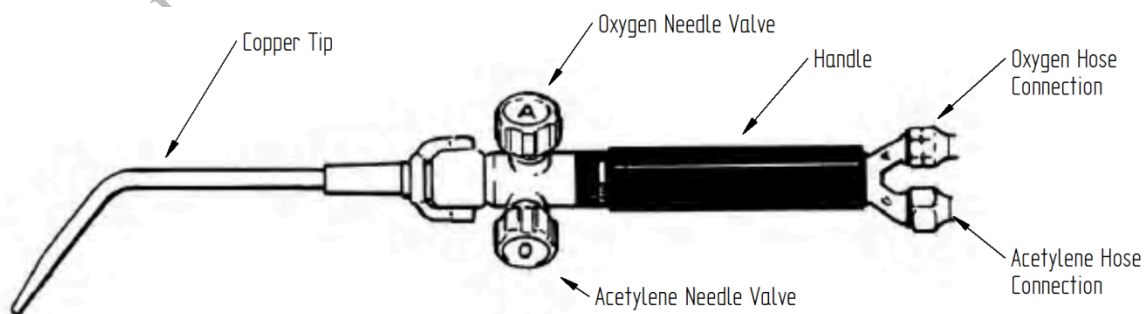
The oxyacetylene welding torch is used to mix oxygen and acetylene in proportions required. The torch controls the volume of these gases burning at the tip, and also controls the type of flame.

The torch consists of a handle or body, which contains the hose connections for the oxygen and the fuel gas. The torch also has two needle valves, one for adjusting the flow of oxygen and other one for acetylene, mixing head and a tip. The tip is made up of hard drawn electrolytic copper or 95 % copper and 5 % tellurium. The diameters of the tip orifices differ in order to control the quantity of heat and the type of flame. Generally, a number is designated to the tip size, smaller the number, smaller the tip orifice. A mixer is provided to ensure correct flow of mixed gas for each size tip. Mixer is situated between the tip and the needle valve.

Welding torch is classified based on pressure available at the outlet. There are two types

1. Injector torch
2. High pressure torch

In high pressure torch the acetylene and oxygen are supplied from the cylinder, mixed in the mixer provided in the torch. Where as in low-pressure torches the oxygen flows into the torch through a central jet producing a suction effect that draws acetylene from the acetylene generator. From there on the gas continues to the mixing section and then to the tip.



## Types of Flames

One volume of Acetylene gas requires 2.5 times the volume of oxygen for its complete combustion. Depending up on the ratio of oxygen supplied for combustion we can classify flames into 3 types. This ratio is known as gas ratio. These flame types have a great effect on the melt pool chemically. There are 3 types of flames and they are-

Neutral Flame,

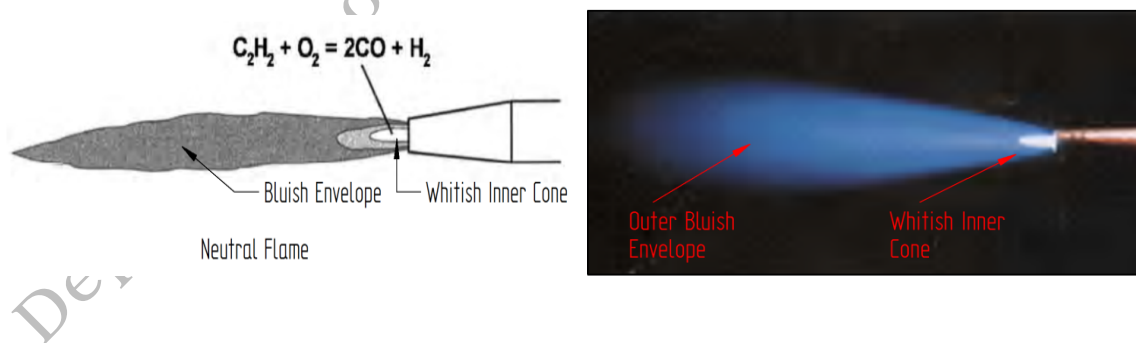
Carburising flame, and

Oxidizing Flame.

### Neutral Flame

Neutral flame is the one that is used the most. This flame is obtained by supplying equal volumes of oxygen and acetylene. The flame can be distinguished by its zones of combustion. The innermost zone, the cone, is white in color and extends a short distance from the tip of the torch. Acetylene burns in this zone to form carbon monoxide and hydrogen. The approximate temperature of the inner cone is  $3200^{\circ}\text{C}$ .

The outer cone or envelope is faintly luminous and bluish is color. Carbon monoxide and hydrogen burn with oxygen in air forming carbon dioxide and water vapour hence preventing the atmospheric oxygen coming in contact with the molten metal pool.

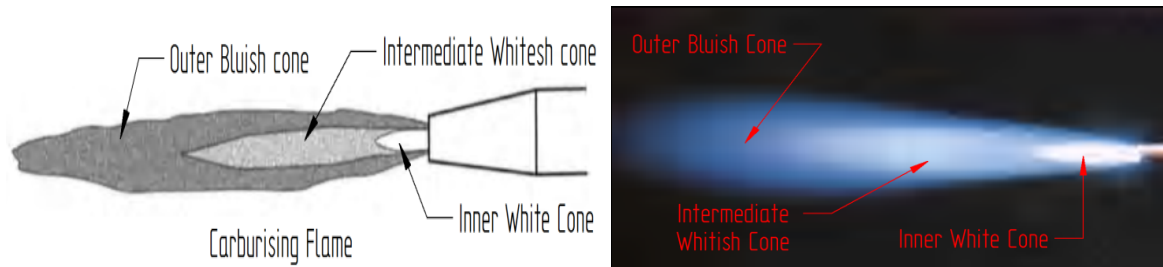


### Carburising Flame

In this flame acetylene proportion is increased in the gas ratio of 0.95 to 1.0. The flame has three zones a) inner white zone b) intermediate whitish cone and c) bluish outer envelope. The inner zone which is sharply defined and white in color. In this zone

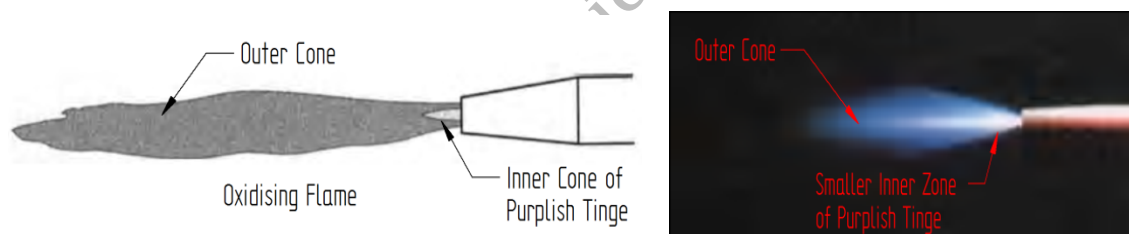
there is insufficient oxygen to burn surplus acetylene. Cone temperature is approximately about  $3100^{\circ}\text{C}$ .

The remaining acetylene continues to the second zone known as the intermediate cone of whitish color. This intermediate zone extends in length depending up on the amount of acetylene present in the gas. The inner



### Oxidizing Flame

In this flame there is excess of oxygen. The flame has a small inner cone and outer envelope. The inner cone is not sharply defined as in neutral flame and carburising flame. The inner cone is purplish in color. The temperature of the inner cone is approximate of  $3350^{\circ} - 3400^{\circ}\text{C}$ .



### 2.1.3.2 Flux

Flux is a material that is expected to react with oxides present and form low density slag which would float on the top of molten metal pool protecting it from further oxidation.

During the process of welding, some metal may get oxidised. Flux is used to dissolve and remove these metal oxides. Flux like borax or a mixture of fluorides and chlorides of sodium, potassium and lithium are most commonly used. The flux reacts with metallic oxides to yield slag, which being lighter floats on top of molten metal pool. After solidification, the welder removes the flux using a chipping hammer and wire brush.

The main functions of flux is given below:



1. Provide a protective atmosphere for welding, thus prevents oxidation
2. Stabilize the arc
3. Reduce spattering.
4. Reduce the viscosity of molten metal.

### 2.1.3.3 Difference between Brazing, Soldering and Welding

|   | <b>Welding</b>                                                                                                                | <b>Soldering</b>                                                                                             | <b>Brazing</b>                                                                                                   |
|---|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| 1 | These are the strongest joints used to bear the load. Strength of a welded joint may be more than the strength of base metal. | These are weakest joint out of three. Not meant to bear the load. Use to make electrical contacts generally. | These are stronger than soldering but weaker than welding. These can be used to bear the load up to some extent. |
| 2 | Temperature required is up to 3800°C of welding zone.                                                                         | Temperature requirement is up to 450°C.                                                                      | It may go to 600°C in brazing.                                                                                   |
| 3 | Workpiece to be joined need to be heated till their melting point.                                                            | No need to heat the workpieces.                                                                              | Work pieces are heated but below their melting point.                                                            |
| 4 | Mechanical properties of base metal may change at the joint due to heating and cooling.                                       | No change in mechanical properties after joining.                                                            | May change in mechanical properties of joint but it is almost negligible.                                        |
| 5 | Heat cost is involved and high skill level is required.                                                                       | Cost involved and skill requirements are very low.                                                           | Cost involved and skill required are in between others two.                                                      |
| 6 | Heat treatment is generally required to eliminate undesirable effects of welding.                                             | No heat treatment is required.                                                                               | No heat treatment is required after brazing.                                                                     |
| 7 | No preheating of workpiece is required before welding as it is carried out at high temperature.                               | Preheating of workpieces before soldering is good for making good quality joint.                             | Preheating is desirable to make strong joint as brazing is carried out                                           |



|  |  |  |                                |
|--|--|--|--------------------------------|
|  |  |  | at relatively low temperature. |
|--|--|--|--------------------------------|

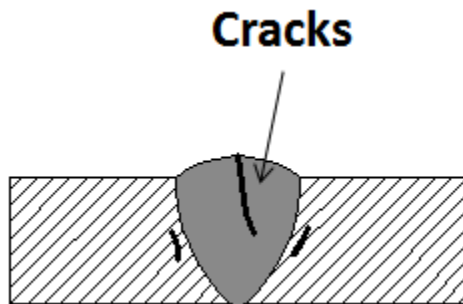
#### 2.1.3.4 Welding Defects

Welding Defects can be defined as the irregularities formed in the given weld metal due to wrong welding process or incorrect welding patterns, etc. The defect may differ from the desired weld bead shape, size, and intended quality. Welding defects may occur either outside or inside the weld metal. Some of the defects may be allowed if the defects are under permissible limits but other defects such as cracks are never accepted.

The various types of welding defects with their causes and remedies are listed below:

##### 1. Weld Crack

The most serious type of welding defect is a weld crack and it's not accepted almost by all standards in the industry. It can appear on the surface, in the weld metal or the area affected by the intense heat.



##### Causes of cracks:

- Use of hydrogen when welding ferrous metals.
- Residual stress caused by the solidification shrinkage.
- Base metal contamination.
- High welding speed but low current.
- No preheat before starting welding.
- A high content of sulphur and carbon in the metal.

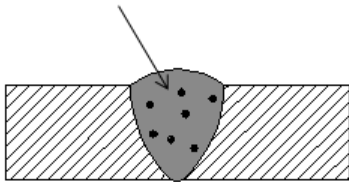
### Remedies:

- Preheat the metal as required.
- Provide proper cooling of the weld area.
- Use proper joint design.
- Remove impurities.
- Use appropriate metal.
- Use proper welding speed and amperage current.

## 2. Porosity

Porosity occurs as a result of weld metal contamination. The trapped gases create a bubble-filled weld that becomes weak and can with time collapse.

### Porosity



### Causes of porosity:

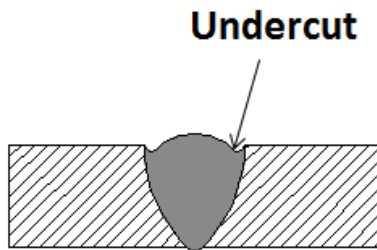
- Using a longer arc.
- The presence of moisture.
- Improper gas shield.
- Use of too high gas flow.
- Contaminated surface.

### Remedies:

- Clean the materials before you begin welding.
- Use dry electrodes and materials.
- Use correct arc distance.
- Check the gas flow meter and make sure that it's optimized as required with proper with pressure and flow settings.
- Reduce arc travel speed, which will allow the gases to escape.

## 3. Undercut

This welding imperfection is the groove formation at the weld toe, reducing the cross-sectional thickness of the base metal. The result is the weakened weld and work piece.



#### **Causes:**

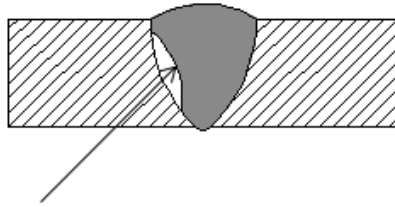
- Too high weld current.
- Too fast weld speed.
- The use of an incorrect angle, which will direct more heat to free edges.
- Incorrect usage of gas shielding.
- Incorrect filler metal.

#### **Remedies:**

- Use proper electrode angle.
- Reduce the arc length.
- Reduce the electrode's travel speed, but it also shouldn't be too slow.
- Choose shielding gas with the correct composition for the material type you'll be welding.
- Use of proper current, reducing it when approaching thinner areas and free edges.

#### **4. Incomplete Fusion**

This type of welding defect occurs when there's a lack of proper fusion between the base metal and the weld metal. It can also appear between adjoining weld beads. This creates a gap in the joint that is not filled with molten metal.



## Incomplete Fusion

### Causes:

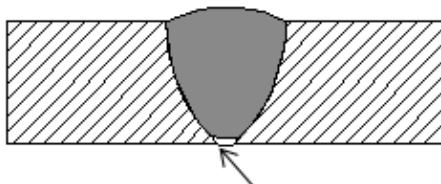
- Low heat input.
- Surface contamination.
- Electrode angle is incorrect.
- Travel speed is too fast.
- The weld pool is too large and it runs ahead of the arc.

### Remedies:

- Use a sufficiently high welding current with the appropriate arc voltage.
- Before you begin welding, clean the metal.
- Avoid molten pool from flooding the arc.
- Use correct electrode diameter and angle.
- Reduce deposition rate.

## 5. Incomplete Penetration

Incomplete penetration occurs when the groove of the metal is not filled completely, meaning the weld metal doesn't fully extend through the joint thickness.



## Incomplete Penetration

### Causes:

- There was too much space between the metal you're welding together.

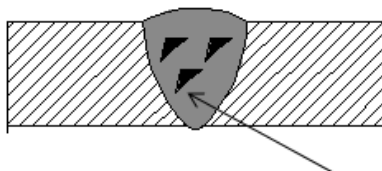
- You're moving the bead too quickly, which doesn't allow enough metal to be deposited in the joint.
- You're using a too low amperage setting, which results in the current not being strong enough to properly melt the metal.
- Large electrode diameter.
- Improper joint.

**Remedies:**

- Use proper joint geometry.
- Reduce arc travel speed.
- Choose proper welding current.
- Check for proper alignment.

## 6. Slag Inclusion

Slag inclusion is one of the welding defects that are usually easily visible in the weld. Slag is a vitreous material that occurs as a by-product of stick welding, flux-cored arc welding and submerged arc welding. It can occur when the flux, which is the solid shielding material used when welding, melts in the weld or on the surface of the weld zone.



**Slag inclusion**

**Causes:**

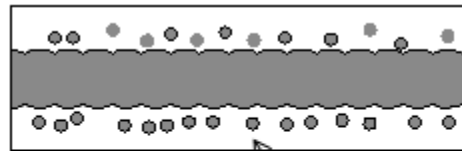
- Improper cleaning.
- The weld speed is too fast.
- Incorrect welding angle.
- The weld pool cools down too fast.
- Welding current is too low.

**Remedies:**

- Increase current density.
- Reduce rapid cooling.
- Adjust the electrode angle.
- Remove any slag from the previous bead.
- Adjust the welding speed.

## 7. Spatter

Spatter occurs when small particles from the weld attach themselves to the surrounding surface. It's an especially common occurrence in gas metal arc welding. No matter how hard you try, it can't be completely eliminated. However, there are a few ways you can keep it to a minimum.



**Spatter**

### Causes:

- The running amperage is too high.
- Voltage setting is too low.
- The work angle of the electrode is too steep.
- The surface is contaminated.
- The arc is too long.
- Incorrect polarity.
- Erratic wire feeding.

### Remedies:

- Clean surfaces prior to welding.
- Reduce the arc length.
- Adjust the weld current.
- Increase the electrode angle.
- Use proper polarity.

## 2.2 Additive Manufacturing -Definition

Additive Manufacturing (AM) refers to a process by which digital 3D design data is used to build up a component in layers by depositing material.

### 2.2.1 Principle of Additive Manufacturing

Additive manufacturing (AM) is a relatively new manufacturing technology that is driven by computer-aided design (CAD) in which the solid shape to be produced is first sliced from the Standard Tessellation Language (STL) file that is submitted, and tool paths are generated using an interface software and submitted to the production system. Based on the information that is received, the production system produces the part layer by layer until the part is completely produced, thereby making it possible for companies to significantly cut design and manufacturing cycle times. AM technologies have revolutionized the way that parts are manufactured because they build parts by adding materials, which is opposite to the traditional way of building parts by removing materials.

### 2.2.2 Need for Additive Manufacturing

Additive manufacturing, or popularly known today as 3D printing, is a technology which is used to manufacture prototype parts by printing them layer by layer. This technology has penetrated every major industry from automotive, aerospace, defence, consumer products to medical and dental. Most common applications include functional prototypes, fit and assembly models, patterns for prototype tooling, patterns for metal casting, presentation models, visual aids, and tooling components. Below are some of the reasons why additive manufacturing is important in the rapid prototyping and manufacturing industry.

**Simplified product design:** Imagine a prototype that starts with a complex polygonal shape and ends with a circular structure at the tail end. The smooth transition from a polygon to a circle is difficult to achieve with conventional manufacturing methods, but can be easily accomplished using additive manufacturing.

**Reduction in weight:** Additive manufacturing systems either cure, extrude, melt or sinter material only where it's producing honeycomb structures with higher



strength-to-weight ratio as compared to completely solid structures. This results in significant weight reduction and improved part performance.

**Complex geometries:** Additive manufacturing brings innovation at the forefront. Designers and engineers are able to try different iterations without the added costs associated with traditional manufacturing methods.

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### 2.2.3 Procedure for product development in additive manufacturing

Most AM processes involve the following eight steps:

#### **Step 1: CAD:**

All AM parts must start from a software model that fully describes the external geometry. This can involve the use of almost any professional CAD solid modelling software, but the output must be a 3D solid or surface representation. Reverse engineering equipment (e.g., laser scanning) can also be used to create this representation.

#### **Step 2: Conversion to STL**

Nearly every AM machine accepts the STL file format and every CAD system can output such a file format. This file describes the external closed surfaces of the original CAD model and forms the basis for calculation of the slices.

#### **Step 3: Transfer to AM Machine and STL File Manipulation**

The STL file describing the part must be transferred to the AM machine. Here, there may be some general manipulation of the file so that it is the correct size, position, and orientation for building.

#### **Step 4: Machine Setup**

The AM machine must be properly set up prior to the build process. Such settings would relate to the build parameters like the material constraints, energy source, layer thickness, timings, etc.

#### **Step 5: Build**



Building the part is mainly an automated process and the machine can largely carry on without supervision. Only superficial monitoring of the machine needs to take place at this time to ensure no errors have taken place like running out of material, power or software glitches, etc.

### **Step 6: Removal**

Once the AM machine has completed the build, the parts must be removed. This may require interaction with the machine, which may have safety interlocks to ensure for example that the operating temperatures are sufficiently low or that there are no actively moving parts.

### **Step 7: Post processing**

Once removed from the machine, parts may require an amount of additional cleaning up before they are ready for use. Parts may be weak at this stage or they may have supporting features that must be removed. This therefore often requires time and careful, experienced manual manipulation.

### **Step 8: Application**

Parts may now be ready to be used. However, they may also require additional treatment before they are acceptable for use. For example, they may require priming and painting to give an acceptable surface texture and finish. Treatments may be laborious and lengthy if the finishing requirements are very demanding. They may also be required to be assembled together with other mechanical or electronic components to form a final model or product.

## **2.2.4 Difference between Additive and Subtractive Manufacturing**

|   | <b>Additive Manufacturing</b>                                                        | <b>Subtractive Manufacturing</b>                                                                             |
|---|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| 1 | In additive manufacturing, layer by layer material is added to develop a 3-D object. | In subtractive manufacturing, layer by layer material is removed from a solid block to develop a 3-D object. |

|   |                                                                                                                                               |                                                                                                                                                       |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 | This manufacturing concept is usually suitable for materials having lower melting point, such as plastic.                                     | This manufacturing concept is suitable for materials having high melting point, such as metallic materials.                                           |
| 3 | Volumetric density (thus weight) of the constructive material of final component can be controlled during operation.                          | Material density cannot be controlled during operation. Density of object remains same with that of the initial solid block (usually a cast product). |
| 4 | No material wastage takes place in these processes.                                                                                           | These processes are associated with material wastage in the form of chips, scraps, dissolved ions, etc.                                               |
| 5 | Certain complex shapes can be easily fabricated using additive manufacturing techniques.                                                      | Subtractive manufacturing processes have limited capability in fabrication of complex shapes.                                                         |
| 6 | Structures containing fully closed internal hollow part can be produced by these processes.                                                   | Structures containing enclosed hollow parts cannot be produced by these processes, unless joining is allowed.                                         |
| 7 | These processes are applicable to a narrow range of materials.                                                                                | These processes can efficiently handle a wide variety of materials.                                                                                   |
| 8 | These processes are time consuming and costly but can provide superior quality and desired property without requiring any further processing. | These processes are time efficient and economic. These are usually suitable for mass production where requirement of product quality is not so tight. |

### 2.2.5 Classification of additive manufacturing process

1. Photo polymerization
2. Material jetting
3. Binder jetting
4. Material extrusion
5. Powder Bed Fusion
6. Sheet Lamination
7. Direct Energy Deposition

### 2.2.6 Additive manufacturing materials

It is possible to use many different materials to create 3D-printed objects. AM technology fabricates jet engine parts from advanced metal alloys, and it also creates chocolate treats and other food items.

#### **Thermoplastics**

Thermoplastic polymers remain the most popular class of additive manufacturing materials. Acrylonitrile butadiene styrene (ABS), polylactic acid (PLA) and polycarbonate (PC) each offer distinct advantages in different applications. Water-soluble polyvinyl alcohol (PVA) is typically used to create temporary support structures, which are later dissolved away.

#### **Metals**

Many different metals and metal alloys are used in additive manufacturing, from precious metals like gold and silver to strategic metals like stainless steel and titanium.

#### **Ceramics**

A variety of ceramics have also been used in additive manufacturing, including zirconia, alumina and tricalcium phosphate. Also, alternate layers of powdered glass and adhesive are baked together to create entirely new classes of glass products.

#### **Biochemicals**

Biochemical healthcare applications include the use of hardened material from silicon, calcium phosphate and zinc to support bone structures as new bone growth occurs. Researchers are also exploring the use of bio-inks fabricated from stem cells to form everything from blood vessels to bladders and beyond.

### 2.2.7 Softwares used

**Softwares for 3D Modelling-** CATIA, CREO, AUTODESK INVENTOR, FUSION 360, UNIGRAPHICS, 3D Slash, Design Spark

**Softwares for Converting 3D CAD Files to GCode files:** Cura, NetFabb, Repetier, Simplify3D, CraftWare, Skeinforge, SliceCrafter etc

**3D Printer Control Softwares:** PRONTERFACE, Firmware, AstroPrint, MatterControl, 3DPrinterOS



### 2.2.8 Advantages of Additive manufacturing

1. **Complexity is free:** It actually costs less to print a complex part instead of a simple cube of the same size. The more complex (or, the less solid the object is), the faster and cheaper it can be made through additive manufacturing.
2. **Variety is free:** If a part needs to be changed, the change can simply be made on the original CAD file, and the new product can be printed right away.
3. **No assembly required:** Moving parts such as hinges and bicycle chains can be printed in metal directly into the product, which can significantly reduce the part numbers.
4. **Little lead time:** Engineers can create a prototype with a 3-D printer immediately after finishing the part's stereo lithography (STL) file. As soon as the part has printed, engineers may then begin testing its properties instead of waiting weeks or months for a prototype or part to come in.
5. **Little-skill manufacturing:** While complicated parts with specific parameters and high-tech applications ought to be left to the professionals, even children in elementary school have created their own figures using 3-D printing processes.
6. **Few constraints:** Anything you can dream up and design in the CAD software, you can create with additive manufacturing.
7. **Less waste:** Because only the material that is needed is used, there is very little (if any) material wasted.
8. **Infinite shades of materials:** Engineers can program parts to have specific colors in their CAD files, and printers can use materials of any color to print them.

### 2.2.9 Limitations of Additive manufacturing

1. High Energy Consumption
2. 3D Printing Technology is Expensive
3. Limited Materials
4. 3D Printers Aren't that User-friendly
5. Harmful Emissions
6. Too Much Reliance on Plastic
7. 3D Printers are Slow and not suitable for mass production



### 2.2.10 Applications of Additive Manufacturing

#### Automotive:

- 3D printed tools and spare parts
- 3D printed brakes
- 3D printed tires

#### Fashion

- stunning jewellery

#### Aerospace

- Fuel nozzles
- Boeing 787 structural parts

#### Medical

- 3D printed jaws, knee replacements braces, aligners, implants, dentures, and crowns.

#### Architecture

- 3D printed bridge
- 3D printed houses.
- **Electronics**
- Circuit boards

### 2.2.11 Principle of 3D Printing

3D printing starts by making a virtual design of the object you want to create. The virtual design is used as a template of the physical object to be created. This virtual design can be made using a 3D modeling program such as CAD (Computer Aided Design) to create a design from scratch. Alternatively a 3D scanner can be used for an existing object. This scanner makes a 3D digital copy of an object and puts it into a 3D modeling program. The model is then sliced into hundreds or thousands of horizontal layers in preparation for printing. This prepared file is thus uploaded in the 3D printer, which will create the object layer by layer.

### 2.2.12 Applications of 3D Printing

1. **Construction:** In China they were able to build 10 one-story houses in a day, a procedure which normally takes weeks to months to complete. 3D printing thus provides a cheaper, faster and safer alternative relative to more traditional construction. Four giant 3D printers were used by WinSun Decoration Design Engineering to build houses in Shanghai; by using a mix of cement and construction waste to construct the walls layer by layer

#### 2. Medicine

- a) **Hearing aids:** Hearing aids have been made using 3D printing technology.
- b) **Bio printers:** Organ printing or body part printing is being printed and some parts being used as implants of actual body parts. Body parts such as titanium pelvic, plastic tracheal splint, titanium jaws to mention but a few have been printed. Tissue engineering has made tremendous strides as they have been able to print 3D blood vessels. This was achieved by combining advances in 3D bio-printing technology and biomaterials through vascularisation of hydrogel constructs.
- c) **Digital Dentistry:** People are getting 3D printed teeth customised for the individual. Dental Implants are being made on a commercial level and making the whole process faster and more efficient. Before fake teeth used to be a one- size- fits all depending with age. Now, people of the same age can have different sized teeth resulting in people getting discomfort with ill-fitting fake teeth. Thus, customised implants have really brought a sigh of relief to the consumers as they are now able to receive teeth suited for them.
- d) **Prosthetics:** A multitude of people are in need of surrogate body parts from people born without limbs to accident victims. The cost of getting surrogate body parts used to be extremely expensive but thanks to 3D printing; the cost has been significantly reduced. Prosthetics has really done wonders to disabled people with the likes of Paralympics champion Oscar Pistorious being a world famous example. Oscar Pistorious had his

legs cut off as a child, but that did not stop him from running, let alone in the Olympics.

- e) **Bionics:** Researchers from Princeton and John Hopkins were able to make a 3D printed bionic ear. The hearing is done through electronics. This development could help deaf people to hear.
  - f) **Artificial Organs:** Additive manufacturing of stem cells has also led to various possibilities in printing artificial organs, although most of the work is still in the experimental stage. For instance, through 3D printing Heriot-Watt University scientists were able to produce clusters of embryonic stem cells. An endless world of possibilities awaits this world with the prospects of printing actual functioning artificial organs.
3. **Manufacturing:** 3D printing has introduced an era of rapid manufacturing. The prototyping phase is now able to be skipped and go straight to the end product. Car and aeroplane parts are being printed using 3D printing technology. The printing of parts is being done in a fast and efficient manner thus contributing immensely to the value chain. Customized products are able to be manufactured as customers can edit the digital design file and send to the manufacturer for productions. Nokia Company has taken the lead in manufacturing in this area by releasing 3D design files of its case to its end users so they can customize it to their specifications and get the case 3D printed.
4. **Domestic Usage:** 3D printers can be used in the home to make small objects such as ornamental objects such as necklaces and rings. Small plastic toys can also be printed in a domestic setting. In the future, people will be able to print their own products at home instead of buying from shops.
5. **Clothing:** The fashion industry has also not been spared. 3D printed clothing is being made. Fashion designers are experimenting with 3D-printed bikinis, shoes, and dresses. Nike made the 2012 Vapor Laser Talon football shoe and New Balance custom-fit shoes for athletes using a 3D prototype. The production was done on a commercial scale.
6. **Academia:** 3D printing is now being integrated in the learning curriculum. With applications from printed molecule models to plastic gears. Students are now

able to print their prototype models in 3D and it helps in the learning process of the students. Students are better able to understand concepts as it can be practically shown to them.

Department of Mechanical Engineering, DSCE





## Unit 3

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**Steam:** Formation of steam, Types of steam, Steam Properties-Enthalpy, dryness fraction, wetness fraction, latent heat, sensible heat, Internal energy, Specific volume, External work of evaporation, degree of superheat, amount of superheat, saturated and superheated temperature, Numericals on steam.

**Boilers:** Classification of Boilers, Babcock and Wilcox Boiler, Lancashire Boiler, Boiler mountings and accessories (no sketches)

**Steam Turbines:** Classification, Principle operation of Impulse and reaction turbines, Delaval's turbine, Parson's turbine

**Water turbines:** Classification, Principles and operations of Pelton wheel, Francis turbine, Kaplan turbine.

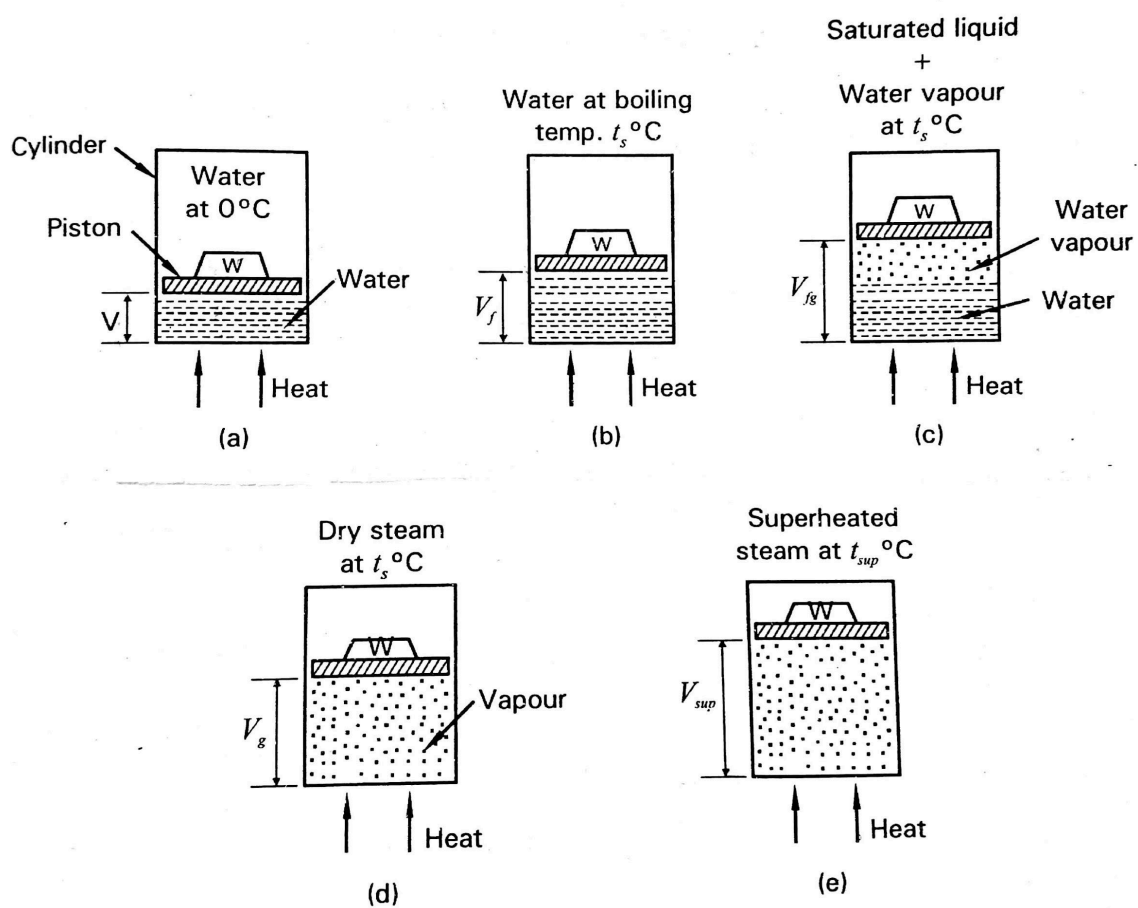
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### 3.1 Steam and its Properties

Water is a pure substance. A pure substance is one which retains its chemical composition even though it undergoes a change in phase during a thermodynamic Process. Water can exist in three different phases viz. Solid Phase as ice, liquid phase as water and gaseous phase as steam. In all its three different phases it retains the same chemical composition.

Steam is the gaseous phase of water. It is the mixture of liquid and gaseous, particles in which liquid particles are present in the form of suspension

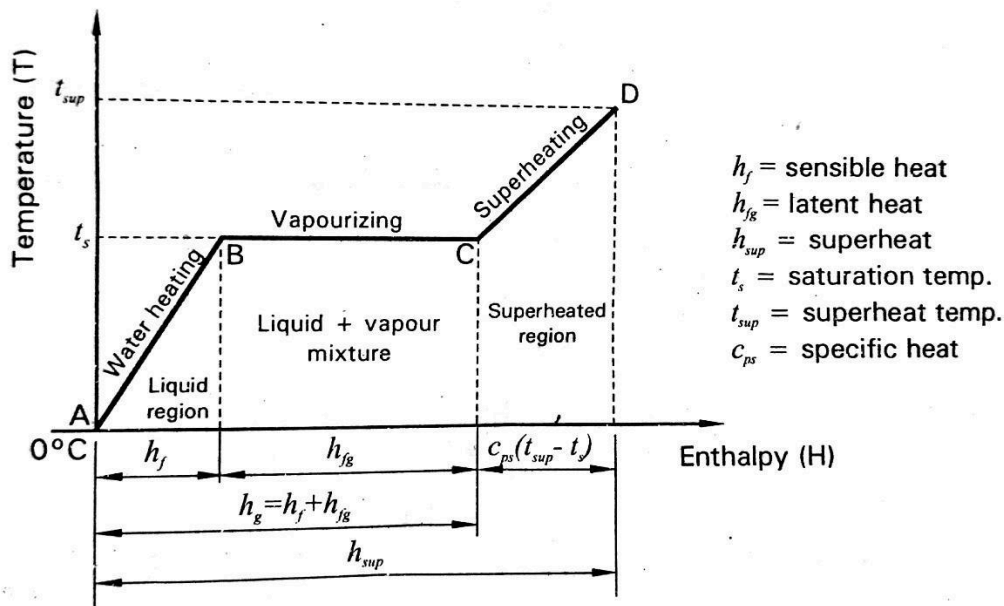


Consider 1 kg of water at 0°C taken in a cylinder fitted with a freely moving frictionless Piston as shown in above Fig The weight on the piston applies a constant pressure on water.

The formation of steam follows the following steps.

- ❖ When the water is heated at constant pressure, its temperature rises till the boiling point is reached. This temperature is called saturation temperature ( $t_s$ ). saturation temperature depends on the pressure at which water is heated. The water at this stage is known saturated water. **The amount of heat supplied to reach this saturation stage from 0 °C TO 100 °C is known as sensible heat or specific enthalpy of saturated water ( $h_f$ ).** at this stage there will be slight increase in the volume of water. It is known as specific volume of saturated water ( $V_f$ ). This process is represented by a line AB on temperature-enthalpy diagram. This process is called as water heating.
- ❖ Further addition of heat initiates the evaporation of water. During this process temperature remains at saturation temperature ( $t_s$ ) This is because the water will be saturated with heat and any further addition of heat changes only the phase from liquid to gas. This evaporation will be continued at ( $t_s$ ) until the whole of water is completely converted into steam. **The amount of heat supplied during this process is called latent heat or specific enthalpy of evaporation ( $h_{fg}$ ).** The initial formation of steam contains water particles is known as **wet steam**. On gradual heating more and more water particles get evaporated. As result, the wet steam starts transforming into dry steam. It is indicated by the line BC on T-H diagram. The steam at C is called as **dry steam**. Between B & C condition of water is wet steam.
- ❖ The total amount of heat added to get the desired wet steam having dryness fraction X from 0 °C is known as specific enthalpy of wet steam ( $H_{wet} = h_f + X h_{fg}$ ). The resulting volume is known as specific volume of wet steam ( $V_{wet}$ )
- ❖ The total dry steam at this stage is known as dry saturated steam (point C). the total amount of heat added to get this dry saturated steam from 0 °C is known as specific enthalpy of dry saturated steam ( $h_g = h_f + h_{fg}$ ). The resulting corresponding volume is known as specific volume of dry saturated steam ( $V_g$ )
- ❖ On heating the steam further, increases its temperature above the saturation temperature. The temperature of the steam above the saturation temperature at a given pressure is called superheated temperature. During this stage the dry steam will be heated from its dry state. The amount of heat supplied during this process is called **superheating**. The resulting steam is known as **superheated**

steam. The amount heat added to convert dry steam to superheated steam is known as enthalpy of super heat ( $h_{sh} = C_p (t_{sup} - t_s)$ ).



### 3.1.1 Types of Steam

Steam can exist in different states. They are

1. Wet steam
  2. Dry steam
  3. Superheated steam
1. **Wet Steam:** It is one in which water molecules and steam coexist to form a two-phase mixture at saturation temperature at a given pressure. The quality of wet steam is indicated by the term dryness fraction.
  2. **Dry Saturated Steam:** It is one which possess only steam particles and no water molecules at saturation temperature and given pressure. For dry steam dryness fraction 1.
  3. **Superheated Steam:** it is defined as the steam is heated beyond its dry saturated state to temperature higher than its saturated temperature at given pressure. The quality of superheated steam is indicated by the term "degree of superheat".

### 3.1.2 Definitions

#### Dryness Fraction

It is defined as the ratio of mass of the actual dry steam present in a known quantity of wet steam to the total mass of the wet steam.

$$\text{Dryness fraction } X = \frac{\text{mass of dry particles}}{\text{total mass of wet steam}}$$

$$= \frac{M_s}{M_s + M_w}$$

$m_s$  = mass of dry steam particles

$M_w$  = mass of water particles.

Dryness fraction of wet steam is less than 1

Dryness fraction of dry steam is 1.

Dryness fraction indicates the quality of wet steam.

**Saturation Temperature:** The boiling point of water at the given pressure is called saturation temperature. It is indicated by  $t_s$  in °C or K. Saturation temperature increases with pressure.

**Sensible heat or Specific enthalpy of saturated water:** it is the amount of heat supplied to 1 kg of water to rise its temperature from 0°C to saturation temperature at the given pressure. It is given by  $h_f = C_{p_w} (t_s - 273)$  in kJ/kg, where  $C_{p_w}$  = Specific heat of water = 4.2 kJ/kg K.

**Latent Heat or Specific enthalpy of vaporization:** It is the amount of heat required to convert 1 kg of liquid into completely dry steam at saturation temperature at the given pressure. It is represented by  $h_{fg}$  in kJ/kg. Latent heat decreases with the increase of pressure.

#### Specific Enthalpy of Superheat



The amount of heat required to increase the temperature of dry steam from its saturation

temperature to any desired higher temperature at the given constant pressure is called enthalpy of superheat. It is superheated by  $H_{sh}$

i.e. Enthalpy of superheat

$$H_{sh} = C_p (T_{sup} - t_s)$$

where,

$C_p$  = specific heat of superheated steam = 2.1 kJ/kg K

$T_{sup}$  = superheated temperature in °C(or) K

$t_s$  = saturated temperature in °C(or) K

### **Enthalpy of wet steam**

It is defined as the total amount of heat supplied to convert 1 kg of water at 0°C to 1 kg of Wet steam at the specified dryness fraction and given constant pressure.

i.e. Enthalpy of Wet Steam

$$h_{wet} = h_f + xh_{fg} \text{ in kJ/kg}$$

### **Enthalpy of dry saturated steam**

It is defined as the total amount of heat supplied to convert 1 kg of water at 0°C into 1 kg of dry steam at saturation temperature and given constant pressure

i. e Enthalpy of dry steam

$$h_g = h_f + h_{fg} \text{ in kJ/kg}$$

$x$  = dryness fraction

### **Enthalpy of Superheated Steam**

It is defined as the total amount of heat supplied to convert 1 kg of water at 0°C into 1 kg of superheated steam at the stated superheated temperature and given constant pressure.

Enthalpy of Superheated steam

$$H_{\text{sup}} = h_f + h_{fg} + H_{\text{sh}}$$

$$H_{\text{sup}} = h_g + C_p (T_{\text{sup}} - t_s) \text{ in kJ/kg}$$

**Specific Volume:** Specific volume is the volume occupied by the unit mass of substance. It is in  $\text{m}^3/\text{Kg}$

**Specific Volume of Saturated Water:** It is defined as the volume occupied by 1 kg of water at the saturation temperature and given pressure. It is denoted by  $V_f$  (or)  $V_w$  in  $\text{m}^3/\text{Kg}$

**Specific Volume of Dry Saturated Steam:** It is defined as the volume occupied by 1 kg of dry steam at given pressure. It is denoted by  $V_g$  (or)  $V_s$ , in  $\text{m}^3/\text{Kg}$

**Specific Volume of Wet Steam:** It is defined as the volume occupied by 1 kg of wet steam at a given pressure.

$$\text{It is given by } V_s^1 = xV_g \text{ in } \text{m}^3/\text{Kg}$$

**Specific Volume of Superheated Steam:** It is defined as the volume occupied by 1 kg of superheated steam at a given pressure and superheated temperature.

Superheated steam behaves like a perfect gas.

Applying Charles's law to the superheating stage, we get

$$\frac{V_s}{t_s} = \frac{V_{\text{sup}}}{T_{\text{sup}}}$$

$$V_{\text{sup}} = V_s * \frac{T_{\text{sup}}}{t_s}$$

## EXTERNAL WORK OF EVAPORATION

In evaporation stage, the amount of heat supplied to the water causes a large increase in volume. This is mainly due to phase conversion of water into dry steam.



This large change in volume, makes the piston of cylinder to move doing some external work.

The fraction of the latent heat of vaporization which does an external work is called the external of evaporation.

It is given by

Work done = pressure \* specific volume

$$E = P v, \text{ bar} \cdot \text{m}^3/\text{Kg} \quad [\text{But } 1 \text{ bar} = 100 \text{ KN/m}^2 \text{ and } 1 \text{ NM} = 1\text{J}]$$

$$E = 100 P v \text{ KJ/Kg} \quad [\text{KN/m}^2 \cdot \text{m}^3/\text{Kg} = \text{KN-M/Kg} = \text{KJ/Kg}]$$

So, external work of evaporation of wet steam  $E_{\text{wet}} = 100 P \cdot X \cdot v_g, \text{ KJ/Kg}$

External work of evaporation of dry steam  $E_{\text{dry}} = 100 P \cdot v_g, \text{ KJ/Kg}$

External work of evaporation of wet steam  $E_{\text{sup}} = 100 P v_u, \text{ KJ/Kg}$

### True Latent Heat (Internal energy) of steam (U):

It is the actual heat energy retained in the system after doing external work on piston during converting the unit mass of water into steam at required temperature.

True Latent Heat (Internal energy) of steam = latent heat of vaporization – work done

$$U = (h - E), \text{ KJ/Kg hence,}$$

Internal energy of wet steam  $U_{\text{wet}} = (h_{\text{wet}} - E_{\text{wet}}), \text{ KJ/Kg}$

Internal energy of dry steam  $U_{\text{dry}} = (h_{\text{dry}} - E_{\text{dry}}), \text{ KJ/Kg}$

Internal energy of super-heated steam  $U_{\text{sup}} = (h_{\text{sup}} - E_{\text{sup}}), \text{ KJ/Kg}$

### Dryness Fraction

It is defined as the ratio of mass of the actual dry steam present in a known quantity of wet steam to the total mass of the wet steam.

Dryness fraction  $X = \text{mass of dry particle} / \text{mass of wet steam}$





$$X = \frac{M_g}{M_g + M_w}$$

$M_g$  = mass of dry steam particles

$M_w$  = mass of water particles.

Dryness fraction of wet steam is less than 1

Dryness fraction of dry steam is 1.

Dryness fraction indicates the quality of wet steam.

### **Wetness Fraction**

It is measure of wetness present in the given quantity of saturated steam. It is calculated by the

Ratio of the mass of water vapor ( $m_w$ ) to the total mass of wet stem.

$$\text{Wetness Fraction} = \frac{M_w}{M_w + M_g}$$

[Note: dryness fraction + wetness fraction = 1]

### **Quality of steam**

It representation of dryness fraction in percentage

$$\text{Quality of steam} = X * 100$$

### **Priming**

It is representation of wetness fraction in percentage

$$\text{Priming} = (1-X) * 100$$

[Note: quality steam + priming = 100 %]

## **3.1.3 Steam tables**

### **Saturated water and steam (temperature) table**

It contains values of pressure, specific volume, specific enthalpy (and specific entropy) corresponding to different temperatures. A sample of the table is given below.



| Temperature<br>in °C (t) | Absolute<br>pressure<br>in bar (p) | Specific volume in<br>m <sup>3</sup> /kg |                | Specific enthalpy in kJ/kg |                 |                |
|--------------------------|------------------------------------|------------------------------------------|----------------|----------------------------|-----------------|----------------|
|                          |                                    | v <sub>f</sub>                           | v <sub>g</sub> | h <sub>f</sub>             | h <sub>fg</sub> | h <sub>g</sub> |
| 20                       | 0.023 37                           | 0.001002                                 | 57.838         | 83.9                       | 2454.3          | 2538.2         |
| 40                       | 0.073 75                           | 0.001008                                 | 19.546         | 167.5                      | 2406.9          | 2574.4         |
| 100                      | 1.0133                             | 0.001044                                 | 1.6730         | 419.1                      | 2256.9          | 2676.0         |
| 175                      | 8.9244                             | 0.001121                                 | 0.21654        | 741.1                      | 2030.7          | 2771.8         |

### Saturated water and steam (pressure) table

It contains values of pressure, specific volume, specific enthalpy (and specific entropy) corresponding to different pressures (absolute). A sample of the table is given below.

| Absolute<br>pressure in<br>bar (p) | Temperature<br>in °C (t) | Specific volume in<br>m <sup>3</sup> /kg |                | Specific enthalpy in kJ/kg |                 |                |
|------------------------------------|--------------------------|------------------------------------------|----------------|----------------------------|-----------------|----------------|
|                                    |                          | v <sub>f</sub>                           | v <sub>g</sub> | h <sub>f</sub>             | h <sub>fg</sub> | h <sub>g</sub> |
| 0.15                               | 54.00                    | 0.001014                                 | 10.023         | 226.0                      | 2373.2          | 2599.2         |
| 1.0133 25                          | 100                      | 0.001044                                 | 1.6730         | 419.1                      | 2256.9          | 2676.0         |
| 10.0                               | 179.9                    | 0.001127                                 | 0.19430        | 762.6                      | 2013.6          | 2776.2         |
| 125                                | 327.8                    | 0.001547                                 | 0.01351<br>8   | 1511.9                     | 1166.5          | 2678.4         |

Choice of the table depends on whether we know the operating temperature or the operating pressure.

### 3.1.4 Numericals

**Problem 1:** Determine the density of 1 kg of steam initially at a pressure of 10 bar absolute, having a dryness fraction of 0.78. If 500 kJ of heat is added at constant pressure, determine the condition and internal energy of the final state of steam. Take specific heat of saturated steam as 2.1 kJ/kgK. (VTU January 2003)

Solution:

Given,  $m=1$  kg,  $p=10$  bar = 1000 kPa,  $x=0.78$ , heat added = 500 kJ.

We know that heat added at constant pressure is nothing but increase in enthalpy.

Let,  $h_1$  = initial enthalpy,

$h_2$  = final enthalpy,

then,  $h_2 = h_1 + \text{heat added}$ .

**Initial enthalpy:**

From steam tables, at 10 bar,

$$h_f = 762.6 \text{ kJ/kg}$$

$$v_g = 0.19430 \text{ m}^3/\text{kg}$$

$$h_{fg} = 2013.6 \text{ kJ/kg}$$

$$h_1 = h_f + x h_{fg} = 762.6 + 0.78 \times 2013.6 = 2333.2 \text{ kJ/kg}$$

**Final enthalpy:**

$$h_2 = h_1 + \text{heat added} = 2333.2 + 500 = 2833.2 \text{ kJ/kg}.$$

From steam tables, at 10 bar

$$h_g = 2776.2 \text{ kJ/kg}, \quad t_{\text{sat}} = 179.9^\circ\text{C}.$$



Since  $h_2$  is greater than  $h_g$ , steam is superheated.

**Enthalpy of superheated steam:**

$$h_2 = h_g + c_{ps} (T_{\text{sup}} - T_{\text{sat}})$$

$$2833.2 = 2776.2 + 2.1 (T_{\text{sup}} - 179.9)$$

$$\text{therefore, } T_{\text{sup}} = 207^\circ\text{C}$$

**Specific volume of steam:**

Specific volume of superheated steam,

$$v_2 = v_g \cdot T_{\text{sup}} / T_{\text{sat}}$$

$$T_{\text{sat}} = 179.9 + 273 = 452.9 \text{ K}$$

$$T_{\text{sup}} = 207 + 273 = 480 \text{ K}$$

$$v_2 = 0.19430 \times 480 / 452.9 = 0.20592 \text{ m}^3/\text{kg}.$$

**Internal energy:**

Internal energy of superheated steam,

$$u_2 = h_2 - pv_2 = 2833.2 - 1000 \times 0.20592 = 2627.2 \text{ kJ/kg}.$$

**Problem 2:** what is the enthalpy of 5 kg of steam under the following conditioning? (i) 0.8 bar absolute pressure and 90% dry, and (ii) 20 bar absolute pressure at  $300^\circ\text{C}$ . Take specific heat of superheated steam as  $2.25 \text{ kJ/kg}$ . (VTU January 2004)

Solution:

$$(i) p = 0.8 \text{ bar, } x = 90\% = 0.9.$$



Enthalpy of wet steam per kg

$$h = h_f + x h_{fg} \quad \text{kJ/kg}$$

For m kg,

$$H = m(h_f + x h_{fg}) = 5(391.7 + 0.9 \times 2274.1) = 12191.95 \text{ kJ.}$$

(ii) Pressure = 20 bar, temperature = 300°C

From steam tables at 20 bar,  $t_{\text{sat}} = 212.4^\circ\text{C}$

Since  $t_{\text{sat}}$  is less than the temperature of steam, the steam is superheated.

From steam tables at 20 bar,  $h_g = 2797.2 \text{ kJ/kg}$ .

Enthalpy of superheated steam per kg

$$h_{\text{sup}} = h_g + c_{\text{ps}}(T_{\text{sup}} - T_{\text{sat}})$$

For m kg,

$$H_{\text{sup}} = m\{h_g + c_{\text{ps}}(T_{\text{sup}} - T_{\text{sat}})\} = 5\{(2797.2 + 2.25(300 - 212.4))\} = 14971.5 \text{ kJ.}$$

**Problem 3:** Find the internal energy of 2.5 kg of steam at 20 bar when (i) it is wet and its dryness fraction is 0.9, (ii) it is superheated to 350°C. Take specific heat of steam as 2.3 kJ/kgK (VTU February 2005)

Solution:

Given,  $m = 2.5 \text{ kg}$ ,  $p = 20 \text{ bar}$ .

(i) steam is wet,  $x = 0.9$

From steam table at 20 bar,

$$h_f = 908.5 \text{ kJ/kg.}$$

$$h_{fg} = 1888.7 \text{ kJ/kg}$$



Enthalpy of wet steam

$$h = h_f + x h_{fg} = 908.5 + 0.9 \times 1887.7 = 2608.33 \text{ kJ/kg.}$$

From steam tables at 20 bar,

$$v_g = 0.09955 \text{ m}^3/\text{kg.}$$

Specific volume of wet steam,

$$v = x v_g = 0.9 \times 0.09955 = 0.089595 \text{ m}^3/\text{kg.}$$

Specific internal energy,

$$u = h - pv = 2608.33 - 2000 \times 0.089595 = 2429.14 \text{ kJ/kg.}$$

Therefore, internal energy for m kg of steam,

$$U = mu = 2.5 \times 2429.14 = 6072.85 \text{ kJ.}$$

(ii) steam is superheated to 350°C:

From steam tables, at 20 bar,

$$t_{\text{sat}} = 212.4^\circ\text{C}, h_g = 2797.2 \text{ kJ/kg}, v_g = 0.09955 \text{ m}^3/\text{kg.}$$

Enthalpy of superheated steam,

$$h_{\text{sup}} = h_g + c_{ps}(T_{\text{sup}} - T_{\text{sat}}) = 2797.2 (350 - 212.4) = 3113.68 \text{ kJ/kg.}$$

Specific volume of superheated steam,

$$v_{\text{sup}} = v_g \cdot T_{\text{sup}} / T_{\text{sat}}$$

$$T_{\text{sup}} = 350 + 273 = 623 \text{ K}$$

$$T_{\text{sat}} = 212.4 + 273 = 485.4 \text{ K}$$

$$v_{\text{sup}} = 0.09955 \times 623 / 485.4 = 0.12777 \text{ m}^3/\text{kg.}$$

Specific internal energy,

$$u_{\text{sup}} = h_{\text{sup}} - pv_{\text{sup}} = 3113.68 - 2000 \times 0.12777 = 2858.13 \text{ kJ/kg.}$$



Therefore, internal energy for m kg of steam,

$$U_{\text{sup}} = m u_{\text{sup}} = 2.5 \times 2858.13 = 7145.34 \text{ kJ}.$$

**Problem 4:** Steam is at 9 bar pressure and dryness fraction 0.98. Find the quality and temperature of steam when (i) the steam loses 50 kJ/kg at constant pressure, and (ii) steam receives 150 kJ/kg at constant pressure. (VTU July 2005)

Solution:

From steam tables, at 9 bar,

$$h_f = 742.6 \text{ kJ/kg},$$

$$h_{fg} = 2029.5 \text{ kJ/kg},$$

$$h_g = 2772.1 \text{ kJ/kg},$$

$$t_{\text{sat}} = 175.4^\circ\text{C},$$

Given,

$$x_1 = 0.98.$$

Initial enthalpy of steam,

$$h_1 = h_f + x_1 h_{fg} = 742.6 + 0.98 \times 2029.5 = 2731.51 \text{ kJ/kg}.$$

(i) Steam loses 50 kJ/kg at constant pressure,

$$h_2 = h_1 - 50 = 2731.51 - 50 = 2681.51 \text{ kJ/kg}.$$

Since  $h_2$  is less than  $h_g$ , steam is wet.

Enthalpy of wet steam,

$$h_2 = h_f + x_2 h_{fg}$$

$$2681.51 = 742.6 + x_2 \times 2029.5$$

$$x_2 = 0.955.$$

(ii) Steam receives 150 kJ/kg of heat.



$$h_2 = h_1 + 150 = 2731.51 + 150 = 2881.51 \text{ kJ/kg.}$$

Since  $h_2$  is greater than  $h_g$ , steam is superheated.

Enthalpy of superheated steam,

$$h_2 = h_g + c_{ps}(t_{\text{sup}} - t_{\text{sat}})$$

$$2881.51 = 2772.1 + 2.1 (t_{\text{sup}} - 175.4)$$

$$t_{\text{sup}} = 227.5^\circ\text{C.}$$

**Problem 5:** 2 kg of wet steam is heated at a constant pressure of 2 bar until its temperature increases to  $150^\circ\text{C}$ . The heat transferred is 2100 kJ. Find the initial dryness fraction of steam. Take the specific heat of steam as 2.1 kJ/kg. The properties of steam at 2 bar pressure are given below-

| p<br>(bar) | $t_s$<br>( $^\circ\text{C}$ ) | $v_f$<br>( $\text{m}^3/\text{kg}$ ) | $v_g$<br>( $\text{m}^3/\text{kg}$ ) | $h_f$<br>(kJ/kg) | $h_g$<br>(kJ/kg) |
|------------|-------------------------------|-------------------------------------|-------------------------------------|------------------|------------------|
| 2          | 120.23                        | 0.001061                            | 0.8857                              | 504.5            | 2706.5           |

(VTU January 2006)

Solution:

Given,  $m=2 \text{ kg}$ ,  $p = 2 \text{ bar} = 2000 \text{ kPa}$ .  $T_2=150^\circ\text{C}$ , heat added = 2100 kJ.

$$h_{fg} = h_g - h_f = 2706.5 - 504.5 = 2202 \text{ kJ/kg.}$$

Since  $t_2$  is greater than  $t_{\text{sat}}$ , steam is superheated in the final state.

Enthalpy of superheated steam,

$$h_2 = h_g + c_p (t_{\text{sup}} - t_{\text{sat}}) = 2706.5 + 2.1 (150-120.3) = 2769.2 \text{ kJ/kg.}$$

Total heat added = 2100 kJ.

Heat added per kg of steam =  $2100/2 = 1050 \text{ kJ/kg}$ .



Final enthalpy of steam,

$$h_2 = h_1 + 1050 \text{ kJ/kg.}$$

Therefore,

$$h_1 = h_2 - 1050 = 2769.2 - 1050 = 1719.2 \text{ kJ/kg.}$$

Since  $h_1$  is less than  $h_g$ , steam is wet.

Enthalpy of wet steam,

$$h_1 = h_f + x_1 h_{fg}$$

$$1719.2 = 504.5 + x_1 \times 2202$$

Initial dryness fraction,

$$x_1 = 0.5516$$

### 3.2 Boilers

Boiler is a closed vessel in which the given fluid is heated and evaporated. Steam boiler is one in which water is evaporated. It is also called as steam generator.

The function of steam boiler is to supply the steam at the required constant pressure with its quality either dry or as nearly dry, or superheated.

#### 3.2.1 Classification of boilers

I. According to types of tubes

- ❖ Fired tube boiler, Ex: Cochran Boiler, Lancashire boiler, Cornish boiler, Scotch boiler
- ❖ Water tube boiler Ex: Bobcock and Wilcox boiler, Stirling boiler.

II. According to the location of furnace

- ❖ Internally fired boiler Ex: Cornish, Cochran boiler
- ❖ Externally fired boilers Ex: Bobcock and Wilcox boilers

III. According to the axis of boiler drum

- ❖ Vertical boilers Ex: Scotch, Cochran boilers
- ❖ Horizontal boiler Ex: Locomotive Boiler, Lancashire boiler

IV. According to circulation of water

- ❖ Natural Circulation Ex: Cochran, Bobcock and Wilcox boiler
- ❖ Forced circulation Ex: Lamont, Benson boiler

V. According to use

- ❖ Stationary boilers Ex: Boiler in power plant
- ❖ Mobile boilers Ex: Locomotive and Marine boilers

VI. According to the pressure

- ❖ Low pressure boilers Ex: Cornish, Cochran
- ❖ High pressure boilers Ex: Lamont, Benson boilers.

### 3.2.1.1 FIRE TUBE BOILERS

In the fire tube boilers, the hot flue gases produced by the combustion of fuels are led through a tube or a nest of tubes around which the water circulates. Fire tube boilers are suitable for steady working pressure up to 20 bars.

Ex: Cochran boiler, Cornish boiler, Lancashire boiler

#### Lancashire Boiler

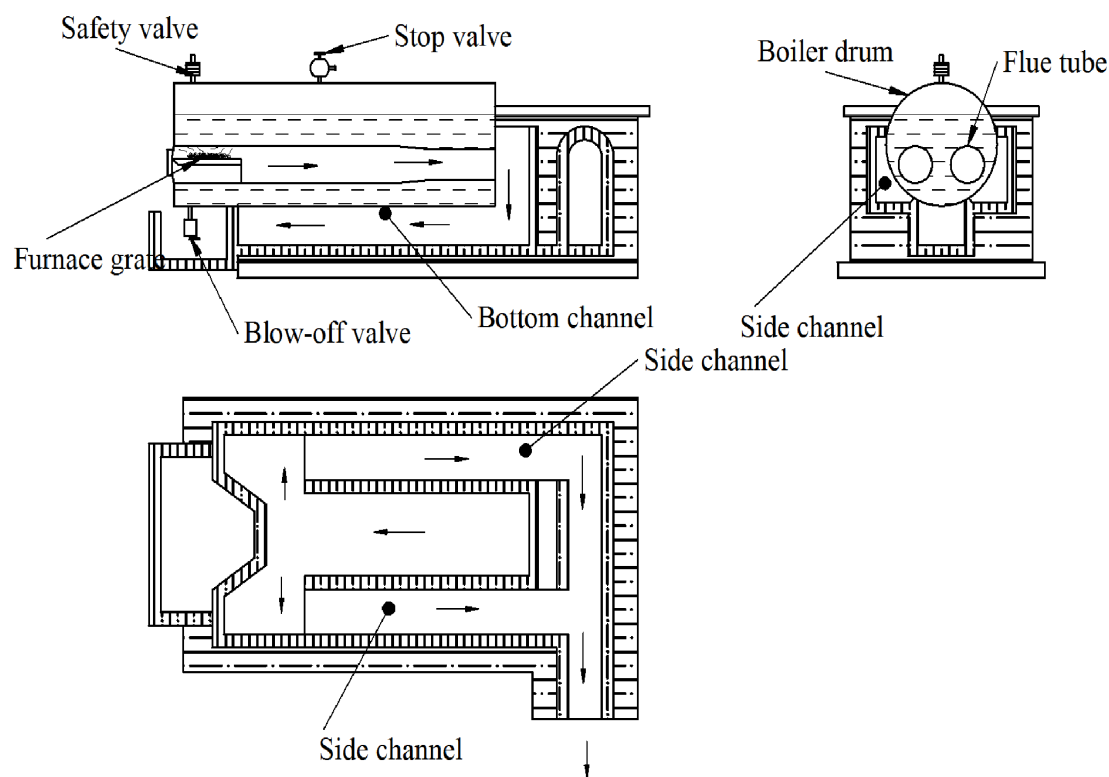


Figure: Lancashire Boiler

Lancashire boiler is a horizontal, internally fired, natural circulation fire tube boiler. This boiler produces steam up to a pressure of 15 bar. Its capacity is 8500 kg of steam per hour. It is used widely in sugar mills and chemical industries. The construction of Lancashire boiler is shown in Fig.

### **WORKING**

The boiler shell is filled with water to  $\frac{3}{4}$  of its volume. The fuel is charged through the furnace door onto the grate and burnt. The hot gases produced from the combustion of fuel pass through the flue tubes from the front end to the rear end of the boiler. This is called first run of flue gases. In the first run about 83% of total heat is transferred to the water. Then the gases pass down ward and unite in the bottom central channel and travel from rear end to the front end of the boiler. This is called second run of the gases. In this run 9.5% of total heat is transferred to the water. Now the hot gases divide at the front end of the boiler and enter into the side channels 1 & 2 and travel from front end to rear end of the boiler. This is called third run of the flue gases. In this run about 7.5% of heat is transferred to the water. Now the gases from channel (1) & (2) unite in the rear passage and make their exit to the atmosphere through a chimney.

Since heat is transferred to the water in all sides of the boiler, water gets evaporated producing steam. Steam is accumulated in the steam space and is taken out through a steam stop valve.

The boiler is mounted with essential mountings and accessories like steam stop valve, safety valve, blow off valve etc.

### **Advantages**

1. It is economical, easy to operate, clean and inspect.
2. Low maintenance lost.
3. Load fluctuation can be easily met due to large water storage.
4. Overall efficiency is high (85%) due to super heater and economizer.

### **Disadvantages**

1. Steam generation is slow
2. Occupies more space

### 3. Low pressure boiler

#### 3.2.1.2 Water tube boilers

In water tube boilers, the water circulates inside the tubes while the hot gases produced by the combustion of fuels pass around them. Water tube boilers are more suitable than the fire tube for the generation of steam at very high pressures

Ex: Babcock and Wilcox boiler, Stirling boiler etc.

#### Babcock and Wilcox Boiler

Babcock and Wilcox boiler is a horizontal, externally fired, naturally circulation, stationary, water tube boiler.

The main parts of the boiler are shown in Fig



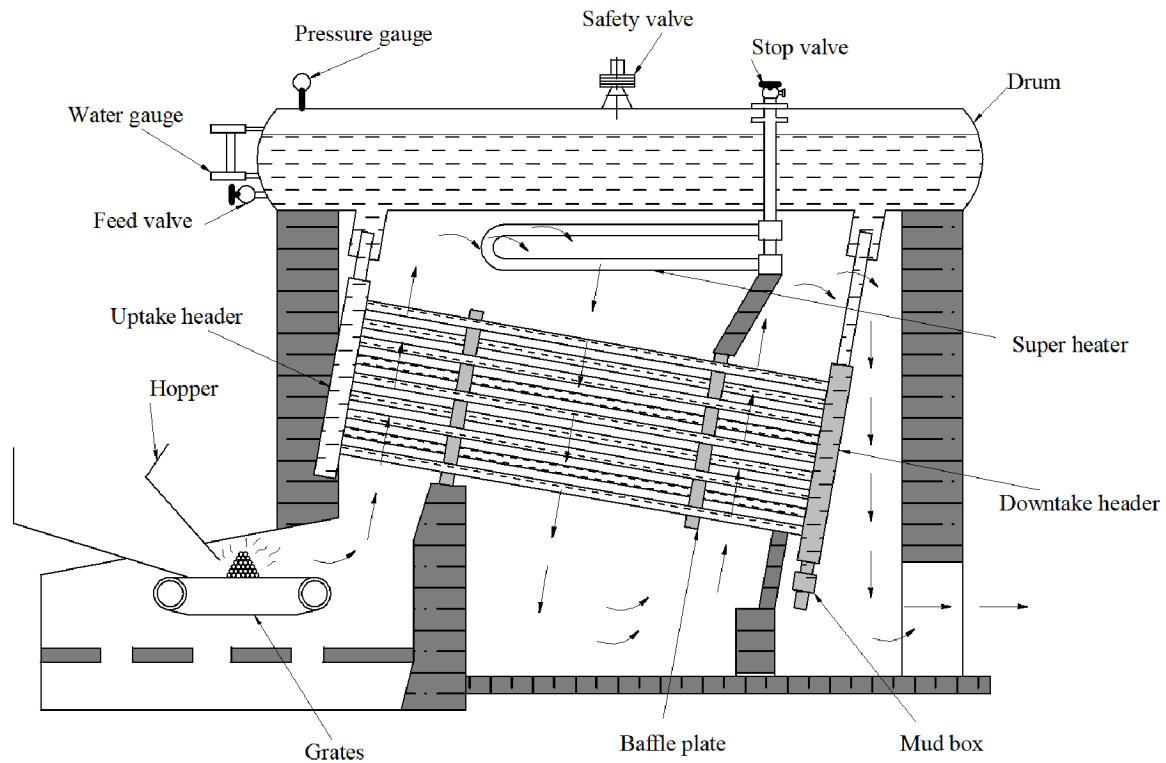


Figure: Babcock & Wilcox Boiler

### Working

The furnace of the boiler is located below the uptake header. The coal is fed through the fire door onto the grate and is fired. Due to combustion of coal, large quantity of gases is generated. These hot gases flow over the steel tubes as shown by arrows in figure.

Baffles are provided to deflect the hot flue gases so that they travel a long way and cover large heating surface. During this path of the hot gases, heat is transferred to the water tubes and makes the water to evaporate.

The water and the steam mixture from these portions ascends through the uptake header and reaches the boiler drum

Due to this, a continuous circulation of water is established between the boiler drum and

the water tubes. The steam gets separated from the surface of the water in the boiler drum and accumulates in the steam space.

A super heater is provided in the path of flue gases which heats the steam again. The boiler is mounted with mountings such as water level indicator, pressure gauge, safety valve etc at shown as their appropriate place.

The capacity of the boiler is 20000 to 40000 kg/hr. Working pressure is 10 to 70 bar

. Advantages:

1. The defective tubes can be replaced easily
2. The draughts loss is comparatively minimum
3. Boiler may expand and contract freely.

### 3.2.2 Boiler mountings

The boiler mountings are necessary for the proper function & safety of a boiler.

- Safety valve
- Water level indicator
- Pressure gauge  Blow off valve
- Steam stop valve
- Feed check valve

### 3.2.3 Boiler accessories

Boiler accessories are auxiliary parts used in steam boilers for their proper functioning and to improve the efficiency of the power plant.

- Super heater
- Economizer
- Air pre-heater
- Steam separator
- Steam trap

#### 1. **Safety Valves:**

**Location:** Top of the boiler shell.



**Function: Protect** the boiler against the high pressure of steam, if the pressure inside the boiler increases beyond the designed pressure, the safety valve automatically opens and allows the excess steam to come out, preventing the explosion of boiler.

## 2. Water Level Indicators:

**Location:** It is fitted at the front end of the boiler to the water space.

**Function:** It indicates the level of water in the boiler drum and guide boiler attendant to maintain a constant level of water in the boiler.

## 3. Fusible Plug:

**Location:** It is fitted over the crown of the furnace or combustion chamber.

**Function: It is used** to extinguish the fire in the furnace of the boiler when the water level falls too much below the normal level.

## 4. Pressure Gauge:

**Location:** It is mounted in the front end at the top of the boiler.

**Function:** It indicates the pressure of the steam in the boiler.

## 5. Feed Check Valve:

**Location:** It is fitted in the feed water pipe line close to the boiler.

**Function:** It is used to feed the water into the boiler continuously. It also regulates the rate of flow of feed water.

## 6. Blow-off cock:

**Location: It** is fitted at the lowest part of the boiler

**Function:** It is used to remove periodically the sediments collected at the bottom of the boiler while in operation. It also used to empty the water in the boiler when required for periodical cleaning and inspection.

## 7. Steam Stop Valve:

**Location:** It is mounted on the boiler shell above the steam space.

**Function:** regulates the flow of steam from the boiler to the steam or turbines and shut the steam flow when not required.

#### 8. Economizer:

**Location:** It is fitted near the chimney.

**Function:** Its function is to heat the feed water using exhaust hot flue gases.

#### 9. Air Pre-heater:

**Location:** It is fitted in between the economizer and chimney

**Function:** It supplies the hot air to the furnace by absorbing the heat from the hot exhaust gases and accelerates the combustion of fuel.

#### 10. Super Heater:

**Location:** It is fitted in the path of hot flue gases from the furnace.

**Function:** It absorbs the heat from the hot flue gases and superheat the saturated steam without changing its pressure.

#### 11. Chimney:

**Location** It is built at the exit end of the hot flue gases.

**Function:** It exhausts the products of combustion to the atmosphere and draws the hot gases from the boiler.

#### 12. Feed Pump:

**Location:** It is provided between feed water tank and boiler drum. **Function:** It pumps the water from the feed water tank into the boiler.

#### 13. Steam Separator:

**Location:** It is fitted very close to the steam turbine on the main supply pipe.

**Function** It separates the water particles from the steam flowing in the pipe line before it enters the turbine.

#### 14. Steam Trap:

**Location:** It is connected to a small by-pass pipe which branches off from the main steam pipe line.



**Function:** It is used to drain off the condensed water accumulating in the steam pipe lined without allowing the high-pressure steam to escape through it.

### 3.3 Steam Turbine

A **steam turbine** is a mechanical device that extracts thermal energy from pressurized steam, and converts it into useful mechanical work. A steam turbine is a prime mover in which rotary motion is obtained by the gradual change of momentum of the steam. Steam turbines are primarily used to run alternators or generators in thermal power plants. It is also used to rotate the propeller of ships through reduction gearing.

#### 3.3.1 Main parts of a steam turbine:-

**Nozzle** :In steam turbines, normally, convergent-divergent type of nozzles is used. When steam flows through the nozzles, there is a pressure drop, which is converted into velocity or kinetic energy. The nozzles also guide the steam in the proper direction to strike the blades. The nozzles are kept very close to the blades to minimize the losses.

**Rotor**: The rotor or runner consists of a circular disc fixed to a horizontal shaft.

**Blade** :On the periphery of the rotor, a large no of blades are fixed. The steam jet from the nozzles impinges on the surface of the blades due to which the rotor rotates.

**Casing**: It is a steam tight steel casing which encloses the rotor, blades etc. In a multistage turbine , the casing also accommodates the fixed blades.

#### 3.3.2 Classification of steam turbine

Steam turbines can be classified in to two types.

i) Impulse turbine

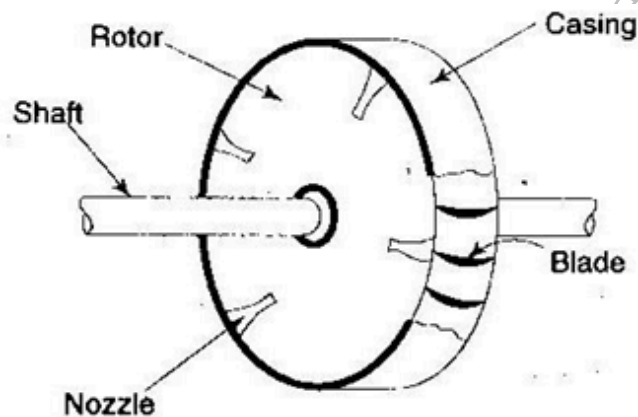
ii) Reaction turbine



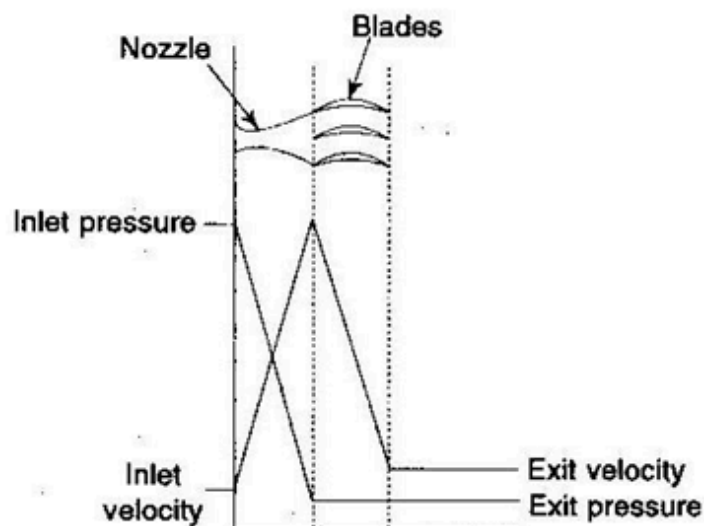
### Single stage impulse turbine (De-Laval turbine)

The pressure energy is converted into velocity energy or kinetic energy by the expansion of steam through a set of nozzles. Normally, in steam turbines, convergent-divergent nozzles are used. The kinetic energy is converted into mechanical energy with the help of moving blades, fixed on the rotor. The rotor is connected to the output shaft. All these parts are enclosed in a casing.

This turbine is not suitable for practical purposes, since high-pressure steam expands in one set of nozzles and get converted to very high velocity steam, due to this the rotor will rotate at a very high speed. So in practice, multistage impulse turbines or compound turbines are used.



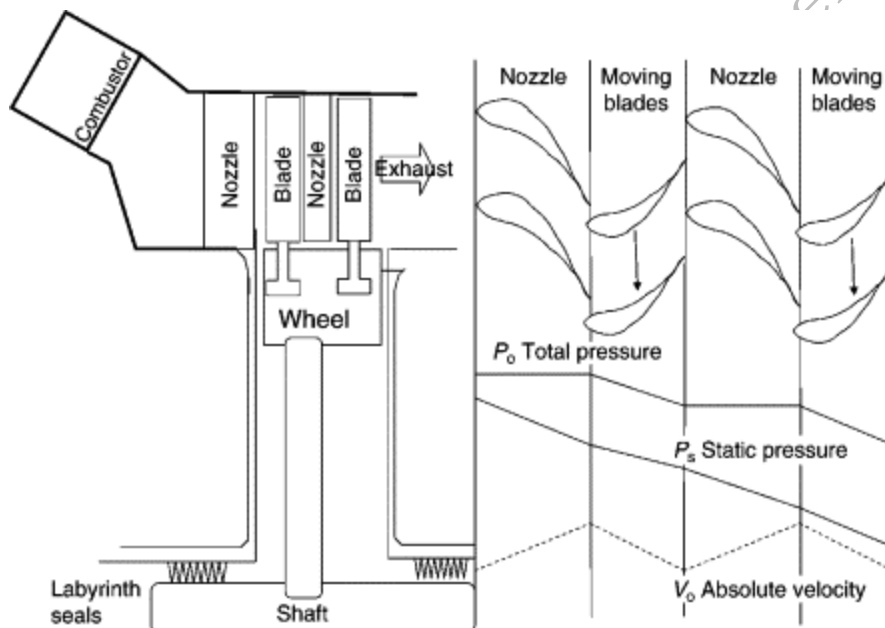
**Fig. 16.2(a)** *Single stage impulse turbine*



**Pressure velocity diagram for a single stage impulse turbine**

## Reaction turbine (Parson turbine)

The reaction turbine, as the name implies, is turned by reactive force rather than by a direct push or impulse. In reaction turbines, the blades that project radially from the periphery of the rotor are formed and mounted so that the spaces between the blades will have the nozzle shape. Since these blades are mounted on the revolving rotor, they are called moving blades. Fixed or stationary blades of the same shape as the moving blades are fastened to the casing in which the rotor revolves. The fixed blades guide the steam into the moving blade. A reaction turbine is moved by three main forces: (1) the reactive force produced on the moving blades as the steam increases in velocity as it expands through the fixed blades. (2) The reactive force produced on the moving blades when the steam changes its direction.



## Difference between impulse turbine and reaction turbine

| Impulse turbine                                       | Reaction turbine                                                                                                   |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Complete expansion of steam takes place in the nozzle | Partial expansion of steam takes place in the fixed blades and further expansion takes place in the moving blades. |

|                                                                               |                                                                   |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Blades are symmetrical in shape                                               | Blades are non symmetrical in shape i.e., aerofoil section        |
| The rotor runs at higher speeds                                               | The rotor runs at relatively low speed.                           |
| The impulse turbines are used for small power generation plant.               | The reaction turbines are used in large power generation plant.   |
| Less floor area is required.(small power plant)                               | More floor area is required.(Medium and large power plant)        |
| The pressure of steam remains constant from inlet to the outlet of the blade. | The pressure of steam drops from inlet to the outlet of the blade |

### 3.4 Water turbines

**Introduction:** Water turbines were developed in the nineteenth century and were widely used for industrial power prior to electrical grids. Now they are mostly used for electric power generation. They harness a clean and renewable energy source.

A hydraulic turbine is a machine, which converts pressure energy in to mechanical energy. It uses the kinetic energy and potential energy of water and sets the rotor in motion by the dynamic action of water flowing from high head.

#### 3.4.1 Classification of hydraulic turbines

The classification of water turbines are as follows.

1. According to the type of energy at inlet

- i) Impulse turbine
- ii) Reaction turbine

2. According to the direction of flow of water through the runner

- i) Tangential flow
- ii) Radial inward flow
- iii) Radial outward flow
- iv) Axial flow
- v) Mixed flow

3. According to the head under which turbine works

- i) High head, Impulse turbine Ex: pelton wheel.
- ii) Medium head, reaction turbine Ex: Francis turbine.
- iii) Low head, reaction turbine: Ex: Kaplan turbine.

4. According to the specific speed of the turbine

- i) Low specific speed turbine, impulse turbine. Ex: pelton wheel.
- ii) Medium specific speed, reaction turbine. Ex: Francis turbine.
- iii) High specific speed, reaction turbine. Ex: Kaplan turbine

5. According to the position of the shaft

- i) Horizontal shaft
- ii) Vertical shaft

#### 3.4.1.1 Impulse turbine

In impulse turbine a high-velocity jet of water hits a series of specially shaped cups on the runner. Impulse turbines change the velocity of a water jet. The jet impinges on the turbine's curved blades which reverse the flow. The resulting change in momentum (impulse) causes a force on the turbine blades. Since the turbine is spinning, the force acts through a distance (work) and the diverted water flow is left with diminished energy. Prior to hitting the turbine blades, the water's pressure (potential energy) is converted to kinetic energy by a nozzle and focused on the turbine. No pressure change occurs at the turbine blades, and the turbine doesn't require a housing for operation

#### Pelton wheel

Pelton wheel, a type of impulse turbine, named after L. A. Pelton who invented it in 1880. Water passes through nozzles and strikes cups arranged on the periphery of a runner, or wheel, which causes the runner to rotate, producing mechanical energy. The runner is fixed on a shaft, and the rotational motion of the turbine is transmitted by the shaft to a generator. Pelton turbines are suited to high head, low flow applications; they are used in storage power stations (dams) with downward gradients above 300 meters.

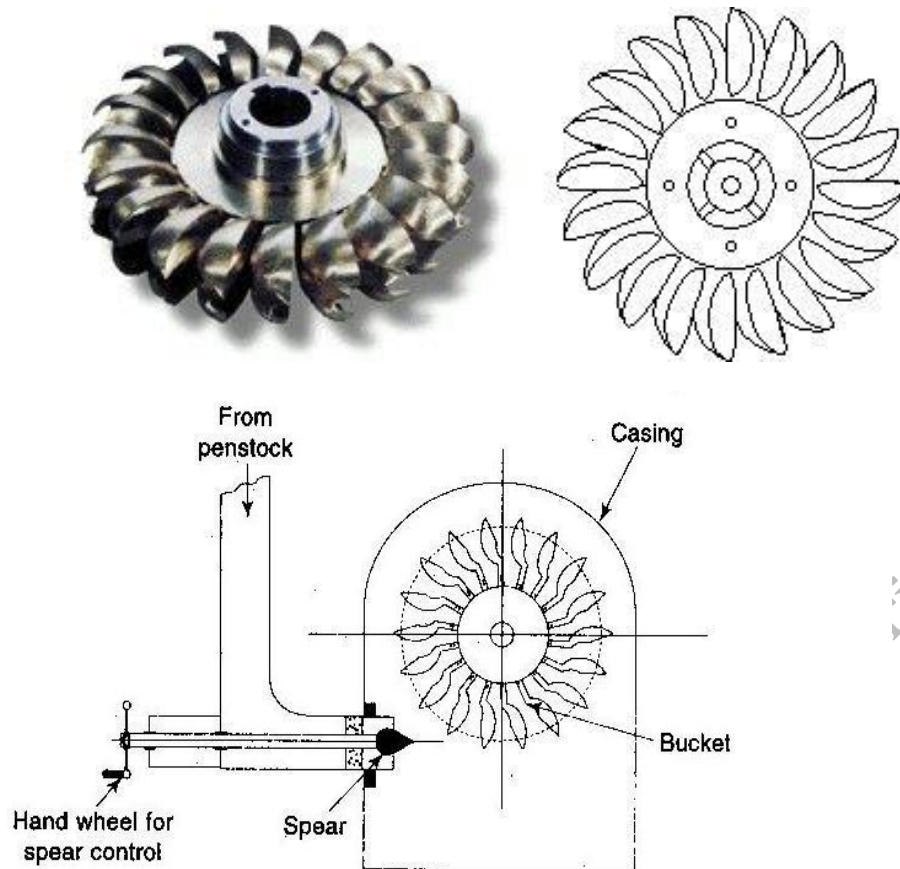


Fig. 17.7(a) Pelton wheel

### 3.4.1.2 Reaction turbines

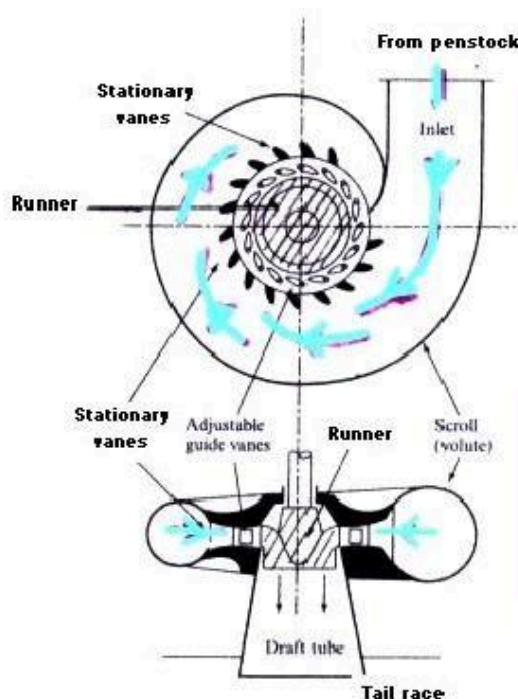
Reaction turbines are acted on by water, which changes pressure as it moves through the turbine and gives up its energy. They must be encased to contain the water pressure (or suction), or they must be fully submerged in the water flow. Newton's third law describes the transfer of energy for reaction turbines. Most water turbines in use are reaction turbines. They are used in low and medium head applications.

#### Francis turbine

It is a reaction turbine working under medium head handling medium quantity of water. Francis turbines can either be volute-cased or open-flume machines. The spiral casing is tapered to distribute water uniformly around the entire perimeter of the runner and the guide vanes feed the water into the runner at the correct angle. Thus, water possessing pressure and kinetic energy enters the runner vanes in the radial

direction and leaves in the axial direction. The runner blades are profiled in a complex manner and direct the water so that it exits axially from center of the runner. In doing so the water imparts most of its pressure energy to the runner before leaving the turbine via a draft tube.

The Francis turbine is generally fitted with adjustable guide vanes. These regulate the water flow as it enters the runner and are usually linked to a governing system which matches flow to turbine loading in the same way as a spear valve or deflector plate in a Pelton turbine. When the flow is reduced the efficiency of the turbine falls away.

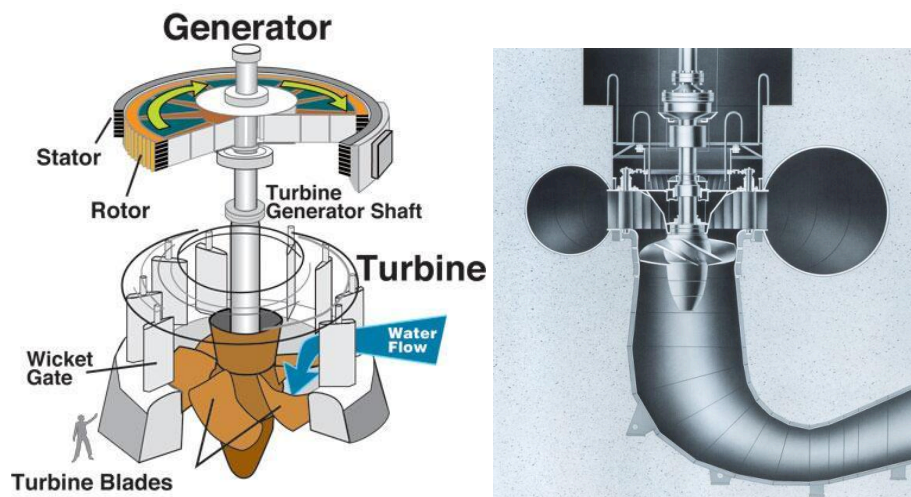


### Kaplan turbine

The Kaplan turbine is an inward flow reaction turbine, which means that the working fluid changes pressure as it moves through the turbine and gives up its energy. The design combines radial and axial features. The inlet is a scroll-shaped tube that wraps around the turbine's wicket gate (guide vanes). Water is directed tangentially, through the guide vanes, and spirals on to a propeller shaped runner, causing it to spin. The outlet is a specially shaped draft tube that helps decelerate the water and recover kinetic energy. The turbine does not need to be at the lowest point of water flow,



as long as the draft tube remains full of water. Variable geometry of the guide vanes and turbine blades allow efficient operation for a range of flow conditions. Kaplan turbine efficiencies are typically over 90%, but may be lower in very low head applications.



### 3.4.2 Comparison of impulse and reaction water turbine

| IMPULSE TURBINE                                                             | REACTION TURBINE                                                                                                       |
|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Available energy is completely converted in to kinetic energy in the nozzle | The available energy is Partially converted in to kinetic energy in the fixed blade and partially in the moving blades |
| The water comes out from the nozzle and directly impinges on the runner.    | In reaction turbine, water first enters the fixed blades and then enters the moving blades.                            |



|                                                                                                               |                                                                                                             |
|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| The pressure of the water flowing from inlet to the outlet of runner remains constant (atmospheric pressure). | The pressure of the water reduces while flowing from inlet to the outlet of runner (less than atmospheric). |
| Turbine can be installed above the tail race.                                                                 | Turbine is submerged in water below tail race.                                                              |
| Casing is provided to prevent splashing of water                                                              | Air tight casing is provided to prevent the pressure leakage.                                               |
| Power developed is mainly due to kinetic energy.                                                              | Power developed is partly by change in kinetic energy and partly due to pressure energy.                    |
| Impulse turbine requires high head                                                                            | Reaction turbine requires low head.                                                                         |

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Department of Mechanical Engineering, DSC



## UNIT-4

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**Internal combustion (I.C) engines:** I.C. Engines parts, 2 Stroke and 4 stroke petrol engines, 4 stroke diesel engines. P-V diagrams of Otto and Diesel cycles, Numericals on indicated power, brake power, indicated thermal efficiency, brake thermal efficiency, mechanical efficiency and specific fuel consumption.

**Refrigeration:** Definitions –Refrigeration, Ton of Refrigeration, Unit of Refrigeration, Refrigerating effect, Ice making capacity, COP, Relative COP, Properties of refrigerants, list of commonly used refrigerants, Principle and working of Vapour Compression Refrigeration and Vapour Absorption Refrigeration

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Department of Mechanical Engineering, DSCE



## 4.1 Internal Combustion Engines

I C engines is called as internal combustion engine combustion (burning) takes inside the closed chamber. E C engine external combustion engine, burning take place outside the engine. I C engine it converts heat (thermal, chemical) energy into mechanical energy.

### 4.1.1. Classification of IC Engine

#### a) Based on thermodynamic cycle

- (1) Otto cycle engine: if the combustion of fuel takes place at constant volume, engine called otto cycle engine
- (2) Diesel cycle engine: if the combustion of fuel takes place at constant pressure, engine called Otto cycle engine
- (3) Dual combustion cycle engine: if the combustion of fuel takes place partially at constant pressure and partially at constant volume.

#### b) Based on the fuel

- (1) Petrol engine: if fuel used is petrol, the engine called petrol engine
- (2) Diesel engine: fuel used is diesel
- (3) Bi-fuel engine: these engines use a mixture of more than one fuel. For example, mixture of diesel and natural gas, mixture diesel and neem oil etc
- (4) Gas engine: gaseous fuels like bio-gas, natural gas, or LPG etc are used as fuels

#### c) Based on strokes

- (1) Two stroke engines: if the engine completes its cycle in two strokes of piston or one revolution of crank shaft
- (2) Four Stroke engine: if the engine completes its cycle in four strokes of piston or two revolution of crank shaft

#### d) Based on the Ignition

- (1) Spark Ignition engine: the fuel gets ignited by electric spark



(2) Compression Ignition engine: the fuel ignites when it comes contact with the hot compressed air.

#### **e) Based on number of Cylinders**

(1) Single cylinder engine: engine consist of only one cylinder

(2) multi cylinder engine: engine consist of more than one ( max 12) cylinder

#### **f) Based on the engine placing**

(1) V-engine: it is combination of two inline engines equally set an angle (2)

vertical engine: cylinder fixed in vertical direction (3) Horizontal engine: cylinder fixed in horizontal direction (4) Opposed engine: cylinders arranged in opposite direction.

(5) Radial engine: the cylinders are arranged along the circumference of the circle

#### **g) Based on the cooling systems**

(1) Air cooled engine: if the heated cylinder walls are cooled by circulating the air

(2) Water cooled engine: if the heated cylinder walls are cooled by circulating the water

#### **h) Based on the application**

(1) Transport (2) Locomotive (3) Marine (4) Power generation (5) Agricultural (6) Earth moving

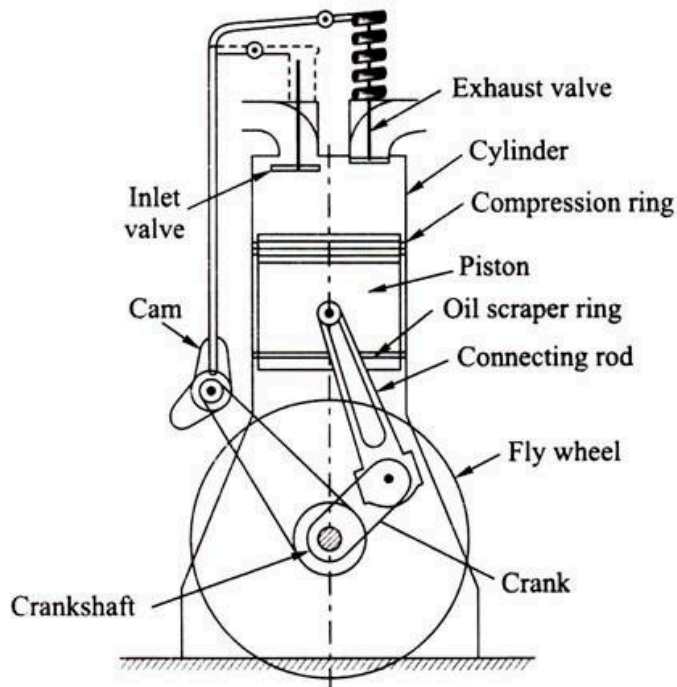
### **4.1.2 Parts of IC engine**

The different parts of internal combustion engine are shown in Fig. above

**Cylinder:** It is the heart of the engine. In this, fuel is burnt and power is developed. The inside diameter of cylinder is called bore.

**Piston:** It is a hollow cylindrical plunger moving to and fro in the cylinder. It transmits the power developed by the combustion of fuel to the crank shaft through the connecting rod.





**Piston Rings:** These are the metallic rings inserted into the circumferential grooves of the piston. Piston rings maintain a gas tight joint between the piston and cylinder while the piston is reciprocating.

**Connecting rod:** It is a link that connects the piston and the crank shaft by means of pins joint. It converts reciprocating motion of the piston into rotary motion of the crankshaft.

**Crank and crank shaft:** Crank is a lever that is connected to the end of connecting rod by a pin joint. Its other end is connected rigidly to a shaft called crankshaft. It causes the connecting rod to oscillate.

**Valves:** These are the devices in 4-stroke engine, which controls the flow of the intake and exhaust gases to and from the engine cylinder. Valves are operated by means of cam driven by the camshaft.

**Ports:** These are the passages in two stroke engines. These ports are covered and uncovered during the piston movements.

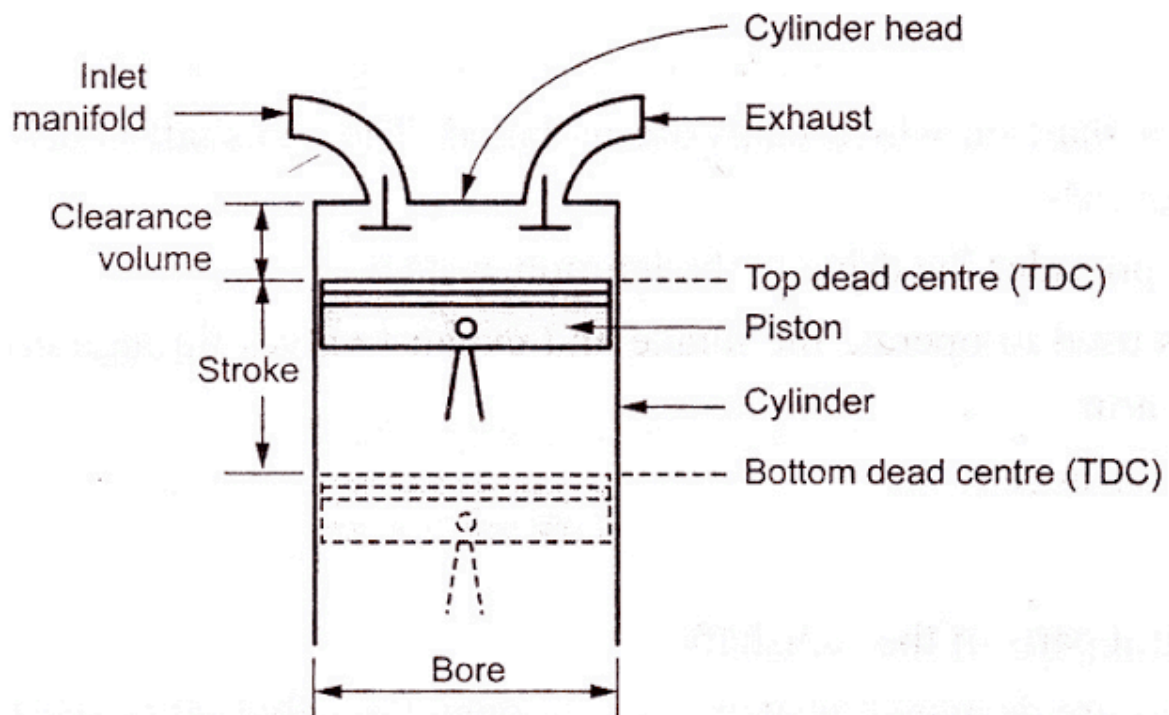
**Flywheel:** It is a heavy wheel mounted on the crank shaft of the engine to maintain uniform rotation of crankshaft.

**Crankcase:** it is lower part of the engine serving as an enclosure for a crankshaft. It also act as oil sump for lubricating oil.

**Sparkplug:** it is component fitted on the cylinder head of petrol engine to initiate the spark for igniting the charge.

**Fuel Injector:** it is component fitted on the cylinder head of diesel engine to spray metered quantity of diesel oil which gets ignited by the high temperature air.

#### 4.1.3 IC Engine Terminology



**Bore:** it is the inside diameter of cylinder.

**Top dead center (TDC):** the extreme position of piston near to the cylinder head is called TDC, TDC is termed as inner dead center when the cylinder is in horizontal position.

**Bottom dead center (BDC):** the extreme position of piston near to the crankshaft is called BDC, BDC is termed as outer dead center, when the cylinder is in horizontal position.

**Stroke or stroke length:** the linear distance travelled by the piston, when it moves from TDC to BDC is called stroke length or stroke. It is denoted by L

**Stroke volume or Swept volume or piston displacement:** the volume swept by the piston when it moves from TDC to BDC. It is denoted by  $V_s$

$$V_s = \frac{\pi}{4} * D^2 L$$

Where D = Bore diameter in mm

L = stroke length in mm

**Clearance volume:** it is the volume between the TDC and cylinder head. It is denoted by ( $V_c$ ).

**Compression ratio:** it is the ratio of the total cylinder volume to the clearance volume. It is given by

$$r_c = \frac{V_c + V_s}{V_c}$$

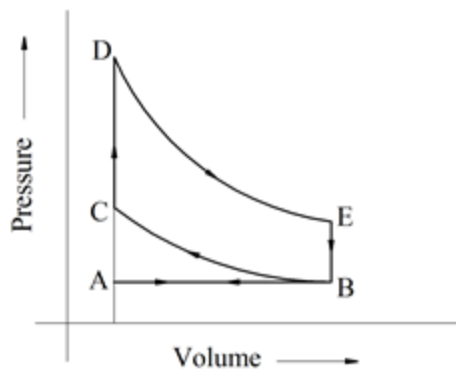
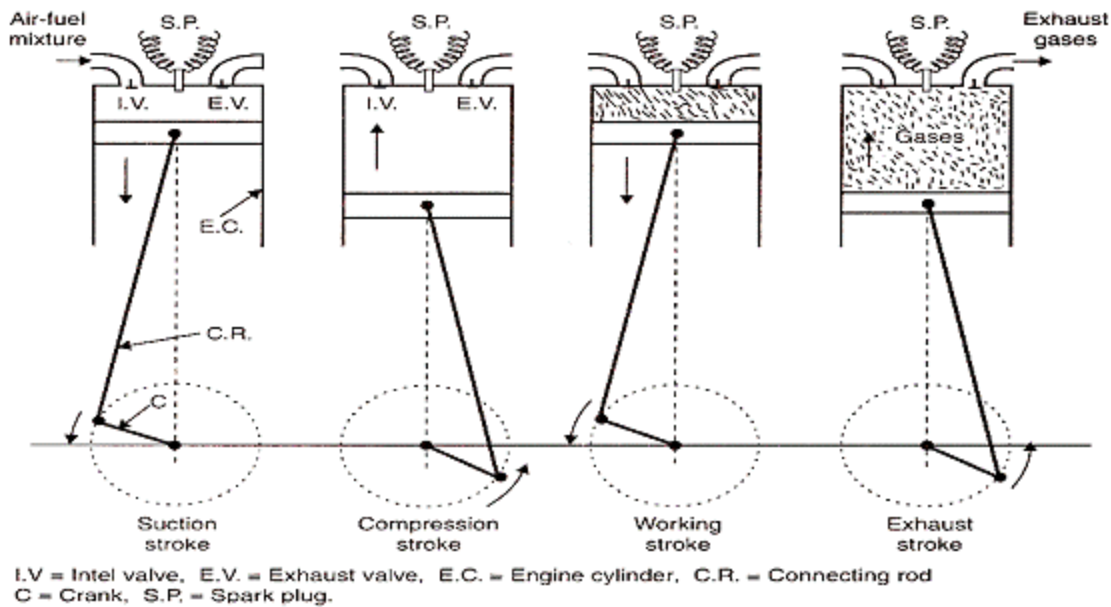
**Piston speed:** it is the distance travelled by the piston per unit time. It is given by  $V_p = 2LN$

N = crank speed in rpm

#### 4.1.4 Four stroke petrol engines (Spark Ignition)

Four stroke petrol engine operates on the theoretical Otto cycle. It is also called constant volume combustion cycle. Since ignition of charge occurs due spark produced by the spark plug, it is also called as spark ignition engine.

It is called four stroke engines as the crank rotates two times and piston moves four times between TDC to BDC to complete one cycle.



### PV diagram

The four-stroke petrol engine works on the principle of Otto (constant volume) cycle. The parts of four-stroke petrol engine are cylinder, piston, head, crankcase, connecting rod, crankshaft, spark plug, inlet and exhaust valve. The four-stroke petrol engine may be air-cooled or water-cooled. The piston performs four strokes to complete one cycle. The four different strokes are i) Suction stroke ii) Compression stroke iii) Power or Expansion stroke iv) Exhaust stroke.



**Suction stroke:** The suction stroke is completed by rotating the crankshaft from  $0^\circ$  to  $180^\circ$ . During suction stroke the inlet valve opens and exhaust valve should be kept in closed condition. When the piston starts moving from TDC to BDC, the volume above the piston increases, results in decrease in pressure (vacuum), This decrease in pressure draws the petrol and air mixture from the carburetor and delivered it to the cylinder, this process is continuous till the pressure inside the cylinder becomes equal to atmosphere. At the end of suction stroke the cylinder is completely filled with petrol and air mixture. At the end of suction stroke the inlet valve closes. The line AB in the PV diagram represents suction stroke (volume of mixture filled in the cylinder).

**Compression stroke:** Rotation of crankshaft from  $180^\circ$  to  $360^\circ$  completes the compression stroke. During compression stroke both inlet and exhaust valves are closed. In this stroke the piston travels from BDC to TDC. When the piston starts moving from BDC to TDC the mixture is compressed, and the pressure increases in the cylinder. The line BC represents the compression stroke.

At or near the before the end of the compression stroke, the spark occurs, this spark ignites the petrol and air mix. The combustion of mixture releases hot gases, which will increase pressure at constant volume. The line CD represents increase in the pressure at constant volume.

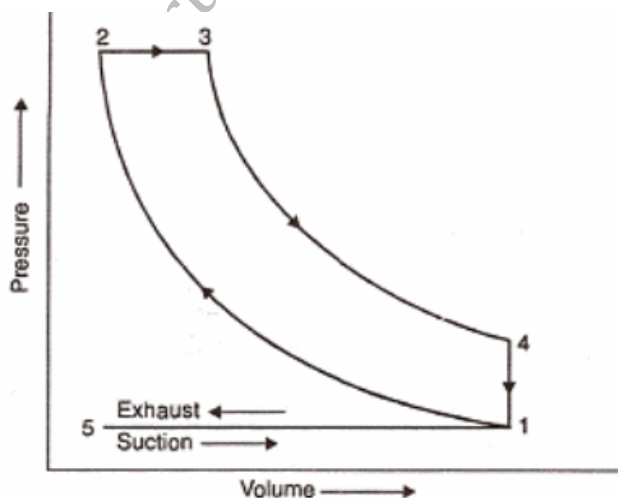
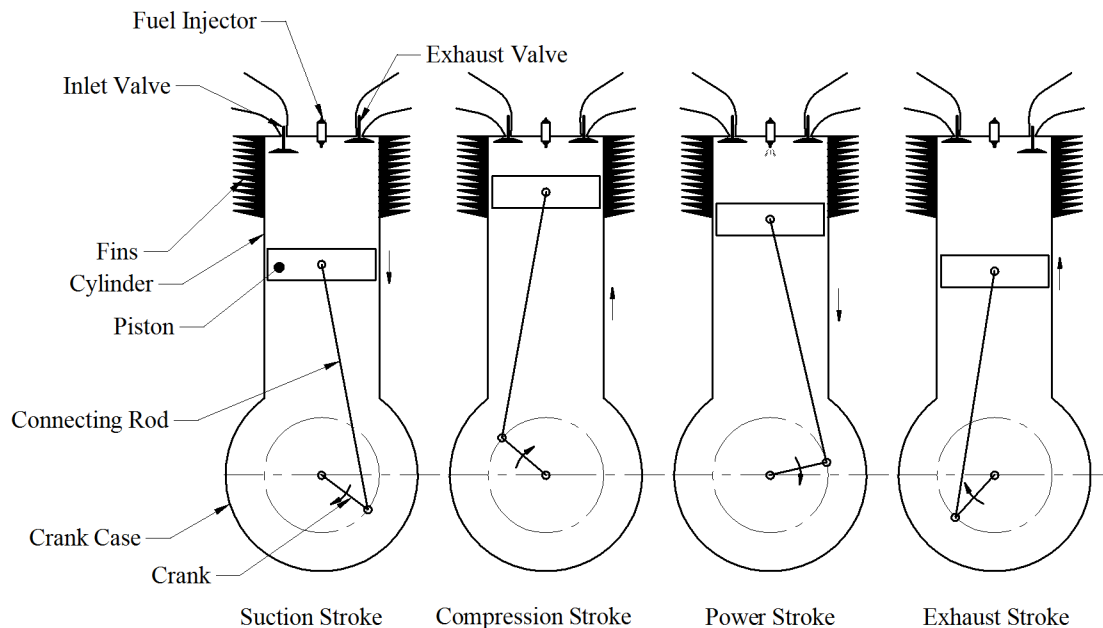
**Power stroke:** Rotation of crankshaft from  $360^\circ$  to  $540^\circ$  completes the power stroke. During power stroke (expansion stroke) both inlet valve and exhaust valve are in closed position. The high-pressure gases produced due to combustion, will exert pressure on the top face of the piston, the piston moves rapidly in the down ward direction performs power stroke.

**Exhaust stroke:** Rotation of crankshaft from  $540^\circ$  to  $720^\circ$  completes the exhaust stroke. At the beginning of exhaust stroke, the exhaust valve opens, and the upward movement of the piston pushes the exhaust gases out the cylinder. At the end of the exhaust stroke

the exhaust valve closes. Thus one cycle completes by rotating the crankshaft from  $0^\circ$  to  $720^\circ$ .

#### 4.1.5 Working of four stroke diesel engine

The four-stroke diesel engine works on the principle of diesel (constant pressure) cycle. The parts of four-stroke diesel engine are cylinder, piston, head, crankcase, connecting rod, crankshaft, fuel injector, inlet and exhaust valve. The four-stroke diesel engine may be air-cooled or water-cooled. The piston performs four strokes to complete one cycle. The four different strokes are i) Suction stroke ii) Compression stroke iii) Power or Expansion stroke iv) Exhaust stroke.



## PV diagram

i) **Suction stroke:** The suction stroke is completed by rotating the crankshaft from  $0^\circ$  to  $180^\circ$ . During suction stroke the inlet valve opens and exhaust valve should be kept in closed position. When the piston starts moving from TDC to BDC, the volume above the piston increases, results in decrease in pressure (vacuum), This decrease in pressure draws the air from atmosphere and fills the air in to the cylinder, this process is continuous till the pressure inside the cylinder becomes equal to atmosphere. At the end of suction stroke the cylinder is completely filled with air. At the end of suction stroke the inlet valve closes. The line AB in the PV diagram represents suction stroke (Volume of air filled in the cylinder).

ii) **Compression stroke:** Rotation of crankshaft from  $180^\circ$  to  $360^\circ$  completes the compression stroke. During compression stroke both inlet and exhaust valves are closed. In this stroke the piston travels from BDC to TDC. When the piston starts moving from BDC to TDC the air is compressed, both pressure and temperature of the air increases. At the end of the compression stroke the temperature of the air reaches the ignition temperature of diesel. The line BC represents the compression stroke.

Before the end of the compression stroke, the fuel injector starts injecting the diesel in to the combustion chamber. The heat of the compressed air burns the injected diesel. The combustion takes place at constant pressure. The line CD represents increase in the pressure at constant pressure.

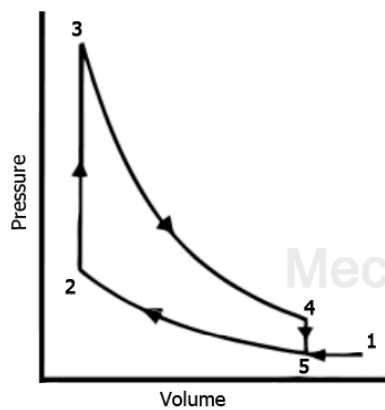
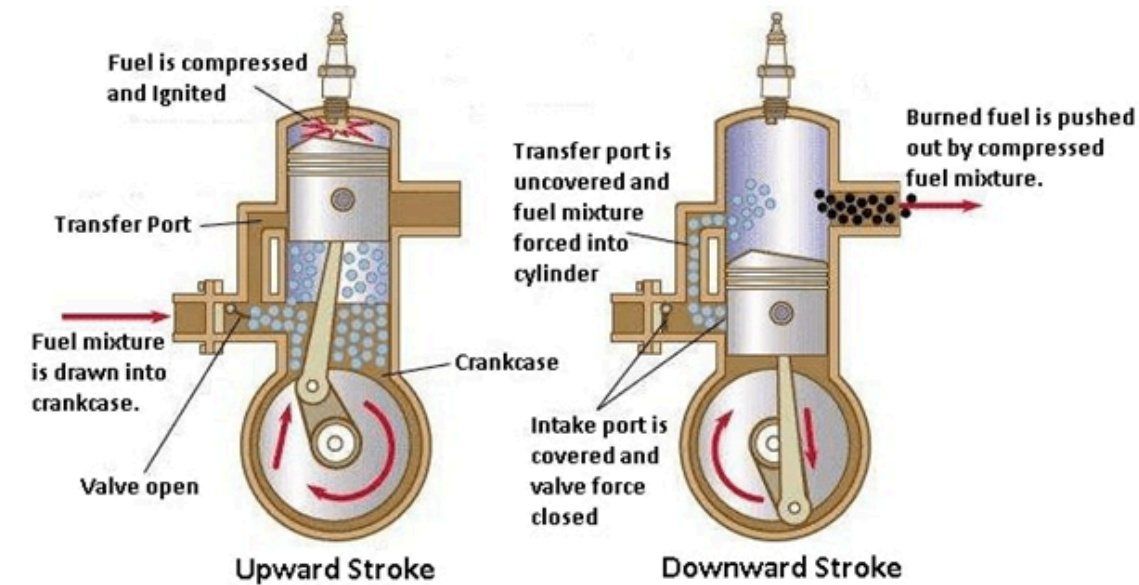
iii) **Power stroke:** Rotation of crankshaft from  $360^\circ$  to  $540^\circ$  completes the power stroke.

During power stroke (expansion stroke) both inlet valve and exhaust valve are in closed position. The high-pressure gases produced during combustion, will exert pressure on the top face of the piston, the piston moves rapidly in the down ward direction performs power stroke. The energy is supplied to the flywheel during power stroke. This energy propels the vehicle

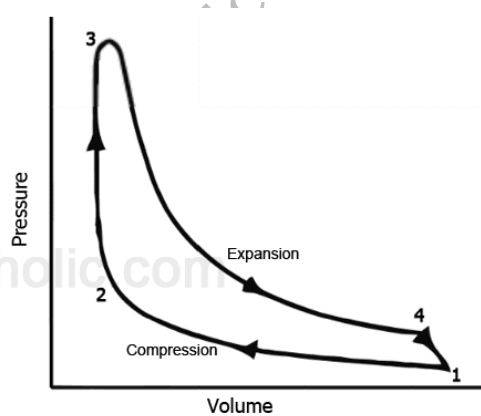
iv) **Exhaust stroke:** Rotation of crankshaft from  $540^\circ$  to  $720^\circ$  completes the exhaust stroke. At the beginning of exhaust stroke, the exhaust valve opens, and the upward movement of the piston pushes the exhaust gases out the cylinder. At the end of the exhaust stroke the exhaust valve closes.

Thus one cycle completes by rotating the crankshaft from  $0^\circ$  to  $720^\circ$ , i.e., two revolutions of the crankshaft.

#### 4.1.6 Two stroke petrol engine



Theoretical P-V Diagram for two-stroke petrol engine.



Actual P-V Diagram for two-stroke petrol engine.

The parts of two-stroke petrol engine are

**Cylinder:** The cylinder liner is made in the form of barrel (hollow cylinder).

The head is connected at the top of the cylinder. A crankcase is connected at the bottom of the cylinder. The cylinder liner is provided with cooling fins for cooling purpose; an exhaust port is located opposite to the transfer port. One end of the transfer port is connected to the cylinder and the other end is connected to the crankcase through which the charge enters.

**Head:** A spark plug is fitted at the top of the head. Fins are provided in the head for cooling purpose.

**Crankcase:** The crankcase is fitted at the bottom of the cylinder. An inlet port is provided in the crankcase to allow the charge from carburetor to the crankcase, another port is known as transfer port, which transfers the charge from transfer port to the cylinder. The crankcase of two-stroke engine should be an airtight chamber, which prevents the leakage of air in or out of the cylinder.

**Piston:** The opening and blocking of the ports are done by the movement of the piston in the cylinder. The piston is fitted with only compression rings (2 or 3 rings)

The two-stroke petrol engine works on the principle of Otto cycle. The parts of two-stroke petrol engine are cylinder, piston, head, crankcase, connecting rod, crankshaft, spark plug, inlet port, transfer port and exhaust port. The piston performs two strokes to complete one cycle. The two strokes are i) First stroke or down ward ii) Second stroke or upward stroke.

**Upward stroke or working stroke:** In two-stroke petrol engine some charge is present either in the cylinder block or in the crankcase. To start a two-stroke engine, power is supplied either by using a kicker or by electric start.

During upward stroke, the piston reciprocates from top dead center to bottom dead center. As the piston moves upward volume below the piston increases results in decrease in pressure in the crankcase. Due to pressure difference charge (petrol & air) is drawn from the carburetor. As the piston moves further upwards covers both exhaust and transfer port, now the charge is subjected to compression. Before the end of the compression stroke the spark (crank angle  $20^\circ$  before TDC) occurs in the combustion chamber. Due to combustion of charge, the pressure increase, which pushes the piston downwards i.e. the working stroke of the piston. As the piston moves rapidly in the down ward direction compress the charge present in the crankcase.

**Down ward stroke or exhaust stroke:** As the piston moves further down wards, first it uncovers the exhaust port. Due to pressure difference the high-pressure gases leaves the

combustion chamber. As the piston moves further down wards, it uncovers transfer port, which allows the compressed charge in the crankcase to the cylinder. The fresh charge is deflected upwards by the deflector provided on the top of the piston and pushes the remaining exhaust gases present in the cylinder. The process of removal of exhaust gases from the cylinder is known as scavenging.

#### 4.1.7 Difference between Petrol and Diesel engines

| S.no | Petrol Engines                                                                                                                                           | Diesel Engines                                                                                                                          |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 1    | A petrol engine draws a mixture of petrol and air during the suction stroke.                                                                             | A diesel engine draws only air during the suction stroke.                                                                               |
| 2    | The carburetor is installed in petrol engines to mix air and petrol in the required proportion and to supply it to the engine during the suction stroke. | The injector or atomizer is installed in diesel engines to inject the fuel at the end of the compression stroke.                        |
| 3    | The pressure at the end of the compression is about 10 bars.                                                                                             | The pressure at the end of the compression is about 35 bars.                                                                            |
| 4    | The charge (i.e. petrol and air mixture) is ignited with the help of spark plug.                                                                         | The fuel is injected in the form of a fine spray. The temperature of the compressed air is about 600° C at a pressure of about 35 bars. |
| 5    | A petrol engine has compression ratio approximately from 6 to 10.                                                                                        | A diesel engine has compression ratio approximately from 15 to 25.                                                                      |
| 6    | The combustion of fuel takes place approximately at constant volume. In other words, it works on <u>Otto cycle</u> .                                     | The combustion of fuel takes place approximately at constant pressure. In other words, it works on <u>Diesel cycle</u> .                |
| 7    | The thermal efficiency is up to about 26%.                                                                                                               | The thermal efficiency is up to about 40%.                                                                                              |
| 8    | Overheating trouble is more in petrol engine due to low thermal efficiency.                                                                              | Overheating trouble is less in diesel engine due to high thermal efficiency.                                                            |
| 9    | The starting of petrol engine is easy due to low compression ratio.                                                                                      | The starting of the diesel engine is slightly difficult due to higher compression ratio compared to a petrol engine.                    |

|    |                                                                                                                                        |                                                                                                                      |
|----|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| 10 | As the compression ratio is low, the petrol engines are cheaper and lighter in weight.                                                 | As the compression ratio is high, the diesel engines are costlier and heavier in weight.                             |
| 11 | The running cost of petrol engine is high because of the higher cost of petrol fuel.                                                   | The running cost of the diesel engine is low because of the lower cost of diesel fuel.                               |
| 12 | The maintenance cost is less.                                                                                                          | The maintenance cost is more.                                                                                        |
| 13 | Petrol engines are high-speed engines.                                                                                                 | Diesel engines are relatively low-speed engines.                                                                     |
| 14 | The petrol engines are installed in light duty vehicles such as scooters, motorcycles, cars. These are also used in <u>airplanes</u> . | The diesel engines are installed in heavy-duty vehicles such as buses, trucks, tractors, earth moving machines, etc. |

#### 4.1.8 Difference between four stroke and stroke engines

| S.No. | <u>Four stroke engines</u>                                                                                                                                                      | <u>Two stroke engines</u>                                                                                                                                |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.    | It has one power stroke for every two revolutions of the <u>crankshaft</u> .                                                                                                    | It has one power stroke for each revolution of the crankshaft.                                                                                           |
| 2.    | Heavy flywheel is required and engine runs unbalanced because turning moment on the crankshaft is not even due to one power stroke for every two revolutions of the crankshaft. | Lighter flywheel is required and engine runs balanced because turning moment is more even due to one power stroke for each revolution of the crankshaft. |
| 3.    | Engine is heavy                                                                                                                                                                 | Engine is light                                                                                                                                          |
| 4.    | Engine design is complicated due to valve mechanism.                                                                                                                            | Engine design is simple due to absence of valve mechanism.                                                                                               |
| 5.    | More cost.                                                                                                                                                                      | Less cost than 4 strokes.                                                                                                                                |
| 6.    | Less mechanical efficiency due to more friction on many parts.                                                                                                                  | More mechanical efficiency due to less friction on a few parts.                                                                                          |
| 7.    | More output due to full fresh charge intake and full burnt gases exhaust.                                                                                                       | Less output due to mixing of fresh charge with the hot burnt gases.                                                                                      |
| 8.    | Engine runs cooler.                                                                                                                                                             | Engine runs hotter.                                                                                                                                      |
| 9.    | Engine is water cooled.                                                                                                                                                         | Engine is air cooled.                                                                                                                                    |

|     |                                                     |                                                                     |
|-----|-----------------------------------------------------|---------------------------------------------------------------------|
| 10. | Less fuel consumption and complete burning of fuel. | More fuel consumption and fresh charge is mixed with exhaust gases. |
| 11. | Engine requires more space.                         | Engine requires less space.                                         |
| 12. | Complicated lubricating system.                     | Simple lubricating system.                                          |
| 13. | Less noise is created by engine.                    | More noise is created by engine.                                    |
| 14. | Engine consists of inlet and exhaust valve.         | Engine consists of inlet and exhaust ports.                         |
| 15. | More thermal efficiency.                            | Less thermal efficiency.                                            |
| 16. | It consumes less lubricating oil.                   | It consumes more lubricating oil.                                   |
| 17. | Less wear and tear of moving parts.                 | More wear and tear of moving parts.                                 |
| 18. | Used in <u>cars</u> , buses, trucks etc.            | Used in mopeds, scooters, <u>motorcycles</u> etc.                   |

#### 4.1.9 Simple calculations in internal combustion engines

i) **Mean effective pressure ( $p_m$ ):** It is defined as the average pressure is acting on the piston during the entire expansion (power stroke) stroke.

$$p_m = \text{Mean effective pressure } \text{N} / \text{m}^2$$

ii) **Indicated power (IP):** The power developed within the piston –cylinder arrangement by the combustion of fuel is known as the indicated power. The pressure acting on the piston varies throughout the working cycle. To record the variation of pressure for one cycle of operation, a device called *piston indicator* is mounted by drilling a small hole on the cylinder cover. It mainly consists of a small plunger and a cylinder. The plunger displacement is proportional to the pressure acting on it from inside against the spring force on the other side. The movement of the plunger transmitted to a stylus through linkages. The stylus traces out a graph on a recording drum, which rotates at a constant speed. The graph thus obtained is called the indicator diagram. The area of the indicator diagram is proportional to the work done in a cycle.

#### Mean effective pressure ( $p_m$ ):

The work done on the piston for one cycle of operation is given by



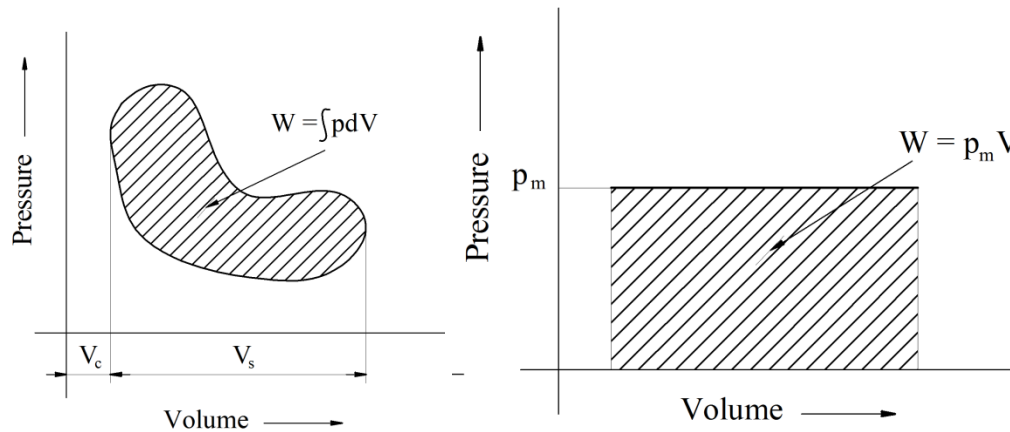


$$W = \int p dV,$$

Where the integration is carried out for one cycle,

$p$  is the pressure,

$V$  is the volume.



*Indicator diagram.*

*Mean effective pressure.*

The right hand side of the equation is nothing but the area within the loop on the pressure-volume diagram.

The mean effective pressure is defined as the equivalent constant pressure which has to be acting on the piston during the expansion stroke, to give the same work output as the varying pressure, in one cycle.

From the indicator diagram, the mean effective pressure can be calculated as,

$$p_m = s \cdot a / l.$$

where,

$s$  = spring constant of the spring used in the piston indicator,

$l$  = length of the indicator diagram,

$a$  = area of the indicator diagram.

*Note that, spring constant is the pressure required to cause unit deflection of the spring.*

i) When  $p_m$  is expressed in  $\text{N/m}^2$

$$\text{IP} = \frac{p_m L A n}{60 \times 1000} \text{ kW}$$

ii) When  $p_m$  is expressed in **bar**

$$\text{IP} = \frac{100 p_m L A n}{60} \text{ kW}$$

Where,

$p_m$  = mean effective pressure,

$L$  = stroke length,

$A$  = area of cross-section of the piston,

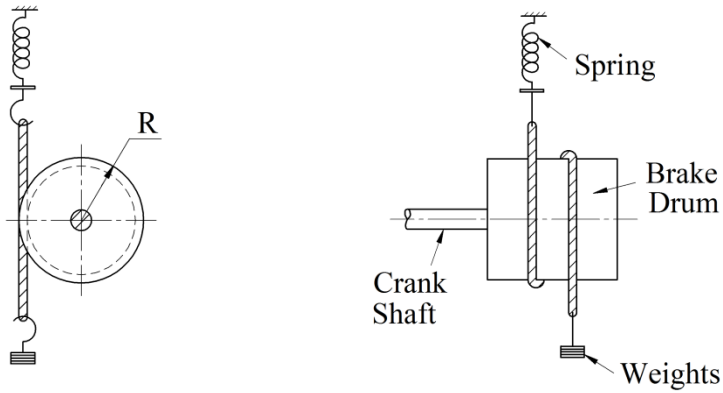
$n$  = number of cycles per minute,

=  $N/2$  for a four stroke engine,

=  $N$  for a two stroke engine.

$N$  = crank shaft speed, rpm

ii) **Brake power (BP)**: The power available at the crank shaft is always less than the power developed within the piston-cylinder arrangement because of frictional losses in the moving parts. The power actually available at the crank shaft is called the brake power. It can be measured using *dynamometers*. One such dynamometer is the *brake-drum* dynamometer.



It consists of a drum, which is mounted on the crankshaft. A rope is wound on the drum. One end of the rope is connected to a spring balance, and the other end, to a weight-loading device.

The torque on the brake drum is given by,

$$T = (W - S) \times R$$

Where,

$W$  = weight on the rope, N.

$S$  = spring balance reading, N.

$R$  = mean radius of brake drum, m.

Brake power is given by,

$$BP = \frac{2 \pi N T}{60 \times 1000} \text{ kW}$$

iii) **Frictional power:** The difference between indicated power and brake power is known as frictional power

$$FP = IP - BP \text{ kW}$$

iv) **Mechanical efficiency:** It is defined as the ratio of brake power to indicated power

$$\eta_{\text{mech}} = \frac{\text{Brake power}}{\text{Indicated power}}$$

v) **Thermal efficiency:** In IC engines, energy is supplied to the engine by burning fuel. But all of the energy that is supplied is not converted into useful mechanical work. Some of the energy supplied is lost through hot exhaust gases, some due to the cooling of the engine and some through radiation and convection heat losses. The fraction of the energy supplied that is available as useful work determines the thermal efficiency of the engine. The thermal efficiency can be calculated either for the indicated power, or for the brake power. Accordingly they are referred to as indicated thermal efficiency, and brake thermal efficiency.

Heat supplied to the engine per sec = mass of fuel burnt x calorific value.

$$\eta_{\text{indicated thermal}} = \frac{\text{Indicated power}}{m_f \times CV}$$

$$\eta_{\text{brake thermal}} = \frac{\text{Brake power}}{m_f \times CV}$$

Where,

$m_f$  = Mass of fuel used in **kg / sec.**

CV = Calorific value of fuel **kJ / kg**

vi) **Specific fuel consumption (SFC):**

It is the mass of fuel supplied per hour in order to get unit power output.

$$\text{SFC} = \frac{m_f}{\text{Power}} \text{ kg/kW-hr}$$



## Power

SFC can be calculated on indicated power basis or on brake power basis.

### Problems

1. Following observations were recorded during a test on a single cylinder 4- stroke oil engine

Bore = 300mm; Stroke = 450mm; Speed = 300rpm; imep = 6bar; Net brake load = 1.5 kN;  
Brake drum diameter = 1.8m; Brake rope diameter = 2 cm.

Calculate: (i) Indicated Power

(ii) Brake Power

(iii) Mechanical Efficiency

### Solution

$$I.P. = n P_{mi} L A N k \times \frac{10}{6} = 1 \times 6 \times 0.45 \times \frac{\pi}{4} (0.3)^2 \times 300 \times \frac{1}{2} \times \frac{10}{6}$$

$$I.P. = 47.71 \text{ kW}$$

$$B.P. = (W - S) \pi (D + d) \frac{N}{60} = 1.5 \times \pi (1.8 + .02) \times \frac{300}{60} = 42.88 \text{ kW}$$

$$\eta_{mean} = \frac{B.P.}{I.P.} = \frac{42.88}{47.71}$$

$$= 0.8987 \text{ or } 89.87 \% \quad (\text{Ans})$$

2. A four cylinder two stroke cycle petrol engine develops 30 KW at 2500rpm. The mean effective pressure on each piston is 8 bars and mechanical efficiency is 80%. Calculate the diameter and stroke of each cylinder with stroke bore ratio 1.5. Also calculate the fuel consumption of the engine, if brake thermal efficiency is 28%. The calorific value of fuel is 43900 kJ/ kg.

Sol:

$$\eta_{mech} = \frac{B.P.}{I.P.}$$

$$I.P. = \frac{30}{0.8} = 37.5 \text{ KW}$$

Also  $1.P = n P_{mi} L A N k \times \frac{10}{6}$

Or

$$37.5 = 4 \times 8 \times 1.5D \times \frac{\pi}{4} D^2 \times 2500 \times 1 \times \frac{10}{6}$$

Or  $D = .062 \text{ m or } 62 \text{ mm (Ans)}$

$$L = 1.5 \times 62 = 93 \text{ mm (Ans)}$$

(ii) Fuel Consumption:

$$\eta_{th(B)} = \frac{B.P}{m_f \times c}$$

or  $0.28 = \frac{30}{m_f \times 43900}$

or  $m_f = 0.00244 \frac{kg}{s}$

or  $8.78 \frac{kg}{hour} \text{ (Ans)}$

3. A 4- cylinder, 4-stroke petrol engine having 90 mm bore and 130mm stroke develops 30 kW of power while running at 1500 r.p.m and using a 20% rich mixture. The theoretical air-fuel ratio is 15:1. Calorific value of petrol is 46000 kJ/ kg. Volumetric efficiency measured at 15°C and 760 mm of mercury as standard temperature and pressure is 70% and mechanical efficiency is 90%.

Find:

- (i) Indicated Thermal efficiency
- (ii) Brake mean effective pressure
- (iii) Brake Thermal efficiency

**Sol:**

Total swept volume of the engine

$$V_s = n \cdot \frac{\pi D^2}{4} \cdot L$$

$$= 4 \times \frac{\pi (.09)^2}{4} \times .13 = 3.3 \times 10^{-3} m^3$$

Volume of air drawn in the engine in the suction stroke

$$= V_s \times \eta_{vol}$$

$$= 3.3 \times 10^{-3} \times 0.7$$

$$= 2.31 \times 10^{-3} m^3$$

Volume flow rate of the 4-stroke engine

$$= \text{Volume sucked during suction stroke} \times \frac{N}{2}$$

$$= 2.31 \times 10^{-3} \times \frac{1500}{2}$$

$$= 1.737 m^3/min$$

This volume flow rate is at pressure 760 mm of mercury (1.013 bar) and temperature 15° C (298K).

So mass flow rate of air

$$m_a = \frac{pV}{R_o T} = \frac{(1.013 \times 10^5) \times 1.737}{287 \times 298}$$

$$= 2.057 \text{ kg/min}$$

$$\text{Mass flow rate of fuel used} = \frac{m'_a}{15}$$

$$= \frac{2.057}{15} = 0.137 \text{ kg/min}$$

Using 20% rich mixture actual mass flow rate of fuel used

$$m'_f = 0.137 \times \frac{120}{100}$$

$$\begin{aligned}
 &= 0.164 \text{ kg/min} \\
 &= \frac{0.164}{60} = 2.74 \times 10^{-3} \text{ kg/sec}
 \end{aligned}$$

Given that, Brake power, B.P = 30 kW

$$\text{Mechanical efficiency } \eta_{\text{mech}} = 90\% = 0.9$$

## 4.2 Refrigeration

### 4.2.1 Introduction:

Refrigeration is the art and science of maintaining a space at a temperature lower than the surrounding temperature. The device, which is used for this purpose, is the refrigerator. Refrigeration is useful in the preservation of foodstuff, medicines, high precision industries, air conditioning, etc.

### 4.2.2 Commonly used refrigerants

Some of the most commonly used refrigerants are listed below-

#### A. Halocarbon compounds:

1. Refrigerant 11 – Trichloromonofluoromethane
2. Refrigerant 12 – Dichlorodifluoromethane
3. Refrigerant 13 – Monochlorotrifluoromethane
4. Refrigerant 14 – Carbontetrafluoride
5. Refrigerant 134a – Tetrafluoroethane

#### B. Cyclic organic compounds:

1. Refrigerant 1112a – Dichlorodifluoroethylene
2. Refrigerant 1113 – Monochlorotrifluoroethylene



### 3. Refrigerant 1114 – Tetraflouroethylene

Apart from these, ammonia is also used extensively because of its excellent thermal properties.

The halocarbon family of refrigerants are obtained by replacing one or more of the hydrogen atoms in methane ( $\text{CH}_4$ ) or ethane ( $\text{C}_2\text{H}_6$ ). But most of the halogenated hydrocarbons contribute to the greenhouse effect and depletion of ozone layer in the earth's stratosphere. Completely halogenated hydrocarbons, commonly referred to as chlorofluorocarbons (CFCs) are the ones which cause the highest damage because of their prolonged atmospheric life.

Another group of halogenated hydrocarbons called hydrochloroflourocarbons (HCFCs) retain one or more of the hydrogen atoms. They have shorter atmospheric life than CFCs, and hence are more environmental friendly.

Still another group of halogenated hydrocarbons called hydroflourocarbons (HFCs) contain no chlorine atom at all, and therefore, do not deplete the ozone layer, and most of them have minimum greenhouse effect. Most significant among them is R-134a.

#### 4.2.3 Terms used in Refrigeration

##### **Coefficient of performance of refrigerator (COP)**

Coefficient of performance of refrigerator is the ratio of heat absorbed in the refrigerated chamber to the work input to the compressor.

$$\text{COP} = \text{Heat absorbed} / \text{work input.}$$

COP as the name indicates is a measure of the performance of the refrigeration cycle. It is similar to the efficiency term associated with any work developing device. Efficiency or COP can be loosely defined as the ratio of the output to input. In a refrigerator, the output is the desired effect which is the heat absorbed in the refrigerated chamber, and input is the work input to the compressor.

### Refrigerating effect

It is the amount of heat absorbed in the refrigerated chamber per unit mass of the refrigerant. More refrigerating effect means less mass of refrigerant required.

### Refrigeration capacity:

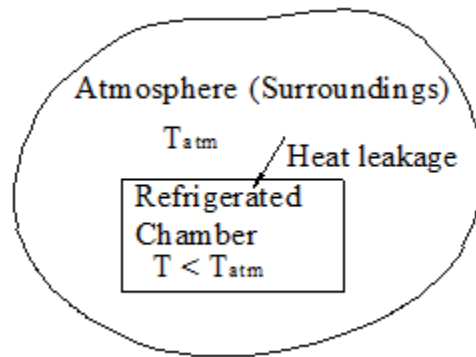
Refrigeration capacity is the rate at which heat is absorbed from the refrigerated chamber. Refrigeration capacity is expressed in tons of refrigeration.

One ton of refrigeration is the rate at which heat is absorbed to convert one ton (1000 kg) of water at 0°C into ice at 0°C, in one day.

$$\begin{aligned} 1 \text{ ton of refrigeration} &= \text{latent heat of ice} \times 1000 \text{ kJ/day.} \\ &= \text{latent heat of ice} \times 1000/24 \text{ kJ/hr.} \\ &= \text{latent heat of ice} \times 1000/(24 \times 60) = 210 \text{ kJ/min.} \\ &= \text{latent heat of ice} \times 1000/(24 \times 60 \times 60) = 3.5 \text{ kJ/s} \end{aligned}$$

### Principle of refrigeration:

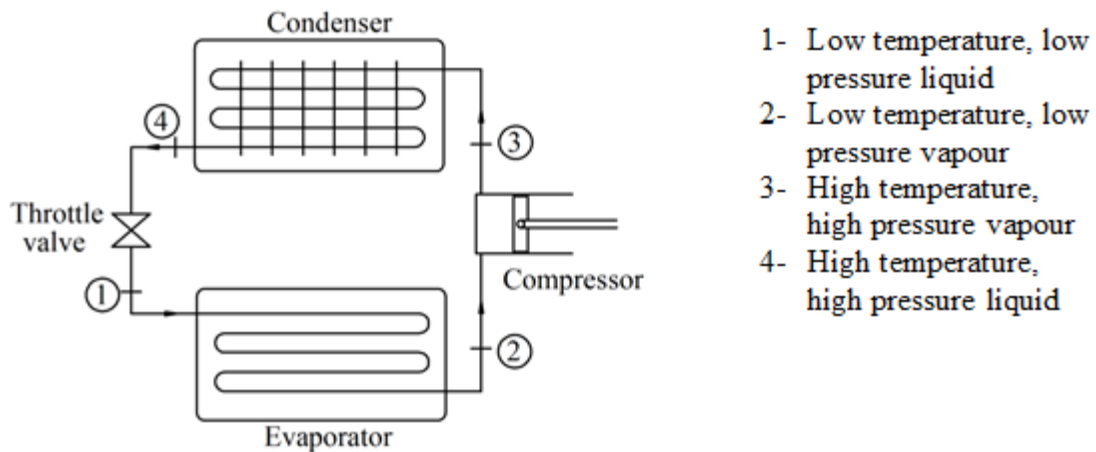
Refrigeration chamber has to be maintained at a temperature lower than the surrounding temperature. Due to temperature difference there is a tendency for the heat to flow from the surrounding to the refrigerated chamber, which result in an increase in the temperature of the chamber. In order to maintain low temperature in the chamber, heat has to be removed from the chamber to the surroundings at a rate equal to the rate at which heat is leaking into it. But heat cannot flow from a hot body to a cold body spontaneously.



We need to have a third body which acts as a medium for heat transfer between the chamber and the surrounding. When the medium comes in contact with the chamber its temperature should be less than the temperature of the chamber so that it can absorb heat from chamber. Similarly when the medium comes in contact with the surrounding, its temperature should be greater than the atmospheric temperature, so that it can reject heat to the atmosphere. The medium is usually a fluid which is referred to as *refrigerant*. Heat transfer usually changes the phase of the refrigerator, i.e., when it absorbs heat it changes its phase from liquid to vapour and when it rejects heat, it changes its phase from vapour to liquid. In other words the refrigerant absorbs or rejects heat in the *latent heat* region.

The temperature of the refrigerant, which comes out of the refrigerated chamber, is well below the atmospheric temperature. In order to reject heat to the atmosphere, the temperature of the refrigerant should be raised above that of the surrounding. We can achieve this either by compressing the refrigerant to a high pressure (vapour compression cycle), or by dissolving the vapour in a liquid, increasing the pressure of the liquid and then heating the liquid to give off vapour at high pressure and temperature (vapour absorption refrigeration cycle).

#### 4.2.4 Vapour Compression Refrigerator



Vapour compression refrigerator consists mainly of four components – (i) Evaporator or refrigerated chamber, (ii) compressor, (iii) condenser or cooler, and (iv) Throttle valve or pressure reducing valve.

A low pressure, low temperature liquid enters the refrigerated chamber. It absorbs heat in the chamber and changes its phase from liquid to vapour. At the exit of the refrigerated chamber we have a low pressure, low temperature vapour. This vapour is taken by the compressor, which compresses it to a high pressure. Due to high compression, the temperature of the vapour rises above that of the atmosphere.

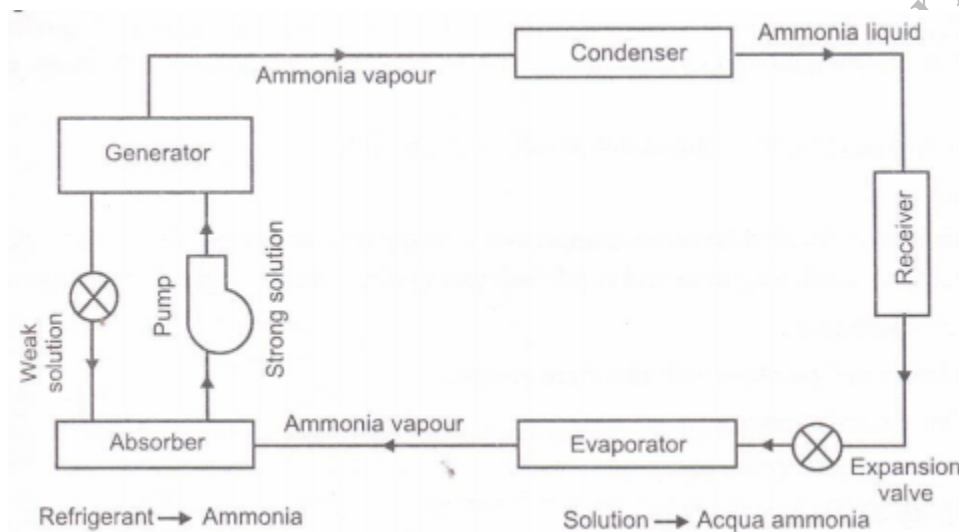
The high temperature, high pressure vapour passes through the condenser. It loses heat to the atmosphere and changes its phase from vapour to liquid. At the exit of the condenser we have high pressure liquid at moderate temperature. The liquid passes through a throttle valve where it suddenly expands to a low pressure. There is a sudden decrease in temperature due to decrease in pressure, and at the inlet to the refrigerated chamber we have a low pressure, low temperature liquid. Thus the cycle is completed.

#### 4.2.5 Vapour Absorption refrigerator

In vapour absorption refrigeration system, we choose a combination of a refrigerant and solvent in such a way that the solvent absorbs large quantities of vapour

refrigerant at low temperatures and gives off vapour when heated to higher temperature. Ammonia and water are such a combination.

Low pressure, low temperature ammonia liquid enters the evaporator chamber. It absorbs latent heat in the evaporator and becomes vapour. The ammonia vapour, which is at low temperature, is absorbed in water in the absorber. A pump pressurizes the ammonia solution and supplies it to the generator. The solution is heated in the generator and water gives off ammonia vapour at high temperature and pressure. This vapour is condensed in the condenser by rejection of heat to the atmosphere. The high-pressure high temperature liquid is suddenly expanded in the throttle valve to a low pressure. Sudden drop in pressure results in drop in temperature of the liquid. Then the low pressure, low temperature liquid ammonia enters the evaporator chamber, and the cycle is repeated.



In the heat exchanger, the pressurized strong ammonia solution passing from the pump to the generator recovers some amount of heat from the weak ammonia solution, which is passing from the generator to the absorber. A pressure-reducing valve is provided between the heat exchanger and the absorber to reduce the pressure of the weak ammonia solution to that of the evaporator pressure.

#### 4.2.6 Comparison between Vapour Compression and Absorption refrigerators

| Sl. No | Parameter.                    | Vapour compression refrigerator.                                                                                                 | Vapour absorption refrigerator.                                                                                                                                            |
|--------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | Energy input.                 | It requires large quantity of mechanical work to run the compressor, because it involves compression of large volumes of vapour. | Pump requires very less mechanical work, because it is pumping liquid. In the generator heating can be achieved either by employing coils or by any other form of heating. |
| 2      | Noise.                        | Noise is more due to the presence of a large compressor.                                                                         | Less noisy due to the presence of pump.                                                                                                                                    |
| 3      | Plant capacity.               | Not suitable for large capacities.                                                                                               | Well suited for large capacities.                                                                                                                                          |
| 4      | Maintenance.                  | Maintenance is more due more number of mechanical parts.                                                                         | Maintenance is less due to less number of moving parts.                                                                                                                    |
| 5      | Contamination of refrigerant. | No contamination of refrigerant because only one substance is used.                                                              | Water vapour some times enters the condenser and other parts along with the refrigerant, which would reduce the refrigerating effect.                                      |

#### 4.2.7 Properties of a good refrigerant

The desirable properties of an ideal refrigerant are as follows-

1. High latent heat of vaporization at evaporator pressure-which would result in more heat absorption for a given mass flow rate of refrigerant.
2. High latent heat of vaporization at the condenser pressure.
3. Low specific volume of vapour-Power input to the compressor is directly related to the volume of vapour compressed. Low specific volume means less work input.
4. Low specific heat in liquid and high specific heat for vapour are desirable in that both increase the refrigerating effect.
5. A high coefficient of conductance-which result in improved heat transfer rates and reduce the size of the condenser.
6. Low freezing point at evaporator pressure-the refrigerant should not free at low temperatures after throttling.
7. It should not corrode the pipes and other parts of the machinery.

8. It should be non-toxic.
  9. It should not react with the lubricant used in the machinery.
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Department of Mechanical Engineering, DSCE

## Unit 5

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### Power Transmission

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#### Power Transmission

**Belt drives**-Terminology of a belt drive, open and cross belt drives, **Gear Drives**: Types of Gears and applications, Advantages and disadvantages of gear drive, Gear Tooth



Nomenclature, Classification of Gear train: Simple, Compound, Reverted & Epicyclic, Velocity ratio of simple and compound gear train, Numerical on Gear drives (only simple and compound gear train).

**Hybrid & Electric vehicles Technology (HEVT)-** Comparison with conventional vehicle technologies, components of EV's, challenges and Key aspects of EV. EV's, Basics of EV, Basics of HEV, Basics of Plug-In Hybrid Electric vehicle (PHEV), Basics of Fuel Cell Vehicle (FCV). Hybrid Electric Vehicles: Classification, Micro, Mild, Full, Plug in, EV.

Department of Mechanical Engineering, DSCE





## 5.1 Power Transmission

### 5.1.1 Introduction:

Power developed in the prime movers has to be transmitted to machinery. The intermediate mechanisms used for transmitting motion from one machine to the other are called *drives* or *transmission systems*.

The drives, which are commonly employed, are belt drives, rope drives, chain drives, gear drives, etc.

### 5.1.2 Belt drives

Belt drives are employed for power transmission when the distance between the shafts is large, and when exact speed of the driven shaft is not a main criterion. Belt drives consist of two discs called *pulleys*, which are mounted on the two shafts. They are connected by an endless belt. Motion is transmitted due to frictional force between the pulleys and the belt surface.

### 5.1.3 Types of belt drives

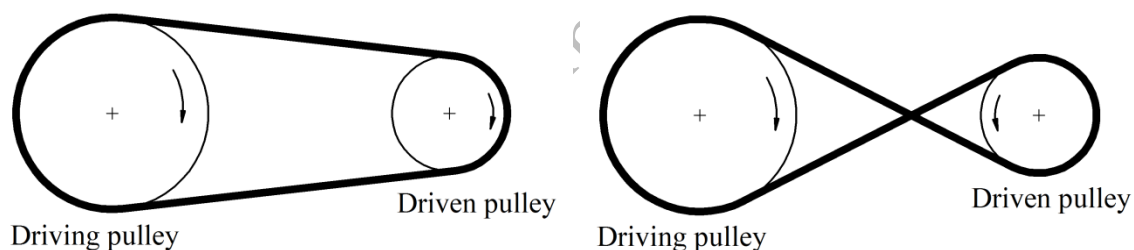


Fig (a) Open belt drive

(b) Cross belt drive

There are two types of belt drives – open belt, and cross belt drives. In open belt drive, the driving and the driven pulley are rotating in the same direction, whereas, in cross belt drives they rotate in opposite directions.

### 5.1.4 Terminologies used in belt drives

In order to transmit power in a belt drive, the belt should be pulling on the driven pulley on one side. This side is known as the *tight side*, and the other side where the tension is comparatively less, is called the *slack side*. There is a gradual increase in tension in the belt from the slack side to the tight side.

The *angle of lap* of belt is the angle subtended at the centre of pulley by the belt, which is overlapping on the pulley surface. In open belt drives, the angle of lap on the smaller pulley is less than the angle of lap on the larger pulley. Power transmitted is mainly due to frictional force, and the angle of lap determines the extent of friction force that can exist between the surfaces of the pulley and belt. Of the two pulleys, the smaller pulley is less capable of transmitting/receiving power owing to less angle of lap on its surface. Hence, it is the angle of lap on the smaller pulley, which is the determining factor in power transmission. In cross belt drives the angle of lap is same on both the pulleys. *Velocity ratio* is the ratio of the speeds of the driven pulley and driving pulley.

*Slip* is the term used to describe the sliding motion between the belt and pulley surfaces. Slip can take place either on the driving pulley or driven pulley or both. The effect of slip is to reduce the velocity ratio, and consequently, power loss in transmission.

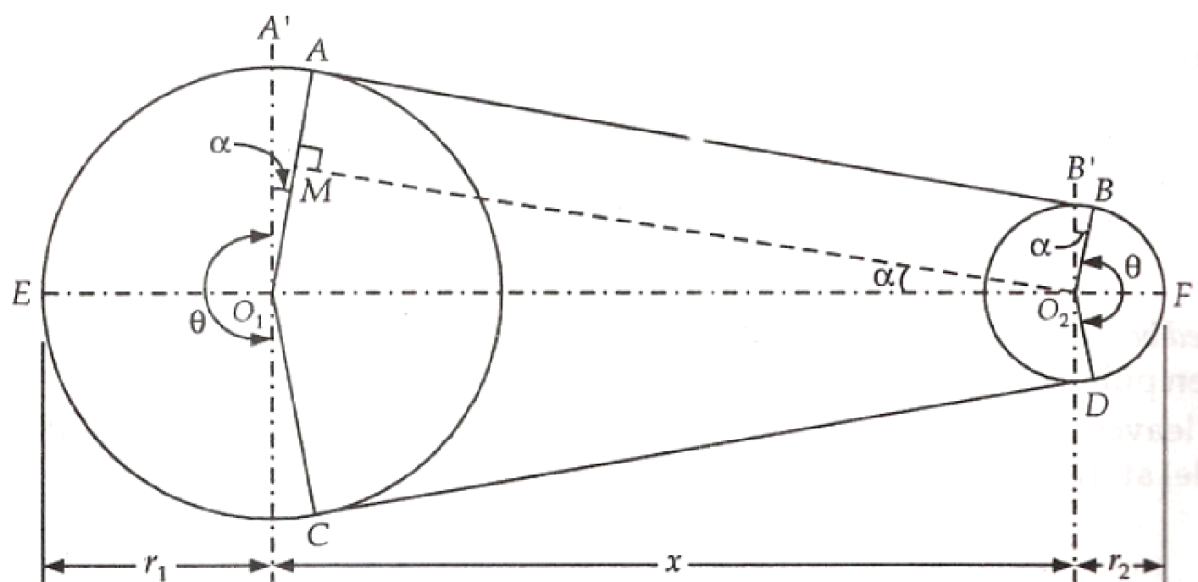
**5.1.5 Open belt system:** the open belt system in which both the driving and driven pulleys rotate in the same direction.

Let  $r_1, r_2$  = radius of the two pulleys

$x$  = distance between  $O_1$  and  $O_2$ ; the centers of the two pulleys.

The belt leaves the bigger pulley at  $A$  and  $C$ , and the smaller pulley at  $B$  and  $D$ .

A line  $O_2M$  drawn parallel to  $AB$  will be perpendicular to  $O_1A$  also.



$$\angle A'O_1A = \angle B'O_2B = \angle O_1O_2M = \alpha$$

$$\sin \alpha = \frac{r_1 - r_2}{x}$$

Since  $\alpha$  is very small,

$$\sin \alpha = \alpha = \frac{r_1 - r_2}{x}$$

Length of the belt,  $l = 2(\text{arc } EA + AB + \text{arc } BF)$

$$\text{arc } EA = r_1 \times \left( \frac{\pi}{2} + \alpha \right) \quad \text{and} \quad \text{arc } BF = r_2 \times \left( \frac{\pi}{2} - \alpha \right)$$

$$AB = MO_2 = \sqrt{(O_1O_2)^2 - (O_1M)^2}$$

$$= \sqrt{x^2 - (r_1 - r_2)^2} = x \sqrt{1 - \left( \frac{r_1 - r_2}{x} \right)^2} = x \left[ 1 - \left( \frac{r_1 - r_2}{x} \right)^2 \right]^{1/2}$$

Since  $\left( \frac{r_1 - r_2}{x} \right)^2$  is very small, Binomial expansion would give

$$AB = x \left[ 1 - \frac{1}{2} \left( \frac{r_1 - r_2}{x} \right)^2 \right] = x \left[ 1 - \frac{(r_1 - r_2)^2}{2x^2} \right]$$

$$\therefore l = 2 \left[ r_1 \left( \frac{\pi}{2} + \alpha \right) + x \left\{ 1 - \frac{(r_1 - r_2)^2}{2x^2} \right\} + r_2 \left( \frac{\pi}{2} - \alpha \right) \right]$$

$$= 2 \left[ \frac{\pi}{2} (r_1 + r_2) + \alpha (r_1 - r_2) + \left\{ x - \frac{(r_1 - r_2)^2}{2x} \right\} \right]$$

$$= \pi(r_1 + r_2) + 2\alpha(r_1 - r_2) + 2x - \frac{(r_1 - r_2)^2}{x}$$

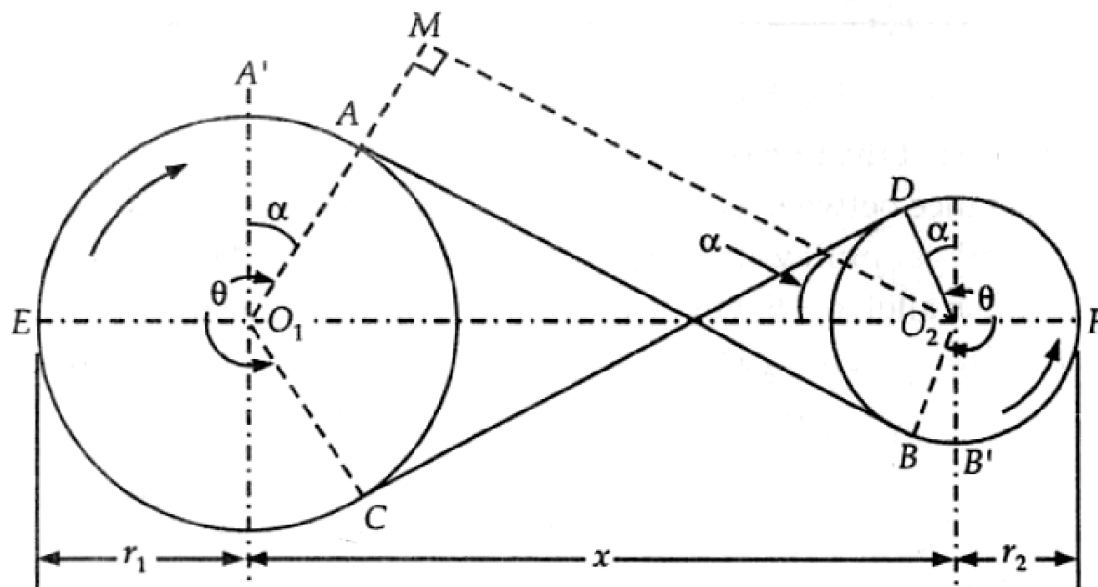
Substituting the value  $\alpha = \frac{r_1 - r_2}{x}$ , we get

$$l = \pi(r_1 + r_2) + 2 \frac{r_1 - r_2}{x} \times (r_1 - r_2) + 2x - \frac{(r_1 - r_2)^2}{x}$$

$$= \pi(r_1 + r_2) + \frac{(r_1 - r_2)^2}{x} + 2x$$

**5.1.6 Crossed belt system:** the crossed belt system in which the driving and the driven pulleys rotate in opposite directions.

The belt leaves the bigger pulley at A and C and the smaller pulley at B and D. A line  $O_2M$  is drawn parallel to AB will be perpendicular to  $O_1A$  also.



let  $r_1$  and  $r_2$  = radius of the two pulleys

$x$  = distance between  $O_1$  and  $O_2$ ; the centres of the two pulleys

angle  $A'O_1A$  = angle  $B'O_2B$  = angle  $O_1O_2M$  =  $\alpha$

$$\sin \alpha = \frac{O_1M}{O_1O_2} = \frac{O_1A + AM}{O_1O_2} = \frac{r_1 + r_2}{x}$$

Since  $\alpha$  is very small,

$$\sin \alpha = \alpha = \frac{r_1 + r_2}{x}$$

length of belt,

$$l = 2(\text{arc } EA + AB + \text{arc } BF)$$

$$\text{arc } EA = r_1 \left( \frac{\pi}{2} + \alpha \right) \quad \text{and} \quad \text{arc } BF = r_2 \left( \frac{\pi}{2} + \alpha \right)$$

$$AB = MO_2 = \sqrt{(O_1O_2)^2 - (O_1M)^2} = \sqrt{x^2 - (r_1 + r_2)^2}$$

$$= x \sqrt{1 - \left(\frac{r_1 + r_2}{x}\right)^2} = x \left[1 - \left(\frac{r_1 + r_2}{x}\right)^2\right]^{1/2}$$

$$AB = x \left[1 - \frac{1}{2} \left(\frac{r_1 + r_2}{x}\right)^2\right] = x \left[1 - \frac{(r_1 + r_2)^2}{2x^2}\right]$$

$$l = 2 \left[ r_1 \left(\frac{\pi}{2} + \alpha\right) + x \left\{1 - \frac{(r_1 + r_2)^2}{2x^2}\right\} + r_2 \left(\frac{\pi}{2} + \alpha\right) \right]$$

$$= 2 \left[ \frac{\pi}{2} (r_1 + r_2) + \alpha (r_1 + r_2) + \left\{x - \frac{(r_1 + r_2)^2}{2x}\right\} \right]$$

$$= \pi (r_1 + r_2) + 2\alpha (r_1 + r_2) + 2x - \frac{(r_1 + r_2)^2}{x}$$

### 5.1.7 Velocity ratio

When there is no slip, the peripheral velocity of the two pulleys should be same, which are in turn equal to the linear velocity of the belt.

On the driving pulley,  $V = \omega_1 r_1$ .

On the driven pulley,  $V = \omega_2 r_2$ .

Therefore,  $V = \omega_1 r_1 = \omega_2 r_2$ .

The velocity ratio,

$$\frac{\omega_2}{\omega_1} = \frac{r_1}{r_2} = \frac{d_1}{d_2}$$

*Velocity ratio is inversely proportional to the ratio of diameters.*

Considering thickness of belt 't',

Effective diameter of the driving pulley =  $d_1 + t$ .

Effective diameter of the driven pulley =  $d_2 + t$ .

$$\frac{\omega_2 d_1 + t}{\omega_1 d_2 + t} = \frac{d_1}{d_2}$$

Velocity ratio considering slip:

Slip is expressed as a percentage of ideal speed.

$$\text{Slip, } s = \frac{(\text{Ideal speed} - \text{actual speed})}{\text{Ideal speed}} \times 100$$

Let  $s_1$  = percentage slip on the driving pulley,  
 $s_2$  = percentage slip on the driven pulley.

Circumferential velocity of the driving pulley =  $\omega_1 r_1$ .

Velocity of belt over the driving pulley =  $\omega_1 r_1 (1 - s_1)/100$ .

$$\begin{aligned} \text{Velocity of the driven pulley} &= \{\omega_1 r_1 (1 - s_1)/100\} \times (1 - s_2)/100 \\ &= \omega_1 r_1 (100 - s_1) \times (100 - s_2) / (100 \times 100) \\ &= \omega_1 r_1 (100 \times 100 - 100s_1 - 100s_2 + s_1s_2) / (100 \times 100) \end{aligned}$$

Since  $s_1$  and  $s_2$  are very small, their product can be neglected.

Velocity of the driven pulley =  $\omega_1 r_1 \{100 - (s_1 + s_2)\} / 100$ .

$$\omega_1 r_2 = \omega_1 r_1 (100 - s) / 100.$$

Where,  $s$  is the total percentage slip.

Velocity ratio,

$$\frac{\omega_2 d_1 (100 - s)}{\omega_1 d_2} = \frac{d_1}{d_2} \times \frac{100 - s}{100}$$

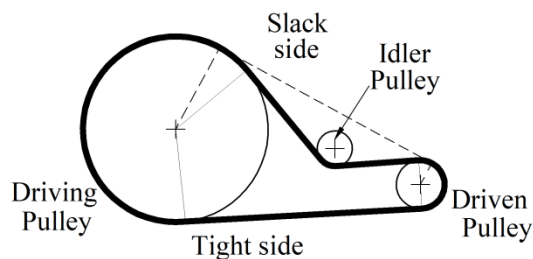
### 5.1.8 Creep

Any material when subjected to tension elongates, and so is the belt material. This elongation is directly proportional to the tension in the belt. Tensions on the two



sides of the belt passing over the pulley are different, on the tight side the tension is more, and on the slack side, the tension is less. On the driven pulley, the tension on the belt entering the pulley is less than the tension on the belt which is leaving the pulley. The belt entering the pulley is stretched less than the belt that is leaving the pulley. Hence, there is a gradual increase in the length of belt over the surface of the pulley. This stretching results in relative motion between the belt and pulley surfaces. This phenomenon is called creep. A similar effect is there on the driving pulley as well. Creep results in loss of power and decrease in velocity ratio.

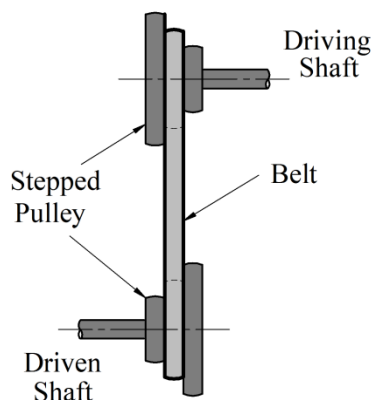
#### 5.1.9 Idler pulley



Power transmitted in a belt drive is directly proportional to the angle of lap on the smaller pulley. In an open belt drive, if one of the pulleys is very small, and the centre distance between the two pulleys is small, then, the angle of lap on the smaller pulley is very less.

This restricts the power transmission between the two shafts. In order to increase the angle of lap, and hence the power that can be transmitted, another pulley is introduced on the slack side of the belt as shown in figure. This pulley does not consume any power, and only helps in increasing the angle of lap. This pulley is called the idler pulley, or the Jackey pulley.

#### 5.1.10 Stepped cone pulley

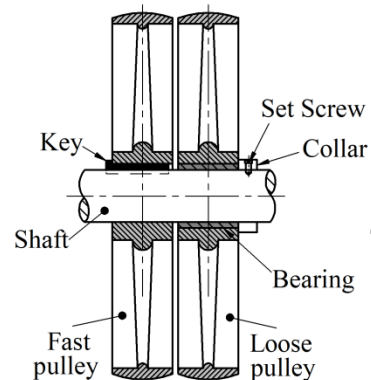


Machines and machine tools under working conditions need to change their speeds frequently. But motors that drive them usually run at a constant speed. In order to drive the machine at different speeds for the same motor speed, a combination of different sizes of pulleys are used. These pulleys used for this purpose are called stepped cone pulleys, or speed cones. Cone pulleys are mounted on both the driving and driven shafts connected by an endless belt. The belt is shifted between different pairs of pulleys to get

varying speeds for the driven pulley. The diameters of the pairs of pulleys are carefully chosen so as to operate all the pairs with one single belt of fixed length.

#### 5.1.11 Fast and loose pulleys

Fast and loose pulley arrangement are helpful in starting and stopping a machine without stopping the driving motor. Two pulleys are mounted on the same shaft. Loose pulley is free to rotate on the driven shaft and fast pulley is keyed to the shaft. When the belt is on the fast pulley, the power is transmitted to the driven shaft, but when the belt is shifted on to the loose pulley, the loose pulley freely rotates on the shaft without transmitting power.



#### 5.1.12 Power transmitted in a belt drive

Power = Net force acting x Linear velocity

$P = (T_1 - T_2) V$  Watts Where  $T_1$  &  $T_2$  are tensions on the tight and slack sides in Newton and  $V$  is speed of the belt in m/ sec.

#### 5.1.13 Initial tension in belt drives

When the belt drive is not running, the belt is kept under tension known as initial tension. When the drive is set into motion, the tension on the tight side of the belt increases due to friction, and that on the slack side decreases. Assuming that the length of belt remains the same before and after the drive was set into operation, the increase in tension on the tight side of the belt over the initial tension should be same as the decrease in tension on the slack side of the belt.

Let  $T_0$  = initial tension in the belt when the drive is not running

$\Delta T$  = increase in tension on the tight side = decrease in tension on the slack side.

$T_1$  = tension on the tight side =  $T_0 + \Delta T$

$T_2$  = tension on the tight side =  $T_0 - \Delta T$

Therefore,  $T_1 + T_2 = (T_0 + \Delta T) + (T_0 - \Delta T)$

$T_1 + T_2 = 2T_0$  Or, initial tension,



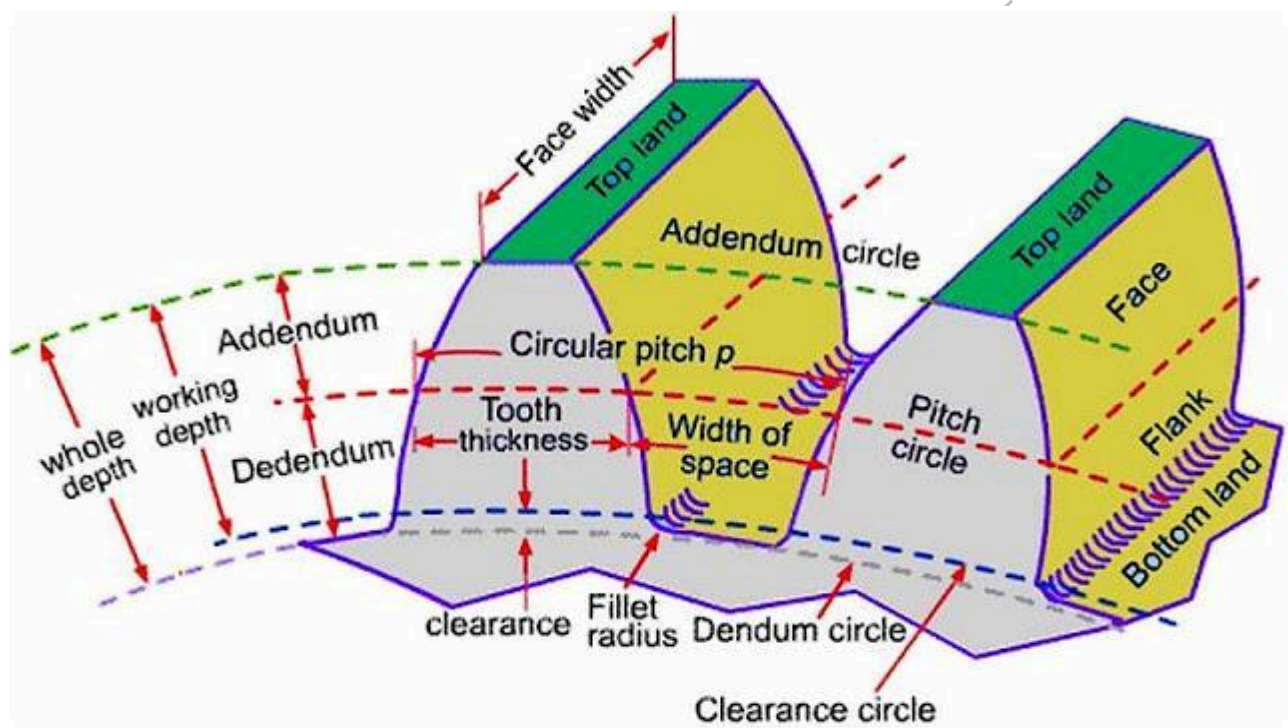
$$T_o = (T_1 + T_2) / 2$$

### 5.1.14 Gear Drives

#### Introduction:

Gear drives are used when the two machine parts are near to each other. Gears give exact velocity ratio, and there is no slip. Hence, they are called positive drives. Gears can be used to transmit power when the two shafts that are to be connected are parallel, intersecting, and non-parallel and non-intersecting.

### 5.1.15 Terminologies of Spur Gear



**Pitch surface:** The surface of the imaginary rolling cylinder (cone, etc.) that the toothed gear may be considered to replace.

**Pitch circle:** A right section of the pitch surface.

**Addendum circle:** A circle bounding the ends of the teeth, in a right section of the gear.

**Root (or dedendum) circle:** The circle bounding the spaces between the teeth, in a right section of the gear.

**Addendum:** The radial distance between the pitch circle and the addendum circle.

**Dedendum:** The radial distance between the pitch circle and the root circle.

**Clearance:** The difference between the dedendum of one gear and the addendum of the mating gear.

**Face of a tooth:** That part of the tooth surface lying outside the pitch surface.

**Flank of a tooth:** The part of the tooth surface lying inside the pitch surface.

**Circular thickness** (also called the **tooth thickness**) : The thickness of the tooth measured on the pitch circle. It is the length of an arc and not the length of a straight line.

**Tooth space:** The distance between adjacent teeth measured on the pitch circle.

**Backlash:** The difference between the circular thickness of one gear and the tooth space of the mating gear. Backlash = Space width – Tooth thickness

**Circular pitch p:** The width of a tooth and a space, measured on the pitch circle.

**Diametral pitch P:** The number of teeth of a gear per inch of its pitch diameter. A toothed gear must have an integral number of teeth. The *circular pitch*, therefore, equals the pitch circumference divided by the number of teeth. The *diametral pitch* is, by definition, the number of teeth divided by the *pitch diameter*.

**Module m:** Pitch diameter divided by number of teeth. The pitch diameter is usually specified in inches or millimetres; in the former case the module is the inverse of diametral pitch.

**Fillet:** The small radius that connects the profile of a tooth to the root circle.

**Pinion:** The smaller of any pair of mating gears. The larger of the pair is called simply the gear.

**Velocity ratio:** The ratio of the number of revolutions of the driving (or input) gear to the number of revolutions of the driven (or output) gear, in a unit of time.

**Pitch point:** The point of tangency of the pitch circles of a pair of mating gears.

**Common tangent:** The line tangent to the pitch circle at the pitch point.

**Base circle:** An imaginary circle used in involute gearing to generate the involutes that form the tooth profiles.

### 5.1.16 Types of gears

Some of the most commonly used types of gears are-

1. Spur gears,
2. Helical gears,
3. Spiral gears,
4. Bevel gears,

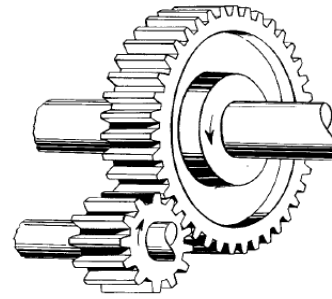


5. Worm and worm-wheel, and
6. Rack and pinion.

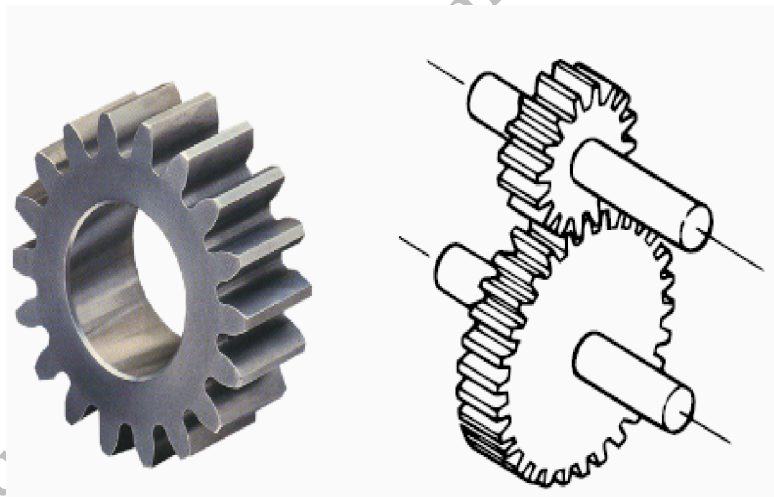
### GEAR DRIVES

A gear is a toothed wheel that engages another toothed mechanism to change speed or the direction of transmitted motion. Gears are generally used for one of four different reasons

1. To increase or decrease the speed of rotation;
2. To change the amount of force or torque;
3. To move rotational motion to a different axis (i.e. parallel, right angles, rotating, linear etc.); and
4. To reverse the direction of rotation. Gears are compact, positive-engagement, power transmission elements capable of changing the amount of force or torque. Sports cars go fast (have speed) but cannot pull any weight.



**SPUR GEARS** Spur gears are used to transmit power between two parallel shafts. The teeth on these gears are cut straight and are parallel to the shafts to which they are attached.



#### Characteristics:

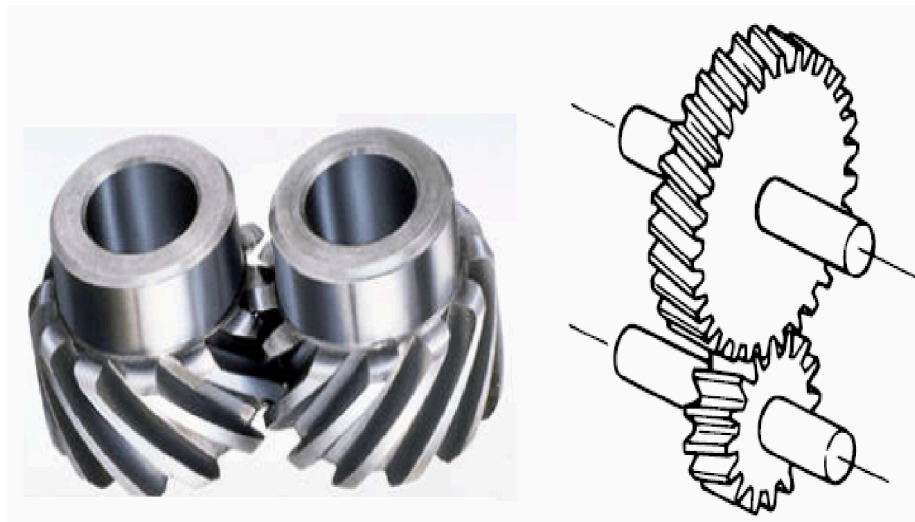
1. Simplest and most economical type of gear to manufacture.
2. Speed ratios of up to 8 (in extreme cases up to 20) for one step (single reduction) Design; up to 45 for two-step design; and up to 200 for three-step design.

#### Limitations:

1. Not suitable when a direction change between the two shafts is required.
2. Produce noise because the contact occurs over the full-face width of the mating Teeth instantaneously.

### HELICAL GEARS

Helical gears resemble spur gears, but the teeth are cut at an angle rather than parallel to the shaft axis like on spur gears. The angle that the helical gear tooth is on is referred to as the helix angle. The angle of helix depends upon the condition of the shaft design and relative position of the shafts. To ensure that the gears run smoothly, the helix angle should be such that one end of the gear tooth remains in contact until the opposite end of the following gear tooth has found a contact. For parallel shafts, the helix angle should not exceed 20 degrees to avoid excessive end thrust.



### Characteristics

The longer teeth cause helical gears to have the following differences from spur gears of the same size:

1. Tooth strength is greater because the teeth are longer than the teeth of spur gear of equivalent pitch diameter.
2. Can carry higher loads than can spur gears because of greater surface contact on the teeth.
3. Can be used to connect parallel shafts as well as non-parallel, non-intersecting shafts.
4. Quieter even at higher speed and are durable.

### Limitations

1. Gears in mesh produce thrust forces in the axial directions.
2. Expensive compared to spur gears.

### BEVEL GEARS

A bevel gear is shaped like a section of a cone and primarily used to transfer power between intersecting shafts at right angles. The teeth of a bevel gear may be straight or spiral. Straight gear is preferred for peripheral speeds up to 1000 feet per minute; above that they tend to be noisy.



#### **Characteristics:**

1. Designed for the efficient transmission of power and motion between intersecting shafts. A good example of bevel gears is seen as the main mechanism for a hand drill. As the handle of the drill is turned in a vertical direction, the bevel gears change the rotation of the chuck to a horizontal rotation.
2. Permit a minor adjustment during assembly and allow for some displacement due to deflection under operating loads without concentrating the load on the end of the tooth.

#### **WORM GEARS**

Worm gears are used to transmit power between two shafts that are at right angles to each other and are non-intersecting. Worm gears are special gears that resemble screws, and can be used to drive spur gears or helical gears. Worm gearing is essentially a special form of helical gearing in which the teeth have line contact and the axes of the driving and driven shafts are usually at right angles and do not intersect.



#### **Characteristics:**

1. Meshes are self-locking. Worm gears have an interesting feature that no other gear set has: the worm can easily turn the gear, but the gear cannot turn the worm. This is because the angle on the worm is so shallow that when the gear tries to spin it, the friction between the gear and the worm holds the worm in place. This feature is useful for machines such as conveyor systems, in which the locking feature can act as a brake for the conveyor when the motor is not turning.

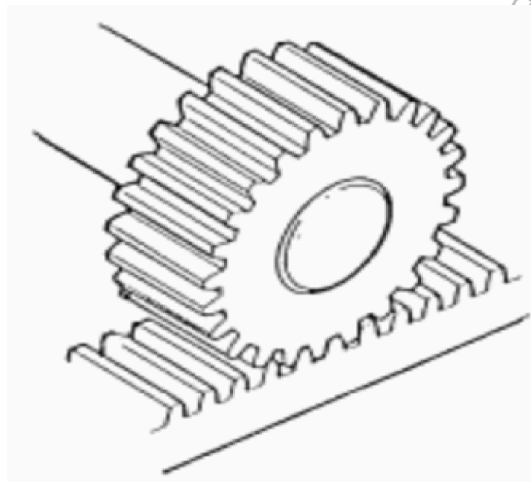
2. Worm gear is always used as the input gear, i.e. the torque is applied to the input end of the worm shaft by a driven sprocket or electric motor.
3. Best suited for applications where a great ratio reduction is required between the driving and driven shafts. It is common for worm gears to have reductions of 20:1, and even up to 300:1 or greater.

**Limitations:**

1. Yield low efficiency because of high sliding velocities across the teeth, thereby causing high friction losses.
2. When used in high torque applications, the friction causes the wear on the gear
3. Teeth and erosion of the restraining surface.

**RACKS (STRAIGHT GEARS)**

The rack is a bar with a profile of the gear of infinite diameter, and when used with a meshing pinion, enables the rotary to linear movement or vice versa.



**Characteristics:**

1. Racks with machined ends can be joined together to make any desired length.
2. The most well known application of a rack is the rack and pinion steering system used on many cars in the past. The steering wheel of a car rotates the gear that engages the rack. The rack slides right or left, when the gear turns, depending on the way we turn the wheel. Windshield wipers in cars are powered by a rack and pinion mechanism.

**5.1.17 Gear Train**

Types of gear trains:

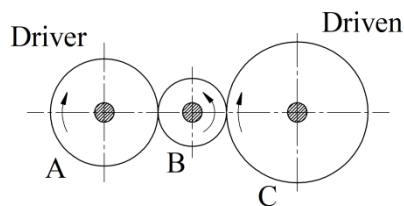
The different types of gear trains commonly used are –

- (i) Simple gear trains,
- (ii) Compound gear trains,

- (iii) Reverted gear trains, and
- (iv) Epicyclic gear trains.

### Simple gear trains:

In a simple gear train, each gear wheel is mounted on an independent shaft, and no two gears are on the same shaft. Figure shows a simple gear train consisting of three gear wheels.



Let A be the driver gear,

C be the driven gear, and

B be the intermediate gear.

Velocity ratio,

$$\frac{N_C}{N_A} = \frac{N_C}{N_B} \times \frac{N_B}{N_A}$$

$$\text{But, } \frac{N_C}{N_B} = -\frac{T_B}{T_C}, \quad \frac{N_B}{N_A} = -\frac{T_A}{T_B}$$

Therefore,

$$\frac{N_C}{N_A} = -\frac{T_B}{T_C} \times -\frac{T_A}{T_B} = \frac{T_A}{T_C}$$

In the above equations, the negative sign indicates that the gears in mesh rotate in opposite directions, and positive sign indicates that the two gears rotate in the same direction.



From the velocity ratio we can see that the size of the intermediate gear does not affect the velocity ratio. The result will be the same even when we have more than one intermediate gear of different sizes. The presence of the intermediate gears determines the direction of rotation of the driven gear with respect to the driving gear. If there are an odd number of intermediate gears, then the driver and the driven rotate in the same direction, and if there are an even number of intermediate gears, the driver and the driven rotate in opposite directions.

### Compound gear trains:

In this type of arrangement, more than one gear is mounted on the same shaft. The gears which are mounted on the same shaft rotate in the same direction with the same speed irrespective of their sizes.

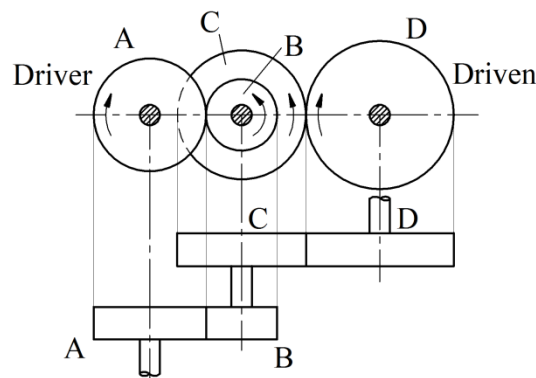


Figure shows a compound gear train arrangement. Gear A is the driver, and gear D is the driven gear. The intermediate gears B and C are mounted on the same shaft. Gear A meshes with gear B, and gear D meshes with gear C.

Velocity ratio,

$$N_D/N_A = N_D/N_C \times N_C/N_B \times N_B/N_A$$

$$\text{But, } \frac{N_D}{N_C} = \frac{T_C}{T_D}, \quad \frac{N_C}{N_B} = 1, \quad \frac{N_B}{N_A} = \frac{T_A}{T_B}$$

Therefore,

$$\frac{N_D}{N_A} = \frac{N_D}{N_C} \times \frac{N_C}{N_B} \times \frac{N_B}{N_A} = \frac{T_C}{T_D} \times 1 \times \frac{T_A}{T_B} = \frac{T_A \cdot T_C}{T_B \cdot T_D}$$

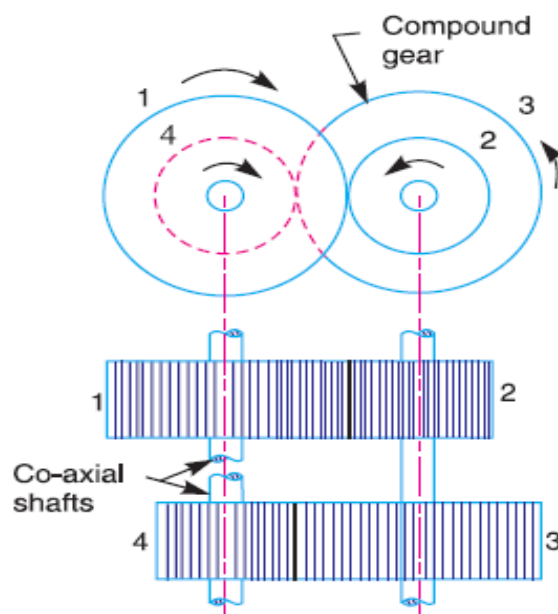


$N_A \quad N_C \quad N_B \quad N_A \quad T_D \quad T_B \quad T_B \cdot T_D$

Gear A is the first driver (drives gear B) and gear C is the second driver (drives gear D).

## REVERTED GEAR TRAIN

When the axes of the first gear (*i.e.*, first driver) and the last gear (*i.e.*, last driven or follower) are co-axial, then the gear train is known as **reverted gear train** as shown in Fig. We see that gear 1 (*i.e.*, first driver) drives the gear 2 (*i.e.*, first driven or follower) in the opposite direction. Since the gears 2 and 3 are mounted on the same shaft, therefore they form a compound gear and the gear 3 will rotate in the same direction as that of gear 2. The gear 3 (which is now the second driver) drives the gear 4 (*i.e.*, the last driven or follower) in the same direction as that of gear 1. Thus, we see that in a reverted gear train, the motion of the first gear and the last gear



Let  $T_1$  = Number of teeth on gear 1,  
 $r_1$  = Pitch circle radius of gear 1, and  
 $N_1$  = Speed of gear 1 in r. p.m.

Similarly,  
 $T_2, T_3, T_4$  = Number of teeth on respective gears,  
 $r_2, r_3, r_4$  = Pitch circle radii of respective gears, and  
 $N_2, N_3, N_4$  = Speed of respective gears in r. p.m.

Since the distance between the centres of the shafts of gears 1 and 2 as well as gears 3 and 4

is same, therefore

$$r_1 + r_2 = r_3 + r_4 \dots (i)$$

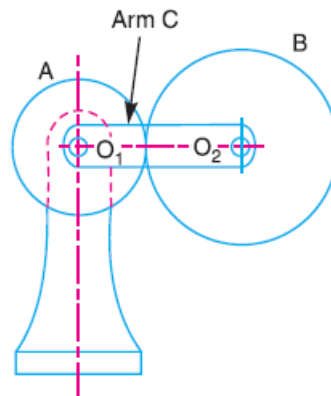
Also, the circular pitch or module of all the gears is assumed to be same, therefore number of

teeth on each gear is directly proportional to its circumference or radius.

$$\therefore T_1 + T_2 = T_3 + T_4 \dots \text{(ii)}$$

## EPICYCLIC GEAR TRAIN

A simple epicyclic gear train is shown in Fig. 13.6, where a gear *A* and the arm *C* have a common axis at *O*<sub>1</sub> about which they can rotate. The gear *B* meshes with gear *A* and has its axis on the arm at *O*<sub>2</sub>, about which the gear *B* can rotate. If the arm is fixed, the gear train is simple and gear *A* can drive gear *B* or **vice-versa**, but if gear *A* is fixed and the arm is rotated about the axis of gear *A* (i.e., *O*<sub>1</sub>), then the gear *B* is forced to rotate **upon** and **around** gear *A*. Such a motion is called **epicyclic** and the gear trains arranged in such a manner that one or more of them



members move upon and around another member are known as **epicyclic gear trains** (**epi.** means upon and **cyclic** means around). The epicyclic gear trains may be **simple** or **compound**. The epicyclic gear trains are useful for transmitting high velocity ratios with gears of moderate size in a comparatively lesser space. The epicyclic gear trains are used in the back gear of lathe, differential gears of the automobiles, hoists, pulley blocks, wrist watches etc.

## 5.2 Introduction to Hybrid Electric Vehicles

### 5.2.1 Introduction:

What is a hybrid? A hybrid vehicle combines any two power (energy) sources. Possible combinations include diesel/electric, gasoline/fly wheel, and fuel cell (FC)/battery. Typically, one energy source is storage, and the other is conversion of a fuel to energy. The combination of two power sources may support two separate propulsion systems. Thus to be a True hybrid, the vehicle must have at least two modes of propulsion.

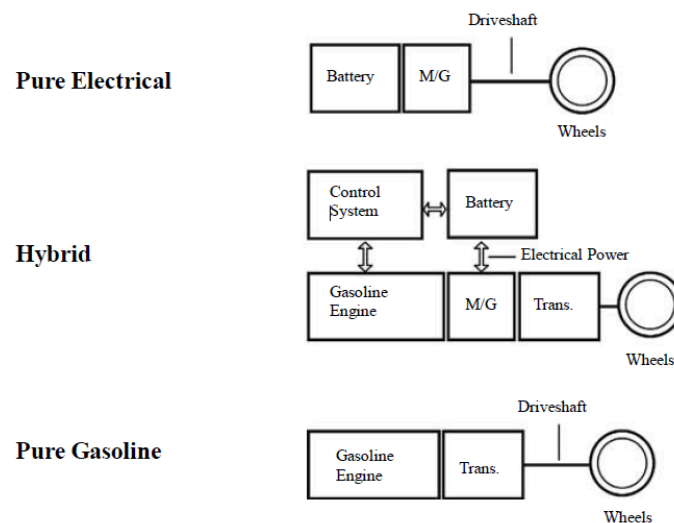
For example, a truck that uses a diesel to drive a generator, which in turn drives several electrical motors for all-wheel drive, is *not a hybrid*. But if the

truck has electrical energy storage to provide a second mode, which is electrical assists, then it is a hybrid Vehicle.

These two power sources may be paired in series, meaning that the gas engine charges the batteries of an electric motor that powers the car, or in parallel, with both mechanisms driving the car directly.

### 5.2.2 Hybrid electric vehicle (HEV)

Consistent with the definition of hybrid above, the hybrid electric vehicle combines a gasoline engine with an electric motor. An alternate arrangement is a diesel engine and an electric motor (figure 1).



**Figure 1: Components of a hybrid Vehicle that combines a pure gasoline with a pure EV. [1]**

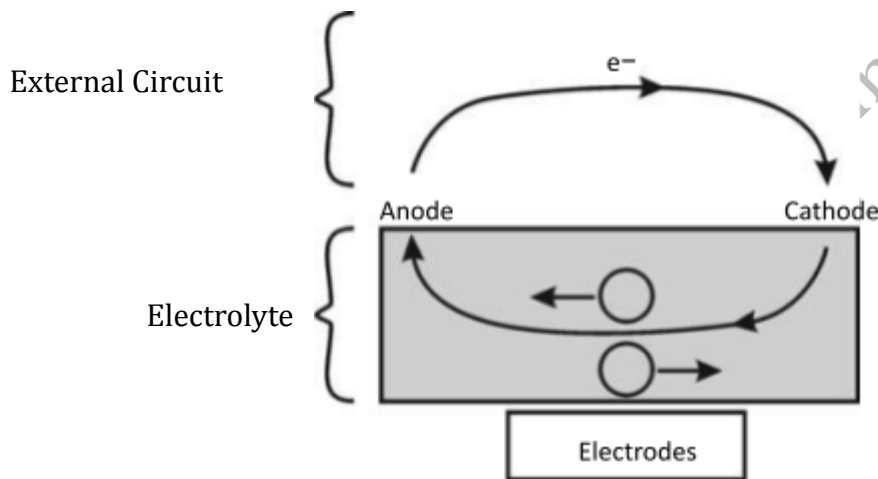
As shown in **Figure 1**, a HEV is formed by merging components from a pure electrical vehicle and a pure gasoline vehicle. The Electric Vehicle (EV) has an M/G which allows regenerative braking for an EV; the M/G installed in the HEV enables regenerative braking. For the HEV, the M/G is tucked directly behind the engine. In Honda hybrids, the M/G is connected directly to the engine. The transmission appears next in line. This arrangement has two torque producers; the M/G in motor mode, M-mode, and the gasoline engine. The battery and M/G are connected electrically.

HEVs are a combination of electrical and mechanical components. Three main sources of electricity for hybrids are batteries, FCs, and capacitors. Each

device has a low cell voltage, and, hence, requires many cells in series to obtain the voltage demanded by an HEV. Difference in the source of Energy can be explained as:

- The FC provides high energy but low power.
- The battery supplies both modest power and energy.
- The capacitor supplies very large power but low energy.

The components of an electrochemical cell include anode, cathode, and electrolyte (shown in fig2). The current flow both internal and external to the cell is used to describe the current loop.



**Figure 2: An electrode, a circuit for a cell which is converting chemical energy to electrical energy. The motion of negative charges is clockwise and forms a closed loop through external wires and load and the electrolyte in the cell. [1]**

A critical issue for both battery life and safety is the precision control of the Charge/Discharge cycle. Overcharging can be traced as a cause of fire and failure.

Applications impose two boundaries or limitations on batteries. The first limit, which is dictated by battery life, is the minimum allowed State of Charge. As a result, not all the installed battery energy can be used. The battery feeds energy to other electrical equipment, which is usually the inverter. This equipment can use a broad range of input voltage, but cannot accept a low voltage. The second limit is the minimum voltage allowed from the battery.

### 5.2.3 Present of Hybrid Electric vehicle

Toyota is the most prominent of all manufacturers when it comes to hybrid cars. As well as the specialist hybrid range they have produced hybrid versions of many of their existing model lines, including several Lexus (now owned and manufactured by Toyota) vehicles. They have also stated that it is their intention to release a hybrid version of every single model they release in the coming decade. As well as cars and SUVs, there are a select number of hybrid motorcycles, pickups, vans, and other road going vehicles available to the consumer and the list is continually increasing.

### 5.2.4 Future of Hybrid electrical vehicle

Since petroleum is limited and will someday run out of supply. In the arbitrary year 2037, an estimated one billion petroleum-fueled vehicles will be on the world's roads. gasoline will become prohibitively expensive. The world need to have solutions for the **"400 million otherwise useless cars"**. So year 2037 "gasoline runs out year" means, petroleum will no longer be used for personal mobility. A market may develop for solar-powered EVs of the size of a scooter or golf cart. Since hybrid technology applies to heavy vehicles, hybrid buses and hybrid trains will be more significant.

### 5.2.5 Comparison of Conventional vehicle with EV's

| IC Engine (ICE) Vehicles                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Electric Vehicles (EV)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Powertrain: IC engine</li> <li>• High specific energy of fuel</li> <li>• Power density: High</li> <li>• Emits greenhouse gases</li> <li>• Travels &gt; 300 miles / fill</li> <li>• Short refilling time (&lt; 5 min.)</li> <li>• Fuel tank takes less space</li> <li>• Fuel weight is very less</li> <li>• Higher maintenance costs</li> <li>• Braking energy not recovered</li> <li>• Running cost: high</li> <li>• Engine efficiency: ~ 30%</li> <li>• Needs complex gear system</li> <li>• Noisy operation</li> <li>• Ample refilling infrastructure</li> <li>• Need to pick up some speed to deliver maximum torque</li> <li>• Uses only hydrocarbons</li> </ul> | <ul style="list-style-type: none"> <li>• Powertrain: Motor (+ Engine)</li> <li>• Low specific energy of battery</li> <li>• Power density: Low</li> <li>• No tailpipe emissions</li> <li>• Travels &lt; 100 miles / charge</li> <li>• Long charging time (0.5-8 hr.)</li> <li>• Battery takes large space</li> <li>• Batteries are very heavy</li> <li>• Lesser maintenance costs</li> <li>• Can recover braking energy</li> <li>• Running cost: low</li> <li>• Motor efficiency: ~ 80%</li> <li>• Needs only one gear</li> <li>• Quiet operation</li> <li>• Lacks charging infrastructure</li> <li>• Produce maximum torque instantly after starting of motor</li> <li>• Uses electricity from many resources</li> </ul> |

### 5.2.6 Basic Architecture of Hybrid Drive

#### The Hybrid Electric Vehicle (HEV)

What exactly is an HEV? The definition available is so general that it anticipates future technologies of energy sources. The term **hybrid vehicle** refers to a vehicle with at least two sources of power. A **hybrid-electric vehicle** indicates that **one source of power** is provided by an **electric motor**. The **other source of motive power** can come from a number of different technologies, but is typically provided by an **internal combustion engine** designed to run on either gasoline or diesel fuel. As proposed by Technical Committee (Electric Road Vehicles) of the International Electrotechnical Commission, **an HEV is a vehicle in which propulsion energy is available from two or more types of energy sources and at least one of them can deliver electrical energy**. Based on this general definition, there are many types of HEVs, such as:

- the gasoline ICE and battery
- diesel ICE and battery
- battery and FC
- battery and capacitor
- battery and flywheel
- battery and battery hybrids.

Most commonly, the propulsion force in HEV is provided by a combination of electric motor and an ICE. The electric motor is used to improve the energy efficiency (improves fuel consumption) and vehicular emissions while the ICE provides extended range capability.

### 5.2.7 Energy Use in Conventional Vehicles

In order to understand how a HEV may save energy, it is necessary first to examine how conventional vehicles use energy. The breakdown of energy use in a vehicle is as follows:

- In order to maintain movement, vehicles must produce power at the wheels to overcome:
  - a. aerodynamic drag (air friction on the body surfaces of the vehicle, coupled with pressure forces caused by the air flow)
  - b. rolling resistance (the resistive forces between tires and the road surface)
  - c. resistive gravity forces associated with climbing a grade
- Further, to accelerate, the vehicle must its inertia. Most of the energy expended in acceleration is then lost as heat in the brakes when the vehicle is brought to a stop.
- The vehicle must provide power for accessories such as heating fan, lights, power steering, and air conditioning.
- Finally, a vehicle will need to be capable of delivering power for acceleration with very little delay when the driver depresses the accelerator, which may necessitate keeping the power source in a standby (energy-using) mode.

A conventional engine-driven vehicle uses its engine to translate fuel energy into shaft power, directing most of this power through the drivetrain to turn the wheels. Much of the heat generated by combustion cannot be used for work and is wasted, both because heat engines have theoretical efficiency

limit. Moreover, it is impossible to reach the theoretical efficiency limit because:

- some heat is lost through cylinder walls before it can do work
- some fuel is burned at less than the highest possible pressure
- fuel is also burned while the engine is experiencing negative load (during braking) or when the vehicle is coasting or at a stop, with the engine idling.

### 5.2.8 Energy Savings Potential of Hybrid Drive trains

In terms of overall energy efficiency, the conceptual advantages of a hybrid over a conventional vehicle are:

- **Regenerative braking.** A hybrid can capture some of the energy normally lost as heat to the mechanical brakes by using its electric drive motor(s) in generator mode to brake the vehicle
- **More efficient operation of the ICE, including reduction of idle.** A hybrid can avoid some of the energy losses associated with engine operation at speed and load combinations where the engine is inefficient by using the energy storage device to either absorb part of the ICE's output or augment it or even substitute for it. This allows the ICE to operate only at speeds and loads where it is most efficient. When an HEV is stopped, rather than running the engine at idle, where it is extremely inefficient, the control system may either shut off the engine, with the storage device providing auxiliary power (for heating or cooling the vehicle interior, powering headlights, etc.), or run the engine at a higher-than-idle (more efficient) power setting and use the excess power (over auxiliary loads) to recharge the storage device. When the vehicle control system can shut the engine off at idle, the drivetrain can be designed so that the drive motor also serves as the starter motor, allowing extremely rapid restart due to the motor's high starting torque.
- **Smaller ICE:** Since the storage device can take up a part of the load, the HEV's ICE can be down sized. The ICE may be sized for the continuous load and not for the very high short term acceleration load.



This enables the ICE to operate at a higher fraction of its rated power, generally at higher fuel efficiency, during most of the driving.

There are counterbalancing factors reducing hybrids' energy advantage, including:

- **Potential for higher weight.** Although the fuel-driven energy source on a hybrid generally will be of lower power and weight than the engine in a conventional vehicle of similar performance, total hybrid weight is likely to be higher than the conventional vehicle it replaces because of the added weight of the storage device, electric motor(s), and other components. This depends, of course, on the storage mechanism chosen, the vehicle performance requirements, and so forth.
- **Electrical losses.** Although individual electric drivetrain components tend to be quite efficient for one-way energy flows, in many hybrid configurations, electricity flows back and forth through components in a way that leads to cascading losses. Further, some of the components may be forced to operate under conditions where they have reduced efficiency. For example, like ICEs, most electric motors have lower efficiency at the low-speed, low-load conditions often encountered in city driving. Without careful component selection and a control strategy that minimizes electric losses, much of the theoretical efficiency advantage often associated with an electric drivetrain can be lost.

## 5.2.9 VEHICLE HYBRIDISATION

### Based on Degree of Hybridization:

Parallel and combined hybrids can be categorized according to degree of hybridization. Degree of hybridization depends upon the power supplied by IC engine and electric motor. In some vehicles, IC engine is dominant; electric motor turns on only when boost is needed. In many vehicles, both IC engine and electric motor share equal loads. Others can run only with electric motor system operating. The ratio of power developed by an electric motor in a

hybrid vehicle to the total power consumed by the vehicle is known as degree of hybridization.

$$\text{Degree of hybridization} = \frac{\text{Motor Power}}{\text{Motor Power} + \text{Engine Power}} * 100$$

| Type Of hybrid                     | Degree of hybridization      |
|------------------------------------|------------------------------|
| Micro                              | <5 %                         |
| Mild                               | Up to 10 %                   |
| Full Hybrid:<br>Parallel<br>Series | 10 % to 50 %<br>50 % to 75 % |
| Electric vehicle                   | 100 %                        |

### 1) Micro Hybrid:

- Electric motor functions to start or stop the system to automatically shut off the engine while idling.
- This motor does not provide additional torque to the vehicle.
- Electric Motor supplies power 2.5kW at 12 volts.
- Energy saving 5 to 10%.
- Example: BMW 1 series, Fortwo Mercedes, etc

### 2) Mild Hybrid:

- Electric motor generator is integrated to provide 10% of maximum engine power.
- These hybrids improve drawbacks of fossil fuel vehicles. Here motor or generator is in parallel with IC engine.
- Electric Motor supplies power 10 to 20 kW at 100-200 volts.
- Energy saving 20 to 30%

- Examples: Chevrolet Malibu, Chevrolet Silverado, Honda Escape etc.

### **3) Full Hybrid:**

- Electric motor provides at least 40% of engine power as additional torque.
- Bigger motor and battery reduces the required size of conventional engine.
- It has improved fuel consumptions and reduced emissions.
- Circa Electric Motor supplies power 50 kW at 200-300 volts.
- Energy saving 30 to 50%
- Example: -Toyota Prius, Camry Hybrid, Ford Escape Hybrid, Ford Fusion Hybrid/Lincoln MKZ Hybrid, Ford C-Max Hybrid, Kia Optima Hybrid etc.

### **4) Plug-in Hybrid:**

- Plug-in hybrid electric vehicles—known as PHEVs—combine a gasoline or diesel engine with an electric motor and a large rechargeable battery.
- Unlike conventional hybrids, these hybrids can be plugged-in and recharged from an outlet, allowing the vehicle to drive extended distances using just electricity.
- When the battery is emptied, the conventional engine turns on and the vehicle operates as a conventional, non-plug-in hybrid.
- Example: Chevrolet Volt, Mitsubishi Outlander P-HEV, Toyota Prius P-HEV, etc.

## **5.2.10 COMPONENTS OF EV**

The basic main elements of electric cars installed in almost all types of electric cars are as follows:

### ***Traction Battery Pack***

The function of the battery in an electric car is as an electrical energy storage system in the form of direct-current electricity (DC). If it gets a signal from the controller, the battery will flow DC electrical energy to the inverter to then be used to drive the motor. The type of battery used is a rechargeable battery that is arranged in such a way as to form what is called a *traction battery pack*.

There are various types of electric car batteries. The most widely used is the type of lithium-ion batteries. Please read the article "[Electric Car Batteries and Their Characteristics](#)" to get a little idea about batteries for electric cars.

### **Power Inverter**

The inverter functions to change the direct current (DC) on the battery into an alternating current (AC) and then this alternating current is used by an electric motor. In addition, the inverter on an electric car also has a function to change the AC current when regenerative braking to DC current and then used to recharge the battery. The type of inverter used in some electric car models is the bi-directional inverter category.

### ***Controller***

The main function of the controller is as a regulator of electrical energy from batteries and inverters that will be distributed to electric motors. While the controller itself gets the main input from the car pedal (which is set by the driver). This pedal setting will determine the frequency variation or voltage variation that will enter the motor, and at the same time determine the car's speed.

In brief, this unit manages the flow of electrical energy delivered by the traction battery, controlling the speed of the electric traction motor and the torque it produces. This component will determine how electric car work.

### ***Electric Traction Motor***

Because the controller provides electrical power from the traction battery, the electric traction motors will work turning the transmission and wheels. Some hybrid electric cars use a type of generator-motor that performs the functions of propulsion and regeneration. In general, the type of electric motor used is the BLDC (brushless DC) motor.



**Charger :** It is a battery charging device. Chargers get electricity from outside sources, such as the utility grid or solar power plants. AC electricity is converted into DC electricity and then stored in the battery. There are 2 types of electric car chargers:

- On-board charger: the charger is located and installed in the car
- Off-board charger: the charger is not located or not installed in the car.

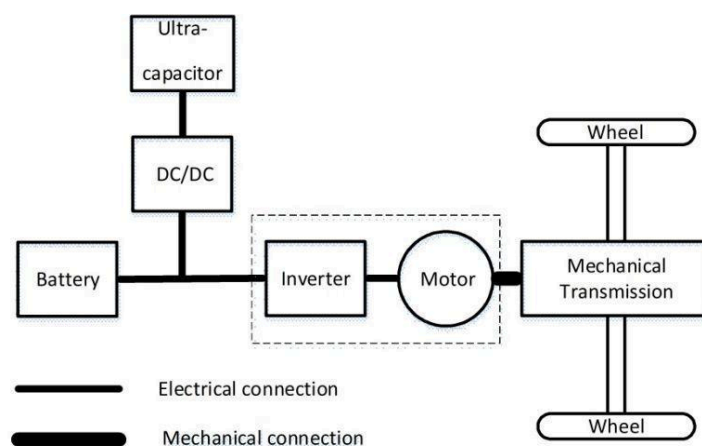
**Transmission:** The transmission transfers mechanical power from the electric traction motor to drive the wheels.

**DC/DC Converter:** This one of electric car parts that to converts higher-voltage DC power from the traction battery pack to the lower-voltage DC power needed to run vehicle accessories and recharge the auxiliary battery.

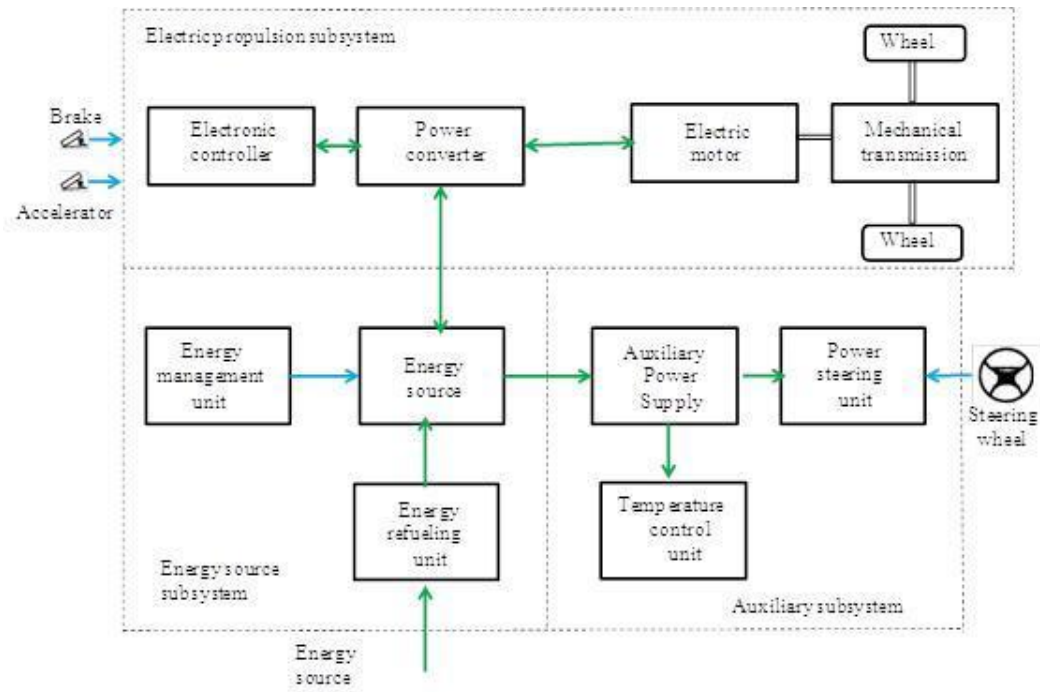
**Battery:** In an electric drive vehicle, the auxiliary battery provides electricity to power vehicle accessories.

**Thermal System – Cooling:** This system maintains a proper operating temperature range of the engine, electric motor, power electronics, and other components.

**Charge Port:** The charge port allows the vehicle to connect to an external power supply in order to charge the traction battery pack.



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Based on the control inputs from the brake and accelerator pedals, the electronic controller provides proper control signals to switch on or off the power converter which in turn regulates the power flow between the electric motor and the energy source. The backward power flow is due to regenerative braking of the EV and this regenerative energy can be stored provided the energy source is receptive. The energy management unit cooperates with the electronic controller to control regenerative braking and its energy recovery. It also works with the energy-refueling unit to control refueling and to monitor usability of the energy source. The auxiliary power supply provides the necessary power with different voltage levels for all EV auxiliaries, especially the temperature control and power steering units.

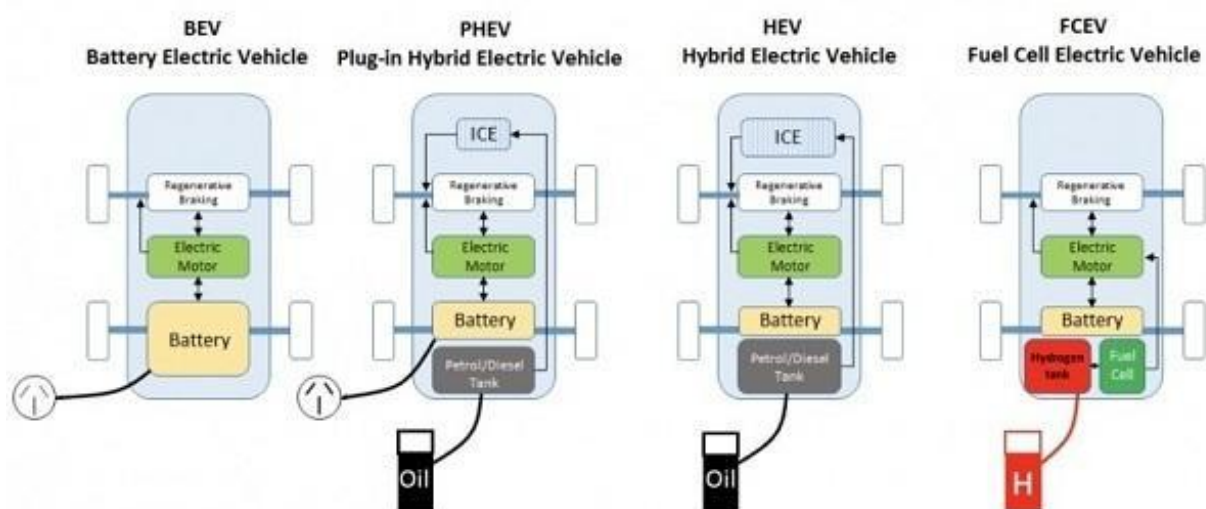
### 5.2.11 Types of EV's

EVs can run solely on electric propulsion or they can have an ICE working alongside it. Having only batteries as energy source constitutes the basic kind of EV, but there are kinds that can employ other energy source modes. These can be called hybrid EVs (HEVs). The International Electrotechnical Commission's Technical Committee 69 (Electric Road Vehicles) proposed that vehicles using two or more types of energy source, storage or converters can be called as an HEV as long as at least one of those provide electrical energy. This definition makes a lot of combinations possible for HEVs

like ICE and battery, battery and flywheel, battery and capacitor, battery and fuel cell, etc. Therefore, the common population and specialists both started calling vehicles with an ICE and electric motor combination HEVs, battery and capacitor ones as ultra-capacitor-assisted EVs, and the ones with battery and fuel cell FCEVs.

These terminologies have become widely accepted and according to this norm, EVs can be categorized as follows:

- (1) Battery Electric Vehicle (BEV)
- (2) Hybrid Electric Vehicle (HEV)
- (3) Plug-in Hybrid Electric Vehicle (PHEV)
- (4) Fuel Cell Electric Vehicle (FCEV)

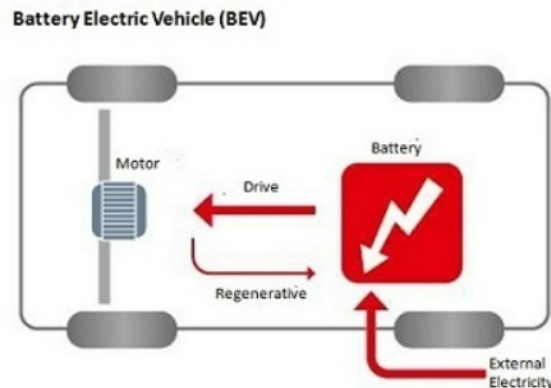


### 5.2.12 Battery Electric Vehicle (BEV)

EVs with only batteries to provide power to the drive train are known as BEVs. BEVs have to rely solely on the energy stored in their battery packs; therefore, the range of such vehicles depends directly on the battery capacity. Typically, they can cover 100 km–250 km on one charge, whereas the top-tier models can go a lot further, from 300 km to 500 km. These ranges depend on driving condition and style, vehicle configurations, road conditions, climate, battery type and age. Once depleted, charging the battery pack takes quite a lot of time compared to refueling a conventional ICE

vehicle. It can take as long as 36 h completely replenish the batteries, there are far less time consuming ones as well, but none is comparable to the little time required to refill a fuel tank. Charging time depends on the charger configuration, its infrastructure and operating power level.

Advantages of BEVs are their simple construction, operation and convenience. These do not produce any greenhouse gas (GHG), do not create any noise and therefore beneficial to the environment. Electric propulsion provides instant and high torques, even at low speeds. These advantages, coupled with their limitation of range, makes them the perfect vehicle to use in urban areas, as urban driving requires running at slow or medium speeds, and these ranges demand a lot of torque. Nissan Leaf and Teslas are some high-selling BEVs these days, along with some Chinese vehicles.

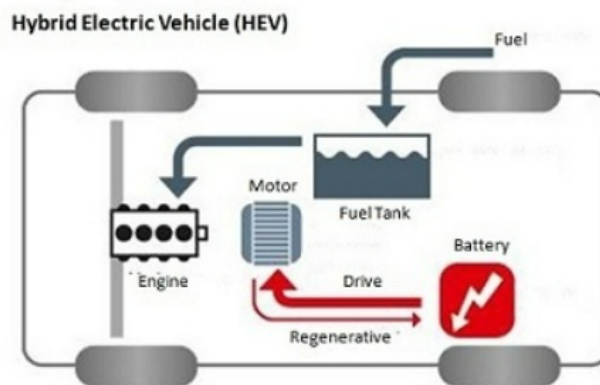


### 5.2.11 Hybrid Electric Vehicle (HEV)

HEVs employ both an ICE and an electrical power train to power the vehicle. The combination of these two can come in different forms. An HEV uses the electric propulsion system when the power demand is low. It is a great advantage in low-speed conditions like urban areas; it also reduces the fuel consumption as the engine stays totally off during idling periods, for example, traffic jams. This feature also reduces the GHG emission. When higher speed is needed, the HEV switches to the ICE. The two drive trains can also work together to improve the performance. Hybrid power systems are used extensively to reduce or to completely remove turbo lag in turbocharged cars, like the Acura NSX. It also enhances performance by filling the gaps between gear shifts and providing speed boosts when required. The ICE can charge up the batteries, HEVs can also retrieve energy by means of regenerative braking. Therefore, HEVs are primarily



ICE driven cars that use an electrical drive train to improve mileage or for performance enhancement. To attain these features, HEV configurations are being widely adopted by car manufacturers. While starting the vehicle, the ICE may run the motor as a generator to produce some power and store it in the battery. Passing needs a boost in speed, therefore the ICE and the motor both drives the power train. During braking the power train runs the motor as generator to charge the battery by regenerative braking. While cruising, ICE runs the both the vehicle and the motor as generator, which charges the battery. The power flow is stopped once the vehicle stops.

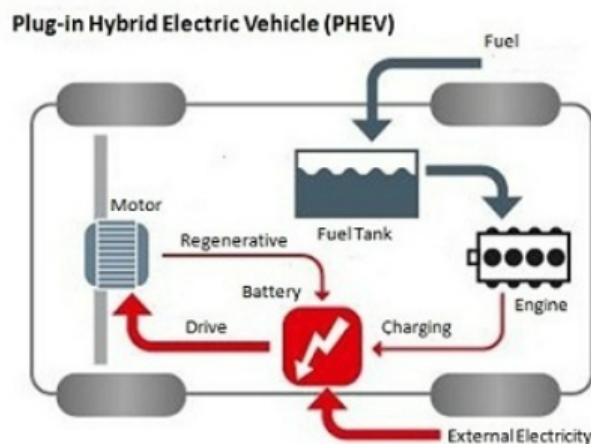


### 5.2.12 Plug-in Hybrid Electric Vehicle (PHEV)

The PHEV concept arose to extend the all-electric range of HEVs. It uses both an ICE and an electrical power train, like a HEV, but the difference between them is that the PHEV uses electric propulsion as the main driving force, so these vehicles require a bigger battery capacity than HEVs. PHEVs start in 'all electric' mode, runs on electricity and when the batteries are low in charge, it calls on the ICE to provide a boost or to charge up the battery pack. The ICE is used here to **extend the range**. Hence the name Range Extender. The range extender has large value for  $H$  (*Degree of hybridization*).

PHEVs can charge their batteries directly from the grid (which HEVs cannot); they also have the facility to utilize regenerative braking. PHEVs' ability to run solely on electricity for most of the time makes its carbon footprint smaller than the HEVs. They consume less fuel as well and thus reduce the associated cost. The vehicle market is now quite populated with these, Chevrolet Volt and Toyota Prius sales show their popularity as well.

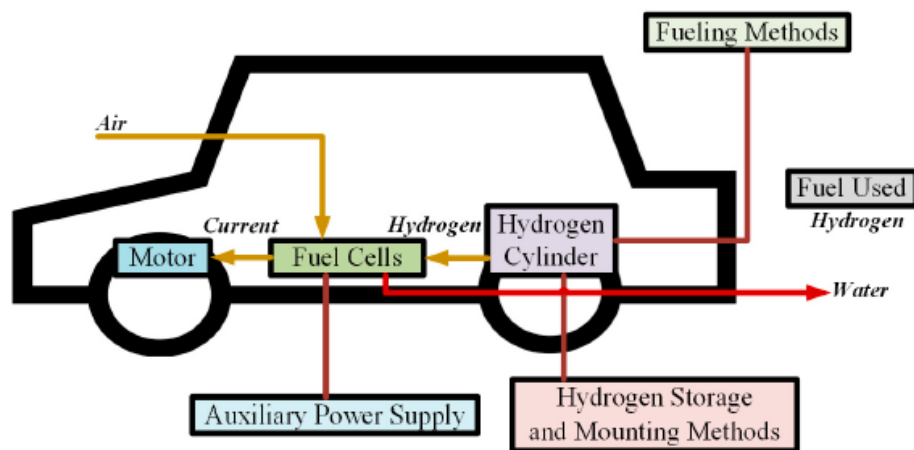
The plug-in hybrid can be viewed as an EV but with a small engine to extend range. Features of a plug-in hybrid include a large, heavy, expensive battery. The comparison with a full hybrid is a battery of a few 45.36 kg instead of the typical 45.36 kg in a full hybrid. Additional equipment is needed to connect to external “wall plug” electrical source for recharging. Since batteries are high voltage, the voltage of the charging source must be even higher. Inductive rechargers prevent exposure to high voltage. The plug-in will likely have small gasoline engine driven generator for on-board charging; this engine separates the plug-in hybrid from the EV. For people willing to undertake the recharging chore, the plug-in offers fantastic mpg. To gain the benefits, the range of hybridness for a plug-in is  $50\% < H < 100\%$  with  $H$  likely to be closer to 100%.



### 5.2.12 Fuel Cell Electric Vehicle (FCEV)

FCEVs also go by the name Fuel Cell Vehicle (FCV). They got the name because the heart of such vehicles is fuel cells that use chemical reactions to produce electricity. Hydrogen is the fuel of choice for FCVs to carry out this reaction, so they are often called ‘hydrogen fuel cell vehicles’. FCVs carry the hydrogen in special high-pressure tanks, another ingredient for the power generating process is oxygen, which it acquires from the air sucked in from the environment. Electricity generated from the fuel cells goes to an electric motor which drives the wheels. Excess energy is stored in storage systems like batteries or supercapacitors. Commercially available FCVs like the Toyota Mirai or Honda Clarity use batteries for this purpose. FCVs only produce water as a by-product of its power generating process which is ejected out of the car through the tailpipes. The configuration of an FCV is shown in Figure. An advantage of such vehicles is they can

produce their own electricity which emits no carbon, enabling it to reduce its carbon footprint further than any other EV. Another major advantage of these are, and maybe the most important one right now, refilling these vehicles takes the same amount of time required to fill a conventional vehicle at a gas pump. This makes adoption of these vehicles more likely in the near future. A major current obstacle in adopting this technology is the scarcity of hydrogen fuel stations.



**Fuel-Cell Electric Vehicle (FCEV)**

